

Global Smoothness for the Three-Dimensional Incompressible Navier–Stokes Equations

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Abstract

We prove global existence and smoothness for the unforced three-dimensional incompressible Navier–Stokes equations on $\mathbb{R}^3$, for every positive viscosity and every divergence-free Schwartz initial velocity. Let (u, p) be the pressure-normalized maximal classical solution on $[0, T_*)$. The main estimate sends the datum to a compatible family of finite mixed rectangles spanning the whole interval $(0, T_*)$: each chosen temporal row is controlled in two adjacent spatial Sobolev spaces, and the differentiated pressure law retains every term generated by transport. If T_* were finite, the adjacent controls would give an endpoint trace at every finite order. Compatibility identifies those traces as one H^∞ velocity and pressure jet, while the limiting momentum and pressure recurrences make that jet the initial jet of a local restart. Uniqueness joins the restart to the original history, contradicting maximality. The proof also keeps the size of its conclusion. Once global smoothness is known, a residual-certificate construction takes u_0 , ν , a finite horizon, and a requested Sobolev order and produces a finite checkable majorant for the solution. For effectively represented data and computable ν and horizon, a fixed dovetailed search terminates with such a verified numerical majorant. Substitution into the rectangle construction gives explicit finite recursions for all mixed velocity and pressure norms, critical-height mass and frequency tails, endpoint moduli, and restart lifespan, with every datum order and horizon dependence displayed. Replacing the base producer while retaining the mixed-row, pressure, endpoint, and restart construction gives quantitative global theorems for three-dimensional axisymmetric flow without swirl, critical hyperdissipation, small critical-data viscous–resistive MHD, and the periodic Navier–Stokes–Voigt equation; the last application also yields an explicit radius of time analyticity.

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Part I

The finite maximal-time problem

1 The whole-space problem and its first possible singular time

The three-dimensional Navier–Stokes regularity problem asks whether a smooth solution can lose its smoothness in finite time. Fix smooth, divergence-free, finite-energy initial data u_0 , fix $\nu > 0$, and let (u, p) be the corresponding maximal classical solution on $[0, T_*)$. We want to prove that T_* cannot be finite.

Fefferman’s formulation of the whole-space Millennium Prize Problem takes u_0 to be smooth and divergence-free and imposes a decay estimate on every spatial derivative at every polynomial order. We write this class as

$$\mathcal{S}_\sigma(\mathbb{R}^3) := \left\{ a \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) : \begin{array}{l} \nabla \cdot a = 0, \\ \forall \alpha \in \mathbb{N}_0^3 \ \forall K \in \mathbb{N}_0 \ \exists C_{\alpha, K} < \infty \ \forall x \in \mathbb{R}^3, \\ |\partial_x^\alpha a(x)| \leq C_{\alpha, K} (1 + |x|)^{-K} \end{array} \right\}.$$

Here $H_\sigma^m(\mathbb{R}^3) := \{v \in H^m(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}$. In particular, $\mathcal{S}_\sigma(\mathbb{R}^3) \subset H_\sigma^m(\mathbb{R}^3)$ for every finite m . The problem asks for velocity and pressure fields satisfying

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0, \quad (1)$$

with

$$u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty))$$

and a kinetic-energy bound uniform in time^[1]. Smoothness here includes the endpoint of every finite interval. It is stronger than being smooth on $\mathbb{R}^3 \times (0, T)$ for every $T < T_*$, because a derivative can remain finite on every shorter interval and still have no finite trace at T_* .

The velocity determines the normalized pressure. Taking the divergence of the momentum equation and using $\partial_i u_i = 0$ gives

$$-\Delta p = \partial_i (u_j \partial_j u_i) = \partial_i \partial_j (u_i u_j).$$

On $\mathbb{R}^3$ we select the Riesz-transform normalization

$$p = R_i R_j (u_i u_j), \quad R_i := \partial_i (-\Delta)^{-1/2}. \quad (2)$$

This removes the additive function of time left by ∇p in (1). It also fixes an identity that will survive every later spatial and temporal differentiation. Pressure is never supplied as an independent field after the velocity has been chosen.

1.1 Smoothness, regularity, and continuation

Three related words will be used throughout the proof, and they should not be asked to do the same job. A velocity–pressure pair is *smooth through* a finite time T when

$$u, p \in C^\infty(\mathbb{R}^3 \times [0, T]).$$

Equivalently, every mixed spatial and temporal derivative of finite order exists on the open spacetime region and extends continuously to both temporal faces. Smoothness for all time means smoothness through every finite T .

Regularity describes the kind and degree of control presently known. A Sobolev statement such as $u(t) \in H^m$, a mixed-norm statement such as $u \in L_t^p L_x^q$, and a pointwise statement such as $u \in C^k$ are different regularity claims. None may be silently replaced by another without the embedding, trace, or equation that connects them.

Suppose a normalized pair (u, p) solves (1) on $[0, T)$. A *continuation through T* is a normalized solution of the same zero-force, fixed-viscosity problem on $[0, T + \varepsilon)$, for some $\varepsilon > 0$, which agrees with (u, p) before T . The Clay problem requires a smooth continuation through every finite time. A finite quantity is not enough by itself: a required derivative can remain finite on every strict subinterval and still fail to approach one value at the endpoint.

1.2 Local existence and the maximal history

An H^3 datum generates a unique pressure-normalized strong solution for a time depending only on ν and its H^3 norm. Every higher integer Sobolev order already present in the datum persists on that same interval, and a finite maximal lifespan forces the H^3 norm to escape. These are the local facts used at both ends of the proof^[2, 3]; the proposition states them together, and Appendix C gives the construction in the form required below.

The precise statement is the following.

Proposition 1.1 (Maximal history and blow-up alternative). *Let $t_0 \geq 0$ and $a \in H_\sigma^3(\mathbb{R}^3)$. There is a time $\delta = \delta(\nu, \|a\|_{H^3}) > 0$ and a unique pressure-normalized solution (w, q) of (1) in the class*

$$\begin{aligned} w &\in C([t_0, t_0 + \delta]; H_\sigma^3) \cap L^2(t_0, t_0 + \delta; H^4), \\ w_t &\in L^2(t_0, t_0 + \delta; H^2), \quad q = R_i R_j (w_i w_j). \end{aligned} \tag{3}$$

with $w(t_0) = a$. If $a \in H_\sigma^m$ for an integer $m > 3$, then, on the same interval,

$$w \in C([t_0, t_0 + \delta]; H_\sigma^m) \cap L^2(t_0, t_0 + \delta; H^{m+1}), \quad w_t \in L^2(t_0, t_0 + \delta; H^{m-1}).$$

Compatible local solutions in these strong classes join by uniqueness into one maximal solution on $[t_0, T_{\max})$. If $T_{\max} < \infty$, then

$$\limsup_{t \uparrow T_{\max}} \|w(t)\|_{H^3} = \infty.$$

Suppose instead that $\|w(t)\|_{H^3} \leq M$ for all $t < T_{\max}$. The local construction then supplies the same positive lifespan $\delta(\nu, M)$ whenever it is restarted from any time $t < T_{\max}$. Choose $t > T_{\max} - \frac{1}{2}\delta(\nu, M)$. The restarted solution crosses $T_{\max}$, and uniqueness identifies it with w on the overlap. This extends the maximal solution. The extension uses the actual value of the same solution at a late time; it does not select a new terminal datum.

For $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, Proposition 1.1 produces a smooth normalized solution on a short interval. Repeated application and uniqueness produce one maximal pair (u, p) on $[0, T_*)$. Every finite spatial order present at time zero persists on each compact subinterval $[0, T] \subset [0, T_*)$. Equation (1) and the pressure identity then recover all finite temporal derivatives there. Thus no regularity is missing before T_* . The unresolved distinction is between bounds on every fixed $[0, T]$, whose constants may diverge as $T \uparrow T_*$, and one bound that survives on the full candidate interval $(0, T_*)$. Proposition 1.1 makes that distinction exact: a whole-terminal H^3 bound supplies a uniform restart time and excludes finite maximality.

The global assertion is the following.

Theorem 1.2 (Whole-space existence and smoothness). *For every $\nu > 0$ and every $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, the unforced three-dimensional incompressible Navier–Stokes equations have a global pressure-normalized classical solution satisfying*

$$u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)).$$

For every $t \geq 0$, this solution obeys

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2,$$

and in particular

$$\sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx \leq \|u_0\|_2^2 < \infty. \quad (4)$$

The energy identity will give the uniform kinetic-energy bound in Theorem 1.2. Its harder conclusion begins one derivative higher. Even if the continuation-grade H^3 norm escapes, the kinetic energy and an integrated first derivative remain controlled for the entire proposed branch. Those are the first quantities that do not end when the classical restart norm does. We derive them next and ask how much of the original evolution they still identify.

2 The first global control: energy and weak continuation

Pair the momentum equation with u and integrate over $\mathbb{R}^3$. The transport, pressure, and viscous terms have three distinct forms, so we record the three integrations separately:

$$\begin{aligned} \int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot u dx &= \frac{1}{2} \int_{\mathbb{R}^3} u \cdot \nabla |u|^2 dx = -\frac{1}{2} \int_{\mathbb{R}^3} (\nabla \cdot u) |u|^2 dx = 0, \\ \int_{\mathbb{R}^3} \nabla p \cdot u dx &= - \int_{\mathbb{R}^3} p \nabla \cdot u dx = 0, \\ \int_{\mathbb{R}^3} \Delta u \cdot u dx &= - \int_{\mathbb{R}^3} |\nabla u|^2 dx. \end{aligned}$$

The decay inherited on compact time intervals justifies the integrations; the same calculation follows by cutoff and density at finite energy. Hence

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 + \nu \|\nabla u(t)\|_2^2 = 0.$$

Integrating from 0 to $t < T_*$ gives the exact energy identity

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2. \quad (5)$$

The two controls in (5) have different strengths in time. The velocity has a pointwise-in-time L^2 ceiling. Its first spatial derivative is controlled only after integration over the interval:

$$\sup_{0 \leq t < T_*} \|u(t)\|_2^2 \leq \|u_0\|_2^2, \quad \int_0^{T_*} \|\nabla u(t)\|_2^2 dt \leq \frac{\|u_0\|_2^2}{2\nu} \quad (T_* < \infty). \quad (6)$$

A finite integral permits high values on intervals whose lengths shrink fast enough. The energy law rules out loss of total kinetic energy control, but it does not place an instantaneous ceiling on the derivative that controls classical continuation.

2.1 What global weak existence does and does not continue

Leray used the bounds in (6) to construct a global weak solution. For the same datum u_0 , there is a Leray–Hopf pair (U, P) on every finite time interval with

$$U \in L^\infty(0, T; L_\sigma^2) \cap L^2(0, T; H_\sigma^1),$$

which solves (1) in distributions and satisfies the energy inequality^[4].

Before T_* , this weak solution is the classical solution already fixed above. To see the identification directly, fix $T < T_*$, set $z = U - u$, and subtract the two equations. Using

$$(U \cdot \nabla)U - (u \cdot \nabla)u = (U \cdot \nabla)z + (z \cdot \nabla)u,$$

pair the difference equation with z . The first transport term cancels because $\nabla \cdot U = 0$, and the pressure difference cancels because $\nabla \cdot z = 0$. The remaining nonlinear term satisfies

$$\left| \int_{\mathbb{R}^3} (z \cdot \nabla)u \cdot z dx \right| \leq \|\nabla u\|_\infty \|z\|_2^2.$$

The difference energy inequality is

$$\frac{1}{2} \|z(t)\|_2^2 + \nu \int_0^t \|\nabla z(s)\|_2^2 ds \leq \int_0^t \|\nabla u(s)\|_\infty \|z(s)\|_2^2 ds.$$

The classical solution belongs to $C([0, T]; H^3)$. The embedding $H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3)$, proved in Appendix A at (386), gives $\|\nabla u\|_\infty \in L^1(0, T)$. Since $z(0) = 0$, Gronwall's inequality gives $z = 0$ on $[0, T]$ ^[5, 6]. As $T < T_*$ was arbitrary,

$$U = u \quad \text{almost everywhere on } \mathbb{R}^3 \times (0, T_*). \quad (7)$$

Thus a finite T_* would not prevent a Leray–Hopf continuation from existing. That continuation agrees with the fixed classical solution before T_* , but uniqueness beyond the endpoint has not been established. The loss at T_* would be the loss of a smooth state that identifies one unique classical continuation, not the disappearance of every distributional solution. The next estimate asks where the first classical derivative can escape while the energy law remains intact.

3 The three-dimensional obstruction in the first derivative

The kinetic-energy identity does not expose the mechanism that makes the three-dimensional problem difficult. The first derivative does. Define the vorticity

$$\omega := \nabla \times u.$$

For a decaying vector field,

$$\int_{\mathbb{R}^3} |\nabla u|^2 dx = \int_{\mathbb{R}^3} (|\nabla \times u|^2 + |\nabla \cdot u|^2) dx,$$

so incompressibility gives

$$\|\nabla u\|_2^2 = \|\omega\|_2^2.$$

The dissipation in the kinetic-energy law is exactly the time-integrated enstrophy.

Taking the curl of the momentum equation removes the pressure gradient. The transport term does not pass through the curl unchanged. In index notation, using $\partial_j u_j = 0$,

$$\nabla \times ((u \cdot \nabla)u) = (u \cdot \nabla)\omega - (\omega \cdot \nabla)u.$$

The vorticity consequently satisfies

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega. \quad (8)$$

Transport carries vorticity through the velocity field, viscosity diffuses its spatial variation, and $(\omega \cdot \nabla)u$ changes vorticity through the velocity gradient already present in that field.

Write

$$S := \frac{1}{2}(\nabla u + (\nabla u)^\top), \quad A := \frac{1}{2}(\nabla u - (\nabla u)^\top).$$

Pairing (8) with ω cancels transport and integrates the Laplacian by parts:

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla \omega\|_2^2 = \int_{\mathbb{R}^3} ((\omega \cdot \nabla)u) \cdot \omega dx.$$

Since $(A\omega) \cdot \omega = 0$, this is

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla \omega\|_2^2 = \int_{\mathbb{R}^3} (S\omega) \cdot \omega dx.$$

The right-hand side has no fixed sign. The strain may shorten vorticity in one region and lengthen it in another, with both actions constrained by the same divergence-free velocity. This fully three-dimensional stretching term prevents the enstrophy balance from inheriting the monotonic form of the kinetic-energy balance.

The absence of an explicit pressure term from (8) does not turn this into a local scalar equation. Since $\nabla \cdot u = 0$,

$$-\Delta u = \nabla \times \omega, \quad \partial_k u_j = \partial_k (-\Delta)^{-1} (\nabla \times \omega)_j.$$

Thus every component of ∇u is an order-zero singular-integral transform of ω ; the required multiplier bounds are (393). The normalized pressure is recovered from u by (2). Stretching, transport, viscous diffusion, pressure response, and incompressibility remain parts of the same evolving field. The point of the vorticity equation is narrower: it locates the sign-indefinite derivative production that the total energy balance cannot see.

3.1 The first differentiated estimate

Set

$$Y(t) := \|\nabla u(t)\|_2^2.$$

Pair (1) with $-\Delta u$. The pressure vanishes because $\nabla \cdot \Delta u = 0$, and integration by parts in the time and viscous terms gives

$$\frac{1}{2}Y'(t) + \nu \|\Delta u(t)\|_2^2 = \int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot \Delta u \, dx. \quad (9)$$

The transport term now survives. Hölder's inequality with exponents $(6, 3, 2)$, followed by the homogeneous Sobolev and interpolation estimates proved in (388), gives the classical enstrophy estimate:

$$\begin{aligned} \left| \int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot \Delta u \, dx \right| &\leq \|u\|_6 \|\nabla u\|_3 \|\Delta u\|_2 \\ &\leq C \|\nabla u\|_2 (\|\nabla u\|_2^{1/2} \|\nabla u\|_6^{1/2}) \|\Delta u\|_2 \\ &\leq C \|\nabla u\|_2^{3/2} \|\Delta u\|_2^{3/2}. \end{aligned}$$

Young's inequality, with the viscosity retained in the absorbing term, yields

$$C \|\nabla u\|_2^{3/2} \|\Delta u\|_2^{3/2} \leq \frac{\nu}{2} \|\Delta u\|_2^2 + C\nu^{-3} \|\nabla u\|_2^6.$$

Substitution in (9) gives

$$Y'(t) + \nu \|\Delta u(t)\|_2^2 \leq C\nu^{-3} Y(t)^3. \quad (10)$$

The energy identity controls $\int_0^T Y(t) \, dt$. The next estimate asks for $\int_0^T Y(t)^3 \, dt$. These are genuinely different demands. A nonnegative function can have finite integral while its cube has infinite integral, and the energy identity contains no additional time information that excludes that pattern. This observation does not assert that a Navier–Stokes solution can realize an arbitrary spike. It identifies the exact point at which the basic estimate stops: the available L_t^1 control of Y cannot be inserted into the cubic term of (10).

The same differential inequality forces a minimum growth rate on any branch where enstrophy becomes unbounded. A nonzero decaying incompressible solution has $Y(t) > 0$ along such a branch. If $Y(t_0) = 0$, then $u(t_0) = 0$, and forward uniqueness forces $u(t) = 0$ for every $t \geq t_0$, excluding any later blow-up. Dropping the nonnegative viscous term from (10) and differentiating Y^{-2} gives

$$\frac{d}{dt} Y^{-2} = -2Y^{-3} Y' \geq -2C\nu^{-3}.$$

Assume $T_* < \infty$ and choose $s_j \uparrow T_*$ with $Y(s_j) \rightarrow \infty$. Integration from t to s_j gives

$$Y(t)^{-2} \leq Y(s_j)^{-2} + 2C\nu^{-3}(s_j - t).$$

Letting $j \rightarrow \infty$ yields

$$T_* < \infty \text{ and } \limsup_{t \uparrow T_*} Y(t) = \infty \implies Y(t) \geq \frac{\nu^{3/2}}{\sqrt{2C}} (T_* - t)^{-1/2}. \quad (11)$$

The rate in (11) is unbounded and still integrable at T_* . It can fit inside the dissipation budget only because the time available above each higher level contracts. The first differentiated estimate has consequently exposed a possible terminal geometry: increasing derivative intensity supported on shrinking time intervals. Since $Y(t)$ is integrated over all of $\mathbb{R}^3$, it says nothing yet about where that intensity sits or what remains when the surrounding spacetime region is magnified.

3.2 From shrinking time to shrinking spacetime

That missing spatial information is exactly what the local energy inequality was built to retain. Caffarelli, Kohn, and Nirenberg inserted a nonnegative spacetime cutoff into the Leray energy argument, keeping both the energy inside a shrinking region and the flux through its boundary. The Leray pair fixed in Section 2 may be chosen suitable; it retains the same normalized pressure and still satisfies (7) before T_* . This pair (U, P) satisfies, for every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times [0, T))$ and almost every $t \in (0, T)$,

$$\begin{aligned} \frac{1}{2} \int_{\mathbb{R}^3} |U(t)|^2 \phi(t) dx + \nu \int_0^t \int_{\mathbb{R}^3} |\nabla U|^2 \phi dx ds \\ \leq \frac{1}{2} \int_{\mathbb{R}^3} |u_0|^2 \phi(0) dx \\ + \frac{1}{2} \int_0^t \int_{\mathbb{R}^3} |U|^2 (\partial_s \phi + \nu \Delta \phi) dx ds \\ + \int_0^t \int_{\mathbb{R}^3} \left(\frac{1}{2} |U|^2 + P \right) U \cdot \nabla \phi dx ds. \end{aligned} \tag{12}$$

Unlike the total energy law, (12) retains the flux through the region selected by ϕ . It is designed to detect a point around which energy and dissipation keep concentrating as the region shrinks.

For $z_0 = (x_0, t_0)$, write

$$Q_r(z_0) := B_r(x_0) \times (t_0 - r^2, t_0).$$

Under the Navier–Stokes scaling

$$U_\lambda(x, t) = \lambda U(\lambda x, \lambda^2 t), \quad P_\lambda(x, t) = \lambda^2 P(\lambda x, \lambda^2 t),$$

a spatial length r is paired with a time length r^2 . If $z_0^\lambda = (x_0/\lambda, t_0/\lambda^2)$, then the change of variables $y = \lambda x$, $s = \lambda^2 t$ gives

$$\frac{1}{r/\lambda} \int_{Q_{r/\lambda}(z_0^\lambda)} |\nabla U_\lambda|^2 dx dt = \frac{1}{r} \int_{Q_r(z_0)} |\nabla U|^2 dy ds.$$

Thus the scaled dissipation used below is genuinely dimensionless. In the Caffarelli–Kohn–Nirenberg terminology, z_0 is a regular point when U is essentially bounded on a spacetime neighbourhood of z_0 . Their epsilon-regularity theorem gives a constant $\varepsilon_{\text{CKN}}(\nu) > 0$ such that^[7]

$$\limsup_{r \downarrow 0} \frac{1}{r} \int_{Q_r(z_0)} |\nabla U|^2 dx dt < \varepsilon_{\text{CKN}}(\nu) \implies z_0 \text{ is a regular point.} \tag{13}$$

At a singular point the contrapositive of (13) gives

$$\limsup_{r \downarrow 0} \frac{1}{r} \int_{Q_r(z_0)} |\nabla U|^2 dx dt \geq \varepsilon_{\text{CKN}}(\nu).$$

Thus the scale-invariant dissipation reaches the threshold along arbitrarily small cylinders. The limsup statement does not assert the same lower bound at every small radius.

The covering argument turns this scale-by-scale lower concentration into a geometric restriction. If $\mathcal{S}(U)$ denotes the set of singular spacetime points of a suitable weak solution, then

$$\mathcal{P}^1(\mathcal{S}(U)) = 0,$$

where $\mathcal{P}^1$ is one-dimensional parabolic Hausdorff measure^[7]. This is far stronger than an almost-everywhere statement. A possible singular set cannot contain a parabolic curve segment, a positive time interval, or any set carrying positive one-dimensional parabolic measure.

Caffarelli–Kohn–Nirenberg have now reduced a possible failure from diffuse loss of control to scale-invariant concentration near a very small singular set. That geometric restriction still leaves an isolated singular point available, and it still supplies no whole-space state at T_* . On $\mathbb{R}^3$, concentration may also move through space, so regularity in each fixed neighbourhood does not by itself bound a global Sobolev norm. The analysis has reached a precise divide: local concentration describes how failure would appear, while continuation requires a global quantity strong enough to propagate the complete spatial state.

4 What would be strong enough to continue the solution

Proposition 1.1 already names one such state: a bounded H^3 norm crosses the proposed endpoint. Differentiating the equation shows something stronger. Once one lower-order transport coefficient is integrable, every fixed Sobolev order propagates on the same interval. Let $m \geq 3$ be an integer and apply ∂^α to (1) for $|\alpha| \leq m$:

$$\partial_t \partial^\alpha u + (u \cdot \nabla) \partial^\alpha u + [\partial^\alpha, u \cdot \nabla] u + \nabla \partial^\alpha p = \nu \Delta \partial^\alpha u.$$

Pair with $\partial^\alpha u$ and integrate. The undifferentiated transport coefficient still cancels,

$$\int_{\mathbb{R}^3} (u \cdot \nabla) \partial^\alpha u \cdot \partial^\alpha u \, dx = 0,$$

and the pressure cancels because $\nabla \cdot \partial^\alpha u = 0$. The commutator contains all the product terms in which at least one derivative lands on the transport field:

$$[\partial^\alpha, u \cdot \nabla] u = \sum_{0 < \beta \leq \alpha} \binom{\alpha}{\beta} (\partial^\beta u \cdot \nabla) \partial^{\alpha-\beta} u.$$

The integer-order commutator estimate (392) gives

$$\sum_{|\alpha| \leq m} |\langle [\partial^\alpha, u \cdot \nabla] u, \partial^\alpha u \rangle| \leq C_m \|\nabla u\|_\infty \|u\|_{H^m}^2.$$

Summing over $|\alpha| \leq m$ produces

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_{H^m}^2 + \nu \|\nabla u(t)\|_{H^m}^2 \leq C_m \|\nabla u(t)\|_\infty \|u(t)\|_{H^m}^2. \quad (14)$$

Gronwall's inequality now states the continuation mechanism in one line:

$$\|u(t)\|_{H^m}^2 \leq \|u_0\|_{H^m}^2 \exp \left(2C_m \int_0^t \|\nabla u(s)\|_\infty \, ds \right), \quad 0 \leq t < T_*. \quad (15)$$

The factor $\|u_0\|_{H^m}$ is finite at every fixed order because $u_0 \in \mathcal{S}_\sigma$. The constant may depend on m ; smoothness does not require one constant uniform over all derivative orders. What must be common to every row is the same finite time interval and the same integral

$$\int_0^{T_*} \|\nabla u(t)\|_\infty dt.$$

If this integral is finite, (15) bounds every fixed H^m norm up to T_* , beginning with H^3 . Proposition 1.1 then continues the solution. Hence

$$T_* < \infty \implies \int_0^{T_*} \|\nabla u(t)\|_\infty dt = \infty. \quad (16)$$

A single uniform Sobolev bound above the Lipschitz threshold also closes the argument. If $s > 5/2$ and

$$\sup_{t < T_*} \|u(t)\|_{H^s} < \infty,$$

then $H^s(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3)$. The finite length of $(0, T_*)$ makes the coefficient in (14) integrable, all higher orders propagate, and Proposition 1.1 restarts the solution. Thus every finite maximal time must force every continuation-grade Sobolev norm to escape.

The calculation has isolated one coefficient that carries the entire spatial ladder. To see how far the energy class lies from paying it, first observe that energy already reaches the spatial exponent that will become critical, but with the wrong time control. Sobolev embedding and interpolation give

$$\|u(t)\|_3 \leq \|u(t)\|_2^{1/2} \|u(t)\|_6^{1/2} \leq C \|u(t)\|_2^{1/2} \|\nabla u(t)\|_2^{1/2}.$$

Raising to the fourth power, integrating in time, and using (5) yields

$$u \in L^4(0, T_*; L^3(\mathbb{R}^3)) \quad (T_* < \infty).$$

Energy already reaches the spatial exponent 3, but only after time integration. The scale-balanced criteria distinguish exactly how much stronger the temporal control must be.

4.1 Scale-balanced continuation and the critical endpoint

Ladyzhenskaya, Prodi, and Serrin found the first scale-balanced way to obtain that control. Their mixed space–time conditions trade spatial intensity against the length of time for which that intensity persists [5, 6, 8, 9]. The first differentiated estimate determines the trade, so its exponents are derived before the criterion is stated.

Fix $q > 3$ and choose $r = 2q/(q - 2)$, so

$$\frac{1}{q} + \frac{1}{r} + \frac{1}{2} = 1.$$

Hölder's inequality and the interpolation formula in (388) give

$$\begin{aligned} \left| \int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot \Delta u \, dx \right| &\leq \|u\|_q \|\nabla u\|_r \|\Delta u\|_2, \\ \|\nabla u\|_r &\leq C \|\nabla u\|_2^{1-3/q} \|\nabla u\|_6^{3/q} \\ &\leq C \|\nabla u\|_2^{1-3/q} \|\Delta u\|_2^{3/q}. \end{aligned}$$

Thus

$$\left| \int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot \Delta u \, dx \right| \leq C \|u\|_q \|\nabla u\|_2^{1-3/q} \|\Delta u\|_2^{1+3/q}.$$

Young's inequality uses the exponent $2/(1+3/q)$ on the last factor and its conjugate $2/(1-3/q) = 2q/(q-3)$ on the remaining product. It follows that

$$\frac{d}{dt} \|\nabla u\|_2^2 + \nu \|\Delta u\|_2^2 \leq C_{\nu,q} \|u\|_q^{2q/(q-3)} \|\nabla u\|_2^2. \quad (17)$$

The exponent is not chosen for appearance. It is exactly the time exponent that makes the coefficient in (17) integrable on the critical line.

Proposition 4.1 (Ladyzhenskaya–Prodi–Serrin continuation criterion). *Let (u, p) be the maximal solution supplied by Proposition 1.1, and let $T \leq T_*$. If*

$$u \in L^p(0, T; L^q(\mathbb{R}^3)), \quad \frac{2}{p} + \frac{3}{q} \leq 1, \quad q > 3,$$

then u remains regular through T . The endpoint $(p, q) = (2, \infty)$ is included.

The proposition is the classical Ladyzhenskaya–Prodi–Serrin continuation theorem^[8, 5, 9]. The calculation above explains its critical exponent in the present notation: $2/p + 3/q \leq 1$ gives $p \geq 2q/(q-3)$, so on a finite interval the coefficient $\|u\|_q^{2q/(q-3)}$ in (17) is integrable. Gronwall bounds enstrophy and the integrated H^2 dissipation; the cited theorem supplies continuation and weak–strong uniqueness through the terminal time.

The continuation criterion also determines when the lower rate (11) applies. If $\sup_{t < T_*} \|\nabla u(t)\|_2 < \infty$, Sobolev embedding gives

$$\sup_{t < T_*} \|u(t)\|_6 < \infty.$$

Since $T_* < \infty$, this places u in $L^4(0, T_*; L^6)$, and

$$\frac{2}{4} + \frac{3}{6} = 1.$$

Proposition 4.1 would continue the solution. A finite maximal time must consequently have unbounded enstrophy, so the Leray lower rate applies.

4.2 Why the endpoint is critical

The Navier–Stokes scaling is

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t). \quad (18)$$

At corresponding times and intervals,

$$\|u_\lambda\|_{L_t^p L_x^q} = \lambda^{1-2/p-3/q} \|u\|_{L_t^p L_x^q}.$$

The equality $2/p + 3/q = 1$ is precisely the scale-invariant line. The spatial exponent q measures the intensity of concentration, while the temporal exponent p measures how long that intensity persists. The

Leray energy controls

$$u \in L_t^\infty L_x^2 \cap L_t^2 L_x^6,$$

and both pairs have $2/p + 3/q = 3/2$, above the continuation line. Scaling expresses the same gap seen in the enstrophy calculation: the global energy class does not forbid activity that becomes stronger as its scale contracts.

As $q \downarrow 3$, the exponent $2q/(q-3)$ in (17) diverges. The direct Gronwall calculation reaches every $q > 3$ and stops at

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3)).$$

The norm is exactly invariant under (18):

$$\|u_\lambda(t)\|_3 = \|u(\lambda^2 t)\|_3.$$

At the same time,

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \|u(\lambda^2 t)\|_2^2.$$

Concentration to a shorter spatial scale preserves the L^3 size while it reduces the amount seen by the L^2 energy. This is why the endpoint cannot be obtained by interpolating the energy law alone.

4.3 Escauriaza–Seregin–Šverák

Escauriaza, Seregin, and Šverák proved that boundedness at the missing endpoint is still enough for regularity^[10]. In the continuation form used here, their theorem states that a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$ satisfying

$$U \in L^\infty(0, T; L^3(\mathbb{R}^3))$$

is regular through T . The proof cannot follow (17), whose exponent has become infinite. It rescales around a presumed concentration, passes to a nontrivial ancient limiting solution, and uses backward uniqueness for vorticity together with unique continuation to rule out that limit.

Its implication for the fixed maximal history is exact. Suppose

$$\limsup_{t \uparrow T_*} \|u(t)\|_3 < \infty.$$

The finite limsup bounds $\|u(t)\|_3$ on some terminal interval. On the earlier compact interval the classical solution is continuous into H^3 and hence bounded in L^3 . Thus

$$u \in L^\infty(0, T_*; L^3).$$

By (7), the suitable Leray–Hopf solution U has the same membership and agrees with u before T_* . The endpoint theorem makes U regular through T_* , and weak–strong uniqueness gives a classical continuation of u . Maximality rules this out. We have proved the necessary condition

$$T_* < \infty \implies \limsup_{t \uparrow T_*} \|u(t)\|_{L^3} = \infty. \quad (19)$$

The conclusion says that the critical norm crosses arbitrarily high levels along times approaching T_* . By itself it allows a sequence of high peaks separated by returns below a fixed level. That distinction matters for any argument that needs a full terminal limit rather than a selected sequence.

4.4 Seregin's terminal divergence and Tao's quantitative rate

Seregin closed the difference between a limsup and a limit in the smooth, compactly supported, divergence-free whole-space class stated in his theorem. For a solution in that class, if $T_* < \infty$, then

$$\lim_{t \uparrow T_*} \|u(t)\|_{L^3} = \infty. \quad (20)$$

Equivalently, for every $M < \infty$ there is a time $t_M < T_*$ such that $\|u(t)\|_3 > M$ for every $t \in (t_M, T_*)$ [11]. A finite singular branch cannot keep returning to a fixed bounded L^3 level.

Compactly supported smooth data form a proper subfamily of the class $\mathcal{S}_\sigma(\mathbb{R}^3)$ fixed in the opening. Within that subfamily, Seregin's theorem replaces arbitrarily late peaks by permanent terminal escape. For the full class, the all-data endpoint condition (19) remains the available conclusion.

The homogeneous Sobolev norm

$$\|u\|_{\dot{H}^{1/2}} = \left\| \Lambda^{1/2} u \right\|_2$$

has the same scaling as L^3 , and the Sobolev embedding $\dot{H}^{1/2}(\mathbb{R}^3) \hookrightarrow L^3(\mathbb{R}^3)$ places it at the same critical height. In the same compact-data setting, Seregin proved the direct terminal condition

$$T_* < \infty \implies \lim_{t \uparrow T_*} \|u(t)\|_{\dot{H}^{1/2}} = \infty. \quad (21)$$

This is a statement about the whole terminal approach, not a subsequence [12].

These theorems specify that the critical norm must diverge, but a qualitative limit does not say how rapidly it must do so. Tao quantified the endpoint argument by replacing its compactness and uniqueness steps with explicit estimates. The change of variables

$$v(x, s) = \nu^{-1} u(x, s/\nu), \quad q(x, s) = \nu^{-2} p(x, s/\nu), \quad S_* = \nu T_*,$$

converts the viscosity- ν equation into the viscosity-one equation. A further Navier–Stokes scaling normalizes S_* to one without changing the L^3 norm. Applied after these two normalizations, Tao's estimate gives an absolute $c > 0$ such that

$$\limsup_{t \uparrow T_*} \frac{\nu^{-1} \|u(t)\|_3}{\left(\log \log \log \left(1000 + \frac{T_*}{T_* - t} \right) \right)^c} = \infty.$$

The time ratio and the normalized amplitude are dimensionless [13]. Since $\dot{H}^{1/2}(\mathbb{R}^3) \hookrightarrow L^3(\mathbb{R}^3)$, the same lower-rate conclusion transfers, up to the Sobolev constant, to excursions of the critical Sobolev norm. The estimate is slow, but it makes the shrinking-time picture quantitative: as the remaining time collapses, critical escape cannot stay below every power of this triple logarithm along all terminal sequences.

5 The question left by the classical theory

The historical picture is now tight enough to state the remaining problem. Energy allows increasing derivative intensity only by concentrating it into less time. Caffarelli–Kohn–Nirenberg retain scale-invariant activity near any singular point. ESS and Tao force arbitrarily late critical excursions; Seregin forces permanent terminal escape in his compact-data class. Each result describes what a finite singular branch would have to do. None gives the upper estimate that would stop it. The missing statement must be generated by u_0 , ν , and the coupled equation, and its constant must remain finite all the way to T_* .

Continuation now has two distinct burdens. Following the solution forward, one needs an estimate strong enough to prevent its H^3 restart norm from escaping before T_* . At the endpoint, any proposed limits for the velocity and its required derivatives must agree wherever they overlap and must still satisfy the original momentum and normalized pressure laws. We call the forward estimate Gold and the endpoint test Silver. Both belong to the same regularity claim.

5.1 Gold: direct continuation control

Fix one exponent $s_0 > 5/2$. A decisive estimate would have the form

$$\sup_{0 \leq t < \min\{T, T_*\}} \|u(t)\|_{H^{s_0}} \leq C_{s_0}(u_0, \nu, T) < \infty \quad \text{for every finite } T > 0. \quad (22)$$

The constant may depend on the finite time and on the chosen Sobolev order. It must be generated by u_0 , ν , and the equation; the finiteness of the norm on the left cannot reappear as an assumption hidden inside the constant. Estimate (22) is the *Gold Standard*.

The threshold $s_0 > 5/2$ makes the estimate continuation-grade because

$$H^{s_0}(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3).$$

For every finite T , it follows that

$$\int_0^{\min\{T, T_*\}} \|\nabla u(t)\|_{\infty} dt < \infty.$$

Equation (15) then propagates every fixed finite Sobolev order present in u_0 . The bound may change with the order. What smoothness needs is one finite bound at every requested order on the same time interval, not one numerical constant for the infinite family.

Once the spatial orders are available, pressure follows from (2), and the momentum equation gives

$$u_t = \nu \Delta u - (u \cdot \nabla)u - \nabla p,$$

so repeated differentiation recovers every finite mixed velocity–pressure derivative. If $T_* < \infty$, take $T = T_*$ in (22). The propagated H^3 bound gives every sufficiently late slice the same positive local lifespan. One such restart crosses T_* , and uniqueness identifies it with the original solution on the overlap. Thus the estimate alone forces $T_* = \infty$.

Gold has now reduced the global problem to one forward task: derive whole-terminal continuation control from the coupled equation without assuming the norm being controlled. The same implication can also be read from its other end.

5.2 Silver: testing the endpoint against the original problem

Silver begins by defining admissibility without placing smoothness inside the definition. A candidate terminal description consists of the fixed classical pair on $[0, T_*)$ together with proposed endpoint distributions. Call that description admissible when it extends the same datum, the same positive viscosity, zero forcing, incompressibility, the momentum equation, the normalized pressure, and finite energy through T_* , while agreeing with the classical pair before the endpoint. The target implication says that every admissible description is smooth. Its contrapositive says that a genuinely nonsmooth candidate must fail at least one of those original requirements. This is the Silver reading.

The definition has not yet supplied a test. Endpoint values alone cannot say whether the momentum and pressure laws still hold. We must first see how the equation acts on the moving fluid, and then distinguish the ways that action could fail at a terminal time.

5.3 The equation acting on the fluid

The equations show how those requirements act in the motion. Let $X(a, t)$ be the trajectory issuing from a :

$$\dot{X}(a, t) = u(t, X(a, t)), \quad X(a, 0) = a.$$

The derivative seen while following the fluid is

$$D_t := \partial_t + u \cdot \nabla.$$

The momentum equation becomes the material acceleration law

$$D_t u = -\nabla p + \nu \Delta u.$$

The velocity of a moving particle changes through the pressure and viscous response of the field that contains it. This is not an additional condition placed on the motion; it is what the momentum equation says the motion is.

The two terms on the right form one stress divergence. With

$$S := \frac{1}{2}(\nabla u + (\nabla u)^T),$$

incompressibility gives

$$\nabla \cdot (2\nu S) = \nu \Delta u + \nu \nabla(\nabla \cdot u) = \nu \Delta u.$$

Consequently,

$$D_t u = \nabla \cdot (-pI + 2\nu S).$$

The same transported velocity supplies both the local viscous curvature and the velocity products from which pressure is recovered.

Jacobi's formula shows what incompressibility preserves along this motion:

$$\frac{d}{dt} \det \nabla_a X(a, t) = (\nabla \cdot u)(t, X(a, t)) \det \nabla_a X(a, t) = 0.$$

Material volume is preserved, while (2) reconstructs pressure from the velocity over the whole field. The equation joins three scales of one event: transport follows the moving particle, viscosity reads local spatial curvature, and pressure enforces the global incompressibility constraint. A candidate endpoint cannot be tested by its velocity value alone; it must retain this response relation. We use *viscosity–pressure–incompressibility response*, abbreviated VPI, for this joined action on the transported velocity. VPI is shorthand for the original momentum, pressure, and divergence equations acting together; it is not an extra equation. Removing viscosity while leaving transport, pressure, and incompressibility in place makes the first possible loss visible.

5.4 A tangential slip sheet at the inviscid boundary

Euler retains incompressibility, nonlinear transport, and pressure but removes the viscous term that reads tangential curvature. Across the interface $\{y = 0\}$, fix constants $a \neq b$ and define

$$u(x, y, z, t) = \begin{cases} (a, 0, 0), & y > 0, \\ (b, 0, 0), & y < 0, \end{cases} \quad p(x, y, z, t) = p_0, \quad (23)$$

where p_0 is constant. Both one-sided velocities are finite. Their normal components agree, while their tangential components do not.

Let $n = (0, 1, 0)$. Since

$$[u \cdot n] = 0,$$

the distributional divergence has no sheet contribution. In conservative form, the possible interface defect in the Euler momentum law is

$$[(u \otimes u + pI)n] = [u(u \cdot n) + pn] = [p]n = 0.$$

Together with $u_t = 0$, this proves that (23) is a stationary weak Euler solution.

The same field fails the fixed-viscosity equation. Writing

$$u_1 = b + (a - b)H(y),$$

gives

$$\partial_y u_1 = (a - b)\delta_{y=0}, \quad \partial_{yy} u_1 = (a - b)\delta'_{y=0}.$$

The viscous term $\nu \Delta u$ contains the nonzero distribution $\nu(a - b)\delta'_{y=0}e_1$. The stationary transport term supplies no matching distribution. Pressure cannot absorb it either: the curl of this viscous defect is nonzero, whereas the curl of every pressure gradient is zero. The sheet lies on the Euler side of the fixed-viscosity boundary.

This infinite sheet is not a finite-energy endpoint generated by the datum in the main theorem. Its role is narrower: it isolates the difference between two finite one-sided values and one compatible viscous state. Euler admits the two tangential traces, while fixed positive viscosity detects their distributional curvature and rejects the sheet. A terminal failure could postpone the same incompatibility to a gradient, acceleration, or higher derivative even when u itself has one limit. The endpoint argument must produce one trace at every required finite order; finite readings along separate approaches do not join themselves.

5.5 What can be lost at the endpoint

The slip sheet and the classical continuation criteria leave three possible endpoint failures. A continuation-grade quantity can become unbounded (*Blown*); finite one-sided readings can fail to join (*Jump*); or the limiting values can cease to satisfy the pressure–viscosity response (*Dead*). These are three obligations of one terminal state—finiteness, compatibility, and the original equation—not three separate solutions.

Dead. A Dead state has lost lawful response. It is not a zero velocity or a zero derivative: the zero solution, a stagnation point, and a steady flow can all satisfy Navier–Stokes exactly. At the base rung the test is

$$D_t u + \nabla p - \nu \Delta u = 0.$$

If a proposed endpoint assigns a material acceleration that does not equal the limiting pressure–viscosity response of its surrounding field, that endpoint is Dead at the base rung. The same mismatch can occur higher when a temporal row no longer answers the transport, diffusion, and pressure terms obtained by differentiating the equation. Excluding Dead requires more than a terminal velocity: the endpoint rows and pressure must satisfy the complete differentiated law and generate the same local restart.

Jump. A Jump is an incompatibility between limiting values of the velocity or of a finite derivative across a point, interface, or time. The tangential slip sheet (23) is the interface example. At a terminal time, the same issue can occur when two regular approaches assign different limits to one spatial or temporal derivative. Both limits may be finite. Separate bounds do not make them equal. Excluding Jump requires a topology in which the evolving derivative has one continuous representative and hence one terminal value. Part II will derive that topology from the equation’s parabolic separation between one time derivative and two spatial derivatives.

Blown. A Blown state has an unbounded velocity, spatial derivative, temporal response, pressure response, or other finite-depth tower readout. The velocity itself need not diverge. The energy identity can remain intact while enstrophy or a stronger continuation norm escapes. Excluding Blown requires a whole-terminal finite bound for every fixed temporal row and every fixed spatial order needed by smoothness. The constants may depend on those finite orders; no norm uniform over the infinite tower is required.

Restart gives these failures their practical significance. At every regular time $t_0 < T_*$, the value $u(t_0)$ is admissible initial data for the local construction in Proposition 1.1. A continuation-grade bound gives a positive lifespan depending only on that bound and ν . A late enough slice then generates a solution beyond the candidate endpoint. Regularity is not a list of finite quantities. It is enough coherent state to let the same equation run again.

6 What a finite-time singularity demands from the response

Dead, Jump, and Blown say what a successful endpoint argument must exclude. They do not yet identify the response that could carry a loss toward that endpoint. A finite singular time need not send the velocity itself to infinity: the energy can remain bounded while a spatial derivative, a pressure response, or another continuation-grade quantity escapes. To follow that change, we return to the acceleration law of the moving fluid.

Along one trajectory, set

$$A_j(t, a) := D_t^j u(t, X(a, t)).$$

Since differentiation along X is D_t , the fundamental theorem of calculus gives, for every $j \geq 0$,

$$A_j(t, a) - A_j(r, a) = \int_r^t A_{j+1}(s, a) ds.$$

If the carried velocity escapes before a finite time, its acceleration must accumulate without a finite L^1 bound. If the acceleration escapes, its material jerk must carry that change. The same implication repeats through

$$u, \quad D_t u, \quad D_t^2 u, \quad D_t^3 u, \quad \dots$$

This is the physical reason to look beyond velocity, but it is not yet a whole-field estimate. A Sobolev norm measures the entire velocity field at one time; its growth does not select a particle whose velocity or acceleration diverges. Spatial redistribution can enlarge that norm while every proposed trajectory remains the wrong witness. The next calculation must begin from the critical quantity that the classical theory already forces to escape, identify the fixed-coordinate response carrying its change, and retain the transport, viscosity, and normalized pressure terms created in passing from the moving fluid to the whole field.

Part II

Critical escape and the response generated by the equation

Part I has reduced a proposed finite maximal time to one exact terminal problem. The kinetic energy remains bounded, while the scale-invariant $\dot{H}^{1/2}$ size must make arbitrarily large excursions arbitrarily close to the endpoint. Yet the proof still has to control every required finite derivative, make the different finite views approach one common state, and preserve the differentiated velocity–pressure laws there. The question is how one fixed evolution could drive a critical quantity through unbounded levels in the time still available. We first make that escape visible in space and time. Its rapid variation forces the Eulerian response generated by the original equation, and that response opens the two independent derivative directions that the terminal estimate must control.

7 The critical height at a finite maximal time

The three-dimensional Navier–Stokes scaling is

$$u_\lambda(t, x) = \lambda u(\lambda^2 t, \lambda x), \quad p_\lambda(t, x) = \lambda^2 p(\lambda^2 t, \lambda x).$$

It preserves the equation and the divergence constraint. The time interval is shortened by the factor λ^{-2} , and the velocity is amplified by the factor λ . A spatial norm left unchanged by this transformation measures the solution at the same scale as the equation.

Let $\Lambda = (-\Delta)^{1/2}$. The rescaled Fourier transform is

$$\widehat{u}_\lambda(t, \xi) = \lambda^{-2} \widehat{u}(\lambda^2 t, \xi/\lambda).$$

Changing variables $\xi = \lambda \eta$ gives, for every real s for which the homogeneous norm is finite,

$$\begin{aligned} \|u_\lambda(t)\|_{\dot{H}^s}^2 &= \int_{\mathbb{R}^3} |\xi|^{2s} \lambda^{-4} |\widehat{u}(\lambda^2 t, \xi/\lambda)|^2 d\xi \\ &= \lambda^{2s-1} \int_{\mathbb{R}^3} |\eta|^{2s} |\widehat{u}(\lambda^2 t, \eta)|^2 d\eta \\ &= \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2. \end{aligned}$$

The scaling exponent vanishes at $s = 1/2$. We accordingly define the critical height

$$H(t) := \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2.$$

The factor $1/2$ will remove an inessential factor of two when the height is differentiated.

Scaling explains why this height belongs at the critical frontier. The all-data endpoint theorem recalled in Part I makes it the exact quantity to follow on a finite maximal branch. The critical Sobolev embedding gives

$$\|u(t)\|_3 \leq C_{\text{Sob}} \left\| \Lambda^{1/2} u(t) \right\|_2 = C_{\text{Sob}} \sqrt{2H(t)}.$$

If H remained bounded near T_* , then the L^3 norm would remain bounded there as well, contrary to the Escauriaza–Seregin–Šverák condition (19). Hence, for the full Schwartz datum class,

$$T_* < \infty \implies \limsup_{t \uparrow T_*} H(t) = \infty. \quad (24)$$

Equivalently, for every finite level L and every $\tau < T_*$, there is a time $t \in (\tau, T_*)$ at which $H(t) > L$. The statement allows the height to return below L between successive excursions.

For the smooth compactly supported data covered by Seregin’s stronger terminal theorem, Part I records the full limit (21). In that historical subcase, every fixed level is eventually left behind. The all-Schwartz argument below rests entirely on the weaker conclusion: every crossing is selected from (24).

The same height remains tied to the kinetic-energy identity by a direct Fourier interpolation. Plancherel and Cauchy–Schwarz give

$$\begin{aligned} 2H(t) &= \int_{\mathbb{R}^3} |\xi| |\widehat{u}(t, \xi)|^2 d\xi \\ &\leq \left(\int_{\mathbb{R}^3} |\widehat{u}(t, \xi)|^2 d\xi \right)^{1/2} \left(\int_{\mathbb{R}^3} |\xi|^2 |\widehat{u}(t, \xi)|^2 d\xi \right)^{1/2} \\ &= \|u(t)\|_2 \|\nabla u(t)\|_2. \end{aligned}$$

After squaring and integrating over $(0, T)$, the energy identity (5) yields

$$\begin{aligned} \int_0^T H(t)^2 dt &\leq \frac{1}{4} \sup_{0 \leq t \leq T} \|u(t)\|_2^2 \int_0^T \|\nabla u(t)\|_2^2 dt \\ &\leq \frac{\|u_0\|_2^4}{8\nu}. \end{aligned}$$

Monotone convergence in T gives the whole-branch budget

$$\int_0^{T_*} H(t)^2 dt \leq \frac{\|u_0\|_2^4}{8\nu} < \infty. \quad (25)$$

Thus the same function satisfies two sharply different requirements on a finite singular branch:

$$H \in L^2(0, T_*), \quad \limsup_{t \uparrow T_*} H(t) = \infty.$$

The first limits the aggregate time available at a large height. The second forces arbitrarily late excursions through every large value before the finite time T_* .

7.1 Critical height as a frequency moment

The difference between kinetic energy and critical height is especially clear in a dyadic decomposition. Choose smooth Littlewood–Paley multipliers P_K , indexed by $K = 2^j$, whose symbols have uniformly finite overlap and are supported where $K/2 \leq |\xi| \leq 2K$ [20]. Put

$$A_K(t) := \frac{1}{2} \|P_K u(t)\|_2^2.$$

On the support of P_K , the multiplier $|\xi|^s$ is comparable to K^s . Plancherel gives

$$\frac{1}{2} \|\Lambda^s P_K u(t)\|_2^2 \simeq K^{2s} A_K(t),$$

with constants independent of K . Summing over the finite overlap gives

$$\frac{1}{2} \|u(t)\|_2^2 \simeq \sum_K A_K(t), \quad H(t) \simeq \sum_K K A_K(t).$$

Kinetic energy is the unweighted shell mass. Critical height is its first frequency moment. A bounded unweighted mass and an unbounded first moment force nonzero contributions at arbitrarily large frequencies: for any fixed K_0 ,

$$\sum_{K \leq K_0} K A_K(t) \leq K_0 \sum_K A_K(t) \lesssim K_0 \|u_0\|_2^2,$$

so the portion below K_0 remains bounded uniformly in time. Whenever $H(t)$ exceeds that bound, frequencies above K_0 must contribute.

This calculation fixes the spatial burden: a bounded amount of kinetic energy must contribute at frequencies beyond every prescribed cutoff. A finite terminal time creates a second burden, because those high-frequency excursions must occur before the remaining time is exhausted. We follow that temporal compression first. The shell weights return after the critical-height balance has identified the response that supplies the rise.

7.2 Finite-time compression

Estimate (25) immediately controls the total time spent above a large height. For every $L > 0$, Chebyshev's inequality gives

$$|\{t \in (0, T_*) : H(t) \geq L\}| \leq \frac{\|u_0\|_2^4}{8\nu L^2}. \quad (26)$$

The terminal limsup and the occupancy estimate (26) perform different work. The limsup supplies arbitrarily late excursions above every high level and permits returns below it. The occupancy estimate bounds the aggregate time available above that level. We will not need the height to stay above a level once it has crossed it. First-passage endpoints alone will give a fixed gain across intervals whose lengths tend to zero.

The crossing times follow directly from continuity on the classical interval.

Lemma 7.1 (First-passage compression). *Assume $T_* < \infty$. Let $L_j \uparrow \infty$ be strictly increasing with $L_1 > H(0)$, and define*

$$t_j := \inf\{t \in (0, T_*) : H(t) = L_j\}.$$

Then every t_j is well defined,

$$t_1 < t_2 < \cdots < T_*, \quad t_j \uparrow T_*.$$

Proof. The height is continuous on every compact subinterval of the classical lifespan. The limsup in (24) places a time $s_j < T_*$ with $H(s_j) > L_j$. Since $H(0) < L_1 \leq L_j$, the intermediate value theorem supplies a point in $[0, s_j]$ where $H = L_j$. The level set is closed inside that compact interval, so it has a first point t_j .

Continuity also fixes the order of the crossings. A path from $H(0) < L_j < L_{j+1}$ to the level L_{j+1} must cross L_j first, so $t_j < t_{j+1}$. The increasing sequence has a limit $\tau \leq T_*$. If $\tau < T_*$, choose $\varepsilon > 0$ with $\tau + \varepsilon < T_*$. Classicality bounds H on the compact interval $[0, \tau + \varepsilon]$, while $H(t_j) = L_j \rightarrow \infty$ inside that interval. This is impossible. Hence $\tau = T_*$. $\square$

Take unit-spaced levels,

$$L_j = L_1 + j - 1,$$

and set

$$\delta_j := t_{j+1} - t_j.$$

The positive gaps fit inside the finite interval and satisfy $\delta_j \rightarrow 0$. On each closed interval $[t_j, t_{j+1}]$, the solution is classical, so the fundamental theorem of calculus gives

$$\begin{aligned} 1 &= H(t_{j+1}) - H(t_j) \\ &= \int_{t_j}^{t_{j+1}} H'(s) ds \\ &\leq \int_{t_j}^{t_{j+1}} |H'(s)| ds. \end{aligned}$$

The average absolute response obeys

$$\frac{1}{\delta_j} \int_{t_j}^{t_{j+1}} |H'(s)| ds \geq \frac{1}{\delta_j} \rightarrow \infty. \quad (27)$$

A fixed gain is being carried across intervals of vanishing length. We call the divergence in (27) the Zeno compression of the critical height. Only the endpoint values are used: the height may fall below L_j inside the interval and after t_j .

The derivative in this calculation is not an abstract scalar source. Since

$$H(t) = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2,$$

classical differentiation gives

$$H'(t) = \text{Re} \left\langle \Lambda^{1/2} u_t(t), \Lambda^{1/2} u(t) \right\rangle.$$

The compressed rise is carried by the first Eulerian response u_t acting against the same velocity field whose critical size is rising. Physical acceleration, however, is the material derivative $D_t u$, not u_t . Understanding the response requires us to separate those two derivatives without discarding the transport correction between them.

The unit ladder measures temporal compression. Repeating the first-passage construction with a dyadic ladder will measure the work performed by the equation. If $L_{j+1} = 2L_j$, then

$$H(t_{j+1}) - H(t_j) = L_j.$$

After the critical-height balance has been derived, integrating it over these same intervals will show that the nonlinear pressure-completed transfer must supply this entire gain together with the intervening viscous loss.

The terminal threat has now become a response problem. The height records the size of the event, while the equation must identify the fields that produce its change. The physical route is velocity, acceleration, and jerk. The analytic route must express that hierarchy at fixed spatial coordinates, where the whole-field Sobolev norms are measured.

8 From acceleration and jerk to the Eulerian tower

Part I fixed the material derivative along particle trajectories and showed that finite-time escape must be accumulated by higher material response. The critical height, however, is a norm of the entire field at one fixed time. We now compute how material acceleration and jerk decompose into the Eulerian rows $V_n = \partial_t^n u$.

The next material response is the jerk

$$D_t^2 u = D_t(D_t u).$$

It is already more involved than a second partial derivative in time. Set

$$V_n := \partial_t^n u, \quad n \in \mathbb{N}_0.$$

Since

$$D_t u = V_1 + (V_0 \cdot \nabla) V_0,$$

one more application of D_t gives

$$\begin{aligned} D_t^2 u &= (\partial_t + V_0 \cdot \nabla)(V_1 + (V_0 \cdot \nabla) V_0) \\ &= V_2 + (V_1 \cdot \nabla) V_0 + (V_0 \cdot \nabla) V_1 + (V_0 \cdot \nabla) V_1 + (V_0 \cdot \nabla)((V_0 \cdot \nabla) V_0) \\ &= V_2 + (V_1 \cdot \nabla) V_0 + 2(V_0 \cdot \nabla) V_1 + (V_0 \cdot \nabla)((V_0 \cdot \nabla) V_0). \end{aligned}$$

The row V_2 is one part of the material jerk. The other terms are the exact transport corrections created by changing from a moving particle to a fixed spatial point. At higher order, this difference keeps growing through the commutator of the material derivative with spatial differentiation. For example,

$$[D_t, \partial_\ell] f = -(\partial_\ell u \cdot \nabla) f.$$

This commutator explains why the proof does not build its mixed array with successive powers of D_t . Material differentiation continually creates new spatial correction terms. By contrast, ∂_t commutes with every fixed spatial derivative, while the differentiated transport sum records all of the corrections explicitly. The Eulerian rows keep temporal depth and spatial depth as separate coordinates without losing the physics that related them.

The acceleration and jerk calculation separates the moving-particle response from the fixed-coordinate derivatives used by the proof. The critical height is a norm of the whole field at one fixed time, and its Fourier multipliers act in fixed spatial coordinates. Global norm divergence leaves every particle-path choice unresolved. We thus work with the Eulerian rows V_n , while the differentiated transport terms retain exactly the corrections created by the change of frame.

8.1 The exact escalation from one Eulerian row to the next

The fundamental theorem of calculus turns the response intuition into a norm statement. Let X be a Banach space, fix $0 \leq r < t < T_*$, and suppose

$$V_j \in W^{1,1}(r, t; X), \quad \partial_t V_j = V_{j+1}.$$

The Bochner fundamental theorem gives

$$\begin{aligned} V_j(t) - V_j(r) &= \int_r^t V_{j+1}(s) ds, \\ \|V_j(t)\|_X &\leq \|V_j(r)\|_X + \int_r^t \|V_{j+1}(s)\|_X ds. \end{aligned} \tag{28}$$

If $V_{j+1} \in L^1(0, T_*; X)$, the right side stays bounded uniformly in $t < T_*$. On the finite interval, Cauchy–Schwarz gives

$$\int_0^{T_*} \|V_{j+1}(s)\|_X ds \leq \sqrt{T_*} \|V_{j+1}\|_{L^2(0, T_*; X)}.$$

Thus one whole-terminal L^2 row supplies finite total variation to the row below it. The implication is conditional at each level: if a chosen row escapes, the next row cannot have finite accumulated variation in the same space. Escape of V_0 alone does not prove that every higher V_j diverges. The equation generates all of those rows for a different reason: smooth continuation must retain every finite time derivative at the endpoint.

Apply this observation at the critical spatial order. Equation (24) is equivalent to

$$\limsup_{t \uparrow T_*} \|V_0(t)\|_{\dot{H}^{1/2}} = \infty.$$

If the hypotheses needed for (28) hold with $X = \dot{H}^{1/2}$, then

$$\int_0^{T_*} \|V_1(s)\|_{\dot{H}^{1/2}} ds = \infty.$$

On every strict classical subinterval the same argument applies to any fixed row for which an escape has been established. At the base row it identifies the exact channel through which critical escape must pass. Whole-terminal control of V_1 in a sufficiently strong space would close that channel; retaining every higher row will later give the complete endpoint jet needed for a smooth restart.

8.2 The pressure-complete temporal recurrence

The Eulerian hierarchy becomes an object of the equation once pressure is differentiated with it. Define

$$Q_n := \partial_t^n p.$$

The zeroth temporal row is the original momentum equation:

$$V_1 + (V_0 \cdot \nabla)V_0 + \nabla Q_0 = \nu \Delta V_0.$$

Differentiating once gives

$$V_2 + (V_1 \cdot \nabla)V_0 + (V_0 \cdot \nabla)V_1 + \nabla Q_1 = \nu \Delta V_1.$$

The two transport terms have different origins. In the first, the derivative lands on the velocity doing the carrying. In the second, it lands on the velocity being carried. Both remain in the same differentiated row.

At order n , Leibniz' rule distributes the time derivatives across the two velocity factors:

$$\begin{aligned}\partial_t^n((u \cdot \nabla)u) &= \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u \\ &= \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a}.\end{aligned}$$

The coefficient $\binom{n}{a}$ counts the choices of the a derivatives landing on the first velocity factor.

Pressure at the same order is fixed by incompressibility. Taking divergence of the differentiated momentum row and using $\nabla \cdot V_a = 0$ gives

$$\begin{aligned}-\Delta Q_n &= \nabla \cdot \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} \\ &= \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}.\end{aligned}$$

With the decaying normalization fixed in (2), inversion by Riesz transforms gives

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}. \quad (29)$$

Substitution into the momentum row produces the temporal recurrence

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n, \quad \nabla \cdot V_n = 0. \quad (30)$$

Equations (30) and (29) preserve the same VPI response at temporal depth n . Viscosity advances the spatial curvature through ΔV_n . Every lower temporal split appears in the transport sum. The pressure equation performs the whole-space correction required to keep V_{n+1} divergence-free. These three operations generate one next row.

Pressure acts simultaneously across the whole field. At the base row, the Riesz formula has the Calderón–Zygmund representation

$$p(t, x) = -\frac{1}{3}|u(t, x)|^2 + \text{p. v.} \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{3z_i z_j - |z|^2 \delta_{ij}}{|z|^5} u_i(t, y) u_j(t, y) dy, \quad z = x - y.$$

The local term is the delta contribution in the distributional Hessian of the Newtonian kernel; together with the principal-value term it is exactly (2). A change in the velocity far from x can still change the pressure gradient at x , and the differentiated Riesz formula carries that same global coupling through every Q_n . Because the Riesz transforms have order zero on Sobolev spaces, the upper velocity rows of a finite rectangle will also determine the size and normalization of every pressure row it contains; no independent pressure datum will be added at the endpoint.

The recurrence also requires separate temporal and spatial indices. The Laplacian differentiates V_n twice in space, transport contains a spatial derivative of one factor, and pressure is obtained from an order-zero nonlocal operator applied to all lower products. Each temporal row opens an entire spatial column. Before constructing the full two-index array, we return to the critical height. The scalar height balance will cancel a pure pressure gradient, but its differentiated interaction terms are still computed from the pressure-complete rows above. This lets the height read spatial information from the response without replacing the response itself.

9 The completed critical-height rows

For $s \geq 0$, define the homogeneous Sobolev energy

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2.$$

On every compact classical subinterval,

$$E'_s(t) = \operatorname{Re} \langle \Lambda^s u(t), \Lambda^s u_t(t) \rangle.$$

Insert the momentum equation. The viscous term gives

$$\begin{aligned} \nu \operatorname{Re} \langle \Lambda^s u, \Lambda^s \Delta u \rangle &= -\nu \|\Lambda^{s+1} u\|_2^2 \\ &= -2\nu E_{s+1}. \end{aligned}$$

The pressure term cancels in this whole-space pairing because the velocity is divergence-free:

$$\begin{aligned} \operatorname{Re} \langle \Lambda^s u, \Lambda^s \nabla p \rangle &= \operatorname{Re} \langle \Lambda^{2s} u, \nabla p \rangle \\ &= -\operatorname{Re} \langle \nabla \cdot \Lambda^{2s} u, p \rangle \\ &= 0. \end{aligned}$$

The cancellation is a consequence of the pressure equation and incompressibility. It removes the pure gradient from this one scalar pairing; it does not remove pressure from the differentiated field. Time derivatives of the interaction below contain the rows V_a and the pressure responses Q_a fixed by (29)–(30). In the Sobolev balance, the VPI law separates into the viscous term above and the signed pressure-complete interaction current

$$\mathcal{P}_s := -\operatorname{Re} \langle \Lambda^s u, \Lambda^s ((u \cdot \nabla)u + \nabla p) \rangle.$$

Equivalently,

$$\mathcal{P}_s = -\operatorname{Re} \langle \Lambda^s u, \Lambda^s \mathbb{P}((u \cdot \nabla)u) \rangle,$$

where $\mathbb{P}$ is the Leray projection. The pressure-explicit and projected forms are two global pairings of the same incompressible interaction. The Sobolev balance is

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s. \tag{31}$$

At $s = 0$, the transport contribution also cancels and (31) is the kinetic-energy identity. At the critical order $s = 1/2$, it becomes

$$H' = -2\nu E_{3/2} + \mathcal{P}_{1/2}. \tag{32}$$

The two terms on the right have clear roles. Viscosity removes critical height through the next spatial energy $E_{3/2}$. The signed interaction $\mathcal{P}_{1/2}$ supplies whatever height gain remains after that loss.

9.1 The first-passage intervals enter the equation

Integrating (32) over an arbitrary classical interval $[r, t]$ gives

$$H(t) - H(r) + 2\nu \int_r^t E_{3/2}(s) ds = \int_r^t \mathcal{P}_{1/2}(s) ds. \quad (33)$$

Return first to the unit-spaced crossings from Lemma 7.1. Since $H(t_{j+1}) - H(t_j) = 1$, the exact balance on the shrinking interval is

$$\int_{t_j}^{t_{j+1}} \mathcal{P}_{1/2}(s) ds = 1 + 2\nu \int_{t_j}^{t_{j+1}} E_{3/2}(s) ds. \quad (34)$$

The transfer supplies the unit height gain and every unit of critical dissipation incurred while that gain is made. The intervals have lengths $\delta_j \rightarrow 0$, so the same fixed gain and dissipation are being supplied on a sequence of vanishing time scales.

The dyadic levels separate the amplitude of that work from the shrinking interval. If $L_{j+1} = 2L_j$, then $H(t_{j+1}) - H(t_j) = L_j$, and (33) gives

$$\int_{t_j}^{t_{j+1}} \mathcal{P}_{1/2}(s) ds = L_j + 2\nu \int_{t_j}^{t_{j+1}} E_{3/2}(s) ds \geq L_j. \quad (35)$$

The integrated transfer must grow with the level being crossed. Equation (35) has converted the shrinking-time picture into an exact demand on the equation: over the interval that raises the height from L_j to $2L_j$, the signed interaction must supply at least L_j , in addition to the viscous loss. The identity determines that demand; an upper estimate must still control it.

At critical order, the response has the same spatial scale as the viscous term. Self-adjointness of $\Lambda^{1/2}$, the pressure cancellation, Hölder's inequality, and the fractional Sobolev embedding give

$$\begin{aligned} |\mathcal{P}_{1/2}| &= |((u \cdot \nabla)u, \Lambda u)_2| \\ &\leq \|u\|_3 \|\nabla u\|_3 \|\Lambda u\|_3 \\ &\leq C \|u\|_{\dot{H}^{1/2}} \left\| \Lambda^{3/2} u \right\|_2^2. \end{aligned} \quad (36)$$

For sufficiently small critical data, this estimate allows the viscous term to absorb the interaction, which is the familiar Fujita–Kato mechanism [2, 3]. The arbitrary datum in the main theorem lies beyond that smallness regime. We cannot close the first balance by estimating its interaction as a small error. The next move is to differentiate the complete identity instead, retaining every response term it creates. The passage identities (34) and (35) remain valid at every critical size and tell us what those completed rows must eventually control.

9.2 One height derivative, two exact descriptions

The first description is kinematic. Since

$$H(t) = \frac{1}{2} \left\langle \Lambda^{1/2} V_0(t), \Lambda^{1/2} V_0(t) \right\rangle,$$

the first derivative is

$$H'(t) = \operatorname{Re} \left\langle \Lambda^{1/2} V_1(t), \Lambda^{1/2} V_0(t) \right\rangle.$$

Repeated Leibniz differentiation gives, for every $n \in \mathbb{N}_0$,

$$H^{(n)}(t) = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{n-a}(t) \right\rangle. \quad (37)$$

The n -th derivative of the scalar height contains the complete finite temporal block

$$V_0, V_1, \dots, V_n.$$

It contracts that block to one number. The block still carries much more information than the scalar contraction.

The second description is dynamic. Differentiate (31) and substitute the same balance at each new spatial order. The first four rows are

$$\begin{aligned} H &= E_{1/2}, \\ H' &= -2\nu E_{3/2} + \mathcal{P}_{1/2}, \\ H'' &= 4\nu^2 E_{5/2} - 2\nu \mathcal{P}_{3/2} + \partial_t \mathcal{P}_{1/2}, \\ H''' &= -8\nu^3 E_{7/2} + 4\nu^2 \mathcal{P}_{5/2} - 2\nu \partial_t \mathcal{P}_{3/2} + \partial_t^2 \mathcal{P}_{1/2}. \end{aligned}$$

Each differentiation advances the leading viscous energy by one full spatial derivative. It also differentiates every interaction current already present and creates the current belonging to the new spatial level. The current terms are part of the identity at the same status as the leading energy.

Lemma 9.1 (Completed critical-height rows). *For every $n \in \mathbb{N}_0$,*

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}, \quad (38)$$

where the sum is empty for $n = 0$. Define the completed row

$$\mathcal{C}_n := \frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n}. \quad (39)$$

Then

$$\mathcal{C}_n = E_{n+1/2}.$$

Proof. For $n = 0$, the assertion is the definition $H = E_{1/2}$. Suppose (38) holds at order n . Differentiate its leading term and apply (31) at $s = n + 1/2$:

$$\begin{aligned} (-2\nu)^n E'_{n+1/2} &= (-2\nu)^n (-2\nu E_{n+3/2} + \mathcal{P}_{n+1/2}) \\ &= (-2\nu)^{n+1} E_{n+3/2} + (-2\nu)^n \mathcal{P}_{n+1/2}. \end{aligned}$$

The derivative of the existing response sum is

$$\sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-j} \mathcal{P}_{j+1/2}.$$

The newly created term $(-2\nu)^n \mathcal{P}_{n+1/2}$ is exactly the missing $j = n$ term. The sum now runs from $j = 0$ through $j = n$, and the leading energy is $(-2\nu)^{n+1} E_{n+3/2}$. This is (38) at order $n + 1$. Solving the identity for its leading energy gives (39). $\square$

The first completed rows read

$$\mathcal{C}_0 = E_{1/2}, \quad \mathcal{C}_1 = E_{3/2}, \quad \mathcal{C}_2 = E_{5/2}, \quad \mathcal{C}_3 = E_{7/2}.$$

The completion is mathematically essential. At the second derivative,

$$E_{5/2} = \frac{H'' + 2\nu\mathcal{P}_{3/2} - \partial_t\mathcal{P}_{1/2}}{4\nu^2}.$$

The spatial energy is isolated only after the full differentiated response has been retained.

The dyadic shell weights from the opening section now return. Since

$$E_{n+1/2}(P_K u) = \frac{1}{2} \left\| \Lambda^{n+1/2} P_K u \right\|_2^2,$$

the Littlewood–Paley calculation gives

$$E_{n+1/2}(u) \simeq \sum_K K^{2n+1} A_K. \quad (40)$$

Equation (40) shows that critical height uses the weight K . The next completed row uses K^3 , then K^5 , and each increase in temporal depth adds the factor K^2 . The high-frequency contributions required by critical escape are being read at progressively stronger spatial weights.

10 The reciprocal temporal–spatial structure

We can now place the two descriptions of $H^{(n)}$ beside each other. The kinematic identity (37) says that the n -th height derivative is built from the temporal block

$$V_0, V_1, \dots, V_n.$$

The dynamic identity (38) says that, once every differentiated interaction current is kept, that same scalar derivative has the leading spatial energy

$$E_{n+1/2} = \frac{1}{2} \left\| \Lambda^{n+1/2} u \right\|_2^2.$$

One index is now doing two visible jobs. It records how far the response has been differentiated in time, and it records the stronger spatial weight exposed in the completed identity.

This is the reciprocal structure. A proposed finite-time singularity would have to lose spatial control as its active scales contract. The temporal response forced by that escape points in the opposite direction: each new height derivative exposes the next stronger spatial energy of the base velocity. The two movements are balanced because they occur inside one exact identity, and opposing because the singular picture needs spatial regularity to disappear while the differentiated response keeps revealing higher spatial orders that a successful estimate could control.

Nothing in the identity has yet supplied that control. Differentiation does not create regularity, and the completed-row formula is not an estimate. It tells us which spatial energy is hidden inside each temporal response after the complete VPI current has been retained. Part III will later provide the whole-terminal bounds; the present calculation identifies the structure those bounds must reach.

There is also an important index distinction. The formula does not say $V_n \in H^{n+1/2}$. Its leading energy belongs to the base velocity u . Each temporal row V_n has its own independently indexed spatial column:

$$V_n, \quad \nabla V_n, \quad \nabla^2 V_n, \quad \dots$$

The completed height sequence reads one diagonal through the spatial hierarchy of u ; it does not retain the independent temporal and spatial indices. The scalar contraction also discards the spatial distribution of the field. To carry smoothness to the endpoint, we need every finite spatial order in every finite temporal row, with the normalized pressure response attached.

Smoothness asks for every entry to the right of a fixed temporal row. Simultaneous membership at every finite spatial order gives

$$V_n \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3).$$

For each integer $k \geq 0$, choose $m > k + 3/2$. Sobolev embedding gives

$$H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3),$$

so membership at every finite m gives membership in every C^k :

$$H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The operation is an intersection. Each finite membership may have its own constant; smoothness does not require one numerical bound uniform over the entire infinite array.

The diagonal has shown why spatial depth matters, but it cannot construct the missing columns. For that we return to the pressure-complete recurrence (29)–(30) and differentiate each temporal row in space. This produces the full mixed jet while keeping the two derivative indices independent.

11 The pressure-complete mixed jet

Fix one row of the recurrence (29)–(30) and apply every finite spatial derivative to it. For a multi-index $\alpha = (\alpha_1, \alpha_2, \alpha_3)$, define

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p.$$

The temporal index n counts fixed-point response. The multi-index α counts spatial differentiation. The two directions remain independent:

	$ \alpha = 0$	$ \alpha = 1$	$ \alpha = 2$	$\dots$
$n = 0$	u	∇u	$\nabla^2 u$	$\dots$
$n = 1$	u_t	∇u_t	$\nabla^2 u_t$	$\dots$
$n = 2$	u_{tt}	∇u_{tt}	$\nabla^2 u_{tt}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Moving down changes temporal depth. Moving right changes spatial depth. The completed-height rows describe one diagonal relation inside this array; the differentiated recurrence generates and retains the array itself.

For $\beta \leq \alpha$, define the multi-index coefficient

$$\binom{\alpha}{\beta} := \prod_{\ell=1}^3 \binom{\alpha_\ell}{\beta_\ell}.$$

Apply ∂_x^α to one transport term in (30). The spatial product rule gives

$$\begin{aligned} \partial_x^\alpha ((V_a \cdot \nabla) V_{n-a}) &= \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (\partial_x^\beta V_a \cdot \nabla) \partial_x^{\alpha-\beta} V_{n-a} \\ &= \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta}. \end{aligned}$$

Insert this expansion into the temporal binomial sum. Since spatial derivatives commute with ∂_t , ∇ , and Δ , every mixed row obeys

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} \\ = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned} \tag{41}$$

The pressure row is fixed by the divergence equation:

$$\begin{aligned} -\Delta \Pi_{n,\alpha} &= \partial_i \partial_j \partial_t^n \partial_x^\alpha (u_i u_j), \\ \Pi_{n,\alpha} &= R_i R_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} J_{a,\beta,i} J_{n-a,\alpha-\beta,j}. \end{aligned} \tag{42}$$

The first binomial factor distributes temporal derivatives. The second distributes spatial derivatives. Every term is produced by the original quadratic transport and the original pressure normalization.

The first spatial face makes the formula concrete. Take $n = 0$ and $\alpha = e_\ell$. Equation (41) becomes

$$\partial_t \partial_\ell u + (u \cdot \nabla) \partial_\ell u + (\partial_\ell u \cdot \nabla) u + \nabla \partial_\ell p = \nu \Delta \partial_\ell u.$$

The spatial derivative lands once on the transporting velocity and once on the transported velocity. At $n = 1$, the same spatial face is

$$\begin{aligned} &\partial_t^2 \partial_\ell u + (u \cdot \nabla) \partial_\ell u_t + (\partial_\ell u \cdot \nabla) u_t \\ &\quad + (u_t \cdot \nabla) \partial_\ell u + (\partial_\ell u_t \cdot \nabla) u + \nabla \partial_\ell p_t \\ &= \nu \Delta \partial_\ell u_t. \end{aligned}$$

The four transport terms are the four ways one time derivative and one spatial derivative can be distributed across the two velocity factors. They remain differentiated faces of the original solution.

11.1 The datum generates the initial jet

The mixed jet begins at the initial datum. Set

$$V_n^0 := V_n(0), \quad Q_n^0 := Q_n(0), \quad V_0^0 = u_0.$$

Evaluating (29) and (30) at $t = 0$ gives

$$Q_n^0 = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^0 V_{n-a,j}^0, \quad (43)$$

$$V_{n+1}^0 = \nu \Delta V_n^0 - \sum_{a=0}^n \binom{n}{a} (V_a^0 \cdot \nabla) V_{n-a}^0 - \nabla Q_n^0. \quad (44)$$

The first row is already determined entirely by u_0 :

$$Q_0^0 = R_i R_j (u_{0,i} u_{0,j}),$$

and

$$V_1^0 = \nu \Delta u_0 - (u_0 \cdot \nabla) u_0 - \nabla Q_0^0.$$

The datum class $\mathcal{S}_\sigma(\mathbb{R}^3)$ is contained in $H_\sigma^\infty(\mathbb{R}^3)$. Suppose inductively that

$$V_0^0, \dots, V_n^0 \in H^\infty.$$

Fix any finite target order m . Choose an auxiliary Sobolev order high enough that it is an algebra and remains above m after every derivative displayed in (43)–(44). The Sobolev multiplication estimate from Appendix A, boundedness of Riesz transforms on every H^m , and closure under differentiation then give

$$Q_n^0, V_{n+1}^0 \in H^m.$$

Since m was arbitrary,

$$Q_n^0, V_{n+1}^0 \in H^\infty.$$

Induction yields

$$V_n^0, Q_n^0 \in H^\infty(\mathbb{R}^3) \quad (n \in \mathbb{N}_0). \quad (45)$$

Applying ∂_x^α gives every

$$J_{n,\alpha}(0), \quad \Pi_{n,\alpha}(0).$$

Equation (45) is the datum-side origin of the mixed jet. The recursion determines every initial coordinate from u_0 . Whole-terminal control of those coordinates is a separate conclusion. A terminal value for V_n requires simultaneous control of V_n and its time derivative V_{n+1} . The differentiated equation separates one time derivative from two spatial derivatives, so the natural finite unit is an adjacent Sobolev pair centred on the space in which the trace is sought.

12 From adjacent rows to finite rectangles

Fix a temporal row n and a spatial order m , and write

$$v := V_n, \quad v_t = V_{n+1}.$$

The finite rectangle will place these two functions in the spaces

$$v \in L^2(0, T_*; H^{m+1}), \quad v_t \in L^2(0, T_*; H^{m-1}). \quad (46)$$

The offset is deliberate. The two spaces sit on opposite sides of H^m :

$$H^{m+1} \hookrightarrow H^m \hookrightarrow H^{m-1}.$$

At $m = 0$, this is the familiar triple

$$H^1 \hookrightarrow L^2 \hookrightarrow H^{-1}.$$

At higher m , the same geometry is shifted up the Sobolev scale.

The middle space appears from an exact Fourier pairing. For $f \in H^{m-1}$ and $g \in H^{m+1}$, define

$$\langle f, g \rangle_m := \int_{\mathbb{R}^3} (1 + |\xi|^2)^m \widehat{f}(\xi) \cdot \overline{\widehat{g}(\xi)} d\xi.$$

Split the weight at the middle exponent:

$$(1 + |\xi|^2)^m = (1 + |\xi|^2)^{(m-1)/2} (1 + |\xi|^2)^{(m+1)/2}.$$

Cauchy–Schwarz gives

$$|\langle f, g \rangle_m| \leq \|f\|_{H^{m-1}} \|g\|_{H^{m+1}}. \quad (47)$$

For a smooth curve v , differentiation of the middle norm yields

$$\frac{d}{dt} \frac{1}{2} \|v(t)\|_{H^m}^2 = \operatorname{Re} \langle v_t(t), v(t) \rangle_m.$$

Integrating from r to t and applying (47) gives

$$\begin{aligned} \left| \|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 \right| &\leq 2 \int_r^t \|v_s(s)\|_{H^{m-1}} \|v(s)\|_{H^{m+1}} ds \\ &\leq 2 \|v_t\|_{L^2(r,t;H^{m-1})} \|v\|_{L^2(r,t;H^{m+1})}. \end{aligned} \quad (48)$$

The upper spatial row and the next temporal row meet exactly at the H^m -energy. This is why the two Sobolev levels differ by two spatial derivatives.

The same offset agrees with the parabolic order of the equation. In (30), one new time derivative is balanced against ΔV_n , which contains two new spatial derivatives. The Hilbert pairing explains how that parabolic separation produces a trace in the middle space.

This is the Lions–Magenes trace mechanism in the shifted Sobolev triple; the proof is included for the exact indices used here^[21].

Lemma 12.1 (Adjacent-row trace). *Let $m \in \mathbb{R}$ and let $I = (0, T)$ with $T < \infty$. Suppose*

$$v \in L^2(I; H^{m+1}), \quad v_t \in L^2(I; H^{m-1})$$

in the distributional sense. Then v has a unique representative in

$$C([0, T]; H^m).$$

For that representative, estimate (48) holds for every $0 \leq r \leq t \leq T$, and

$$\begin{aligned} \|v\|_{C([0,T];H^m)}^2 &\leq \frac{1}{T} \|v\|_{L^2(I;H^m)}^2 \\ &\quad + 2 \|v_t\|_{L^2(I;H^{m-1})} \|v\|_{L^2(I;H^{m+1})}. \end{aligned} \quad (49)$$

Proof. Let S_R be a smooth Fourier cutoff to $|\xi| \leq 2R$ that equals one on $|\xi| \leq R$, and set

$$v_R := S_R v.$$

The cutoff commutes with ∂_t . At fixed R , it raises spatial regularity by any finite amount, so

$$v_R \in W^{1,2}(I; H^{m+1}) \hookrightarrow C([0, T]; H^{m+1}).$$

The smooth-row energy identity holds for v_R .

Choose a Lebesgue time $t_0 \in I$ at which $v(t_0) \in H^{m+1}$. Such times have full measure. For two cutoffs R, S , apply the energy identity to $w = v_R - v_S$. For every $t \in [0, T]$,

$$\begin{aligned} \|w(t)\|_{H^m}^2 &\leq \|w(t_0)\|_{H^m}^2 \\ &\quad + 2 \|w_t\|_{L^2(I; H^{m-1})} \|w\|_{L^2(I; H^{m+1})}. \end{aligned}$$

As $R, S \rightarrow \infty$, the first term tends to zero because $v(t_0) \in H^{m+1}$. The two tail norms also tend to zero because Fourier cutoffs converge strongly in the given Bochner spaces. Hence $(v_R)_R$ is Cauchy in $C([0, T]; H^m)$. Its uniform limit $\tilde{v}$ is continuous into H^m .

On the other hand, $v_R \rightarrow v$ in $L^2(I; H^{m+1})$, hence in $L^2(I; H^m)$. The continuous limit $\tilde{v}$ agrees with v almost everywhere and is its required representative. Passing to the limit in the cutoff energy identity proves (48) for $\tilde{v}$. Two continuous representatives that agree almost everywhere agree everywhere, which gives uniqueness.

The continuous representative belongs to $L^2(I; H^m)$, so some $r \in [0, T]$ satisfies

$$\|v(r)\|_{H^m}^2 \leq \frac{1}{T} \|v\|_{L^2(I; H^m)}^2.$$

Apply (48) on the interval between this r and an arbitrary $t \in [0, T]$, then take the supremum over t . This gives (49). $\square$

The lemma controls the whole vector in the middle space, while the displayed estimate controls the change of its squared norm. The cutoff construction joins those two facts in one continuous representative. In particular, if the adjacent pair holds on $(0, T_*)$, then applying the lemma with $T = T_*$ produces one value $v(T_*) \in H^m$, and every sequence $t_k \uparrow T_*$ converges to that same value. The topology selects one compatible terminal reading and resolves the failure called Jump in Part I.

To use this at every depth, the adjacent pair must be retained for each temporal row and each middle Sobolev order, together with the pressure produced by those rows. The smallest continuation-grade example is the pair for u centred on H^2 .

12.1 The first concrete rectangles

Choose temporal cutoff $N = 0$ and middle spatial order $M = 2$. The resulting rectangle has velocity size

$$\mathcal{N}_{0,2}(u; T) := \|u\|_{L^2(0, T; H^3)}^2 + \|u_t\|_{L^2(0, T; H^1)}^2. \quad (50)$$

The norm (50) is the adjacent pair (46) placed inside the mixed jet. Each coordinate pays a different part of the terminal argument.

First, the two coordinates control the total variation of the critical height. The kinematic identity at $n = 1$ gives

$$H'(t) = \operatorname{Re} \left\langle \Lambda^{1/2} u_t(t), \Lambda^{1/2} u(t) \right\rangle.$$

Since $H^1 \hookrightarrow \dot{H}^{1/2}$ and $H^3 \hookrightarrow \dot{H}^{1/2}$, Cauchy–Schwarz in time gives

$$\begin{aligned} \int_0^{T_*} |H'(t)| dt &\leq \|u_t\|_{L^2(0, T_*; \dot{H}^{1/2})} \|u\|_{L^2(0, T_*; \dot{H}^{1/2})} \\ &\leq C \|u_t\|_{L^2(0, T_*; H^1)} \|u\|_{L^2(0, T_*; H^3)}. \end{aligned}$$

Finiteness of the whole-terminal rectangle would make the right side finite. Then

$$H(t) \leq H(0) + \int_0^{T_*} |H'(s)| ds$$

would rule out the terminal escape in (24).

The same two coordinates also pay the signed passage bill derived earlier. The upper row gives

$$\int_0^{T_*} E_{3/2}(t) dt \leq \frac{1}{2} \|u\|_{L^2(0, T_*; H^3)}^2 < \infty,$$

and the critical balance (32) then gives

$$\mathcal{P}_{1/2} = H' + 2\nu E_{3/2} \in L^1(0, T_*).$$

For the dyadic first-passage intervals, (35) requires

$$L_j \leq \int_{t_j}^{t_{j+1}} \mathcal{P}_{1/2}(s) ds \leq \int_{t_j}^{T_*} |\mathcal{P}_{1/2}(s)| ds.$$

The last tail tends to zero because $t_j \uparrow T_*$, while $L_j \rightarrow \infty$. Thus the first rectangle closes not only the scalar variation estimate but the exact VPI work requirement that arose from Zeno compression.

Second, the upper coordinate produces the continuation-grade Lipschitz coefficient. Sobolev embedding gives

$$H^3(\mathbb{R}^3) \hookrightarrow W^{1, \infty}(\mathbb{R}^3),$$

and Cauchy–Schwarz on the finite interval gives

$$\int_0^{T_*} \|\nabla u(t)\|_\infty dt \leq C \sqrt{T_*} \|u\|_{L^2(0, T_*; H^3)}.$$

This is the integrable coefficient required by the higher Sobolev continuation estimate in Part I.

Third, the two rows surround the middle space H^2 . Applying Lemma 12.1 with $m = 2$ gives

$$u \in C([0, T_*]; H^2).$$

The first rectangle consequently controls critical variation, produces the Lipschitz coefficient, and gives the original velocity curve a strong terminal trace. One finite pair produces all three consequences together.

Raise the temporal cutoff to $N = 1$ while keeping $M = 2$. The next rectangle has size

$$\begin{aligned}\mathcal{N}_{1,2}(u; T) := & \|u\|_{L^2(0,T;H^3)}^2 + \|u_t\|_{L^2(0,T;H^3)}^2 \\ & + \|u_t\|_{L^2(0,T;H^1)}^2 + \|u_{tt}\|_{L^2(0,T;H^1)}^2.\end{aligned}$$

The shared coordinate u_t appears in both spatial rows. In the first adjacent pair it is the derivative of u ; in the second it is the evolving field whose derivative is u_{tt} . Both appearances refer to the same distributional derivative generated by the equation. Raising N extends this overlap through higher temporal rows. Raising M places the same rows around a stronger middle trace space.

For arbitrary finite $N, M \in \mathbb{N}_0$, define

$$\begin{aligned}\mathcal{N}_{N,M}(u; T) := & \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ & + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2.\end{aligned}\tag{51}$$

When $M = 0$, the lower space is H^{-1} . For every M , the index names the middle trace space H^M ; pointwise derivative counts are a different notion. Each cutoff pair carries its own constant, and the family is combined coordinatewise.

A bound uniform in the time cutoff T , for this fixed pair (N, M) , has two immediate consequences. It supplies every adjacent pair in (51), so Lemma 12.1 gives

$$V_n \in C([0, T_*]; H^M), \quad \sup_{0 \leq t \leq T_*} \|V_n(t)\|_{H^M} < \infty \quad (0 \leq n \leq N).$$

At the chosen cutoff, every traced row remains bounded in its middle Sobolev topology, which is the finite-depth control demanded by Blown. The family ranges over every finite M , so every finite spatial readout is reached by a larger rectangle. The same continuous representative forces each row to approach one value in H^M , resolving Jump at that cutoff. The rectangle keeps both consequences attached to the same derivatives at every depth.

12.1.1 The pressure-complete finite rectangle space

Let $I = (0, T)$, where $0 < T \leq T_*$. The velocity space belonging to the cutoff pair (N, M) is

$$\mathfrak{V}_{N,M}(I) := \left\{ (v_0, \dots, v_{N+1}) : \begin{array}{ll} v_n \in L^2(I; H_\sigma^{M+1}), & 0 \leq n \leq N, \\ v_{n+1} \in L^2(I; H_\sigma^{M-1}), & 0 \leq n \leq N, \\ \partial_t v_n = v_{n+1} \text{ in } \mathcal{D}'(I \times \mathbb{R}^3), & 0 \leq n \leq N \end{array} \right\}.$$

The norm is the finite sum displayed in (51). Each v_n , $0 \leq n \leq N$, has an H^M -continuous representative by the adjacent-row lemma and has a well-defined trace at $t = 0$.

The pressure-complete rectangle space $\mathscr{R}_{N,M}(u_0; I)$ contains the tuples

$$(v_0, \dots, v_N; v_{N+1}; q_0, \dots, q_N)$$

in

$$\mathfrak{V}_{N,M}(I) \times [L^1(I; H^M)]^{N+1}$$

that satisfy four finite conditions:

1. $\partial_t v_n = v_{n+1}$ and $\nabla \cdot v_n = 0$ for $0 \leq n \leq N$;
2. the pressure recurrence (29) holds with $V_a = v_a$ and $Q_n = q_n$;
3. the momentum recurrence (30) holds in $\mathcal{D}'(I \times \mathbb{R}^3)$;
4. the traces of $v_0, \dots, v_N$ at the initial time equal the datum-generated values from (43)–(44).

At this cutoff, v_{N+1} has only its lower-space membership. Its H^M trace appears when the temporal cutoff is raised and the same function is retained as an upper row. The pressure coordinate lies in $L_t^1 H_x^M$ and inherits its value from the velocity rows through the Riesz formula.

For the maximal classical solution, the finite rectangle realized on $I_T = (0, T)$ is

$$\mathcal{R}_{N,M}[u_0; T] := (V_0, \dots, V_N; V_{N+1}; Q_0, \dots, Q_N)|_{I_T} \in \mathcal{R}_{N,M}(u_0; I_T).$$

The notation u_0 records the origin of every entry. The rectangle is a finite restriction of the jet generated by u_0 and realized by the maximal solution.

12.1.2 The velocity rows determine the pressure topology

The product estimates explain why the pressure coordinate belongs to $L^1(I; H^M)$. When $M = 0$, interpolation between $H^1 \hookrightarrow L^2$ and $H^1 \hookrightarrow L^6$ gives

$$\|f\|_4 \leq \|f\|_2^{1/4} \|f\|_6^{3/4} \leq C \|f\|_{H^1}.$$

Consequently,

$$\|fg\|_2 \leq \|f\|_4 \|g\|_4 \leq C \|f\|_{H^1} \|g\|_{H^1}.$$

When $M \geq 1$, the order $M + 1$ is greater than $3/2$, and $H^{M+1}(\mathbb{R}^3)$ is a Banach algebra. In both cases,

$$\|fg\|_{H^M} \leq C_M \|f\|_{H^{M+1}} \|g\|_{H^{M+1}}. \quad (52)$$

The Riesz transforms are bounded on H^M . Applying (52) to every finite product in (29), then using Cauchy–Schwarz in time, gives

$$\|Q_n\|_{L^1(J; H^M)} \leq C_{n,M} \sum_{a=0}^n \|V_a\|_{L^2(J; H^{M+1})} \|V_{n-a}\|_{L^2(J; H^{M+1})} \quad (53)$$

for every subinterval $J \subset (0, T_*)$ on which the displayed velocity norms are finite.

The differentiated momentum equation also has a common topology. Since every V_a is divergence-free,

$$(V_a \cdot \nabla) V_b = \nabla \cdot (V_a \otimes V_b).$$

At $M = 0$, the $H^1 \hookrightarrow L^4$ estimate puts $V_a \otimes V_b$ in $L_t^1 L_x^2$, so its divergence lies in $L_t^1 H_x^{-1}$. At $M \geq 1$, the algebra estimate puts the tensor product in $L_t^1 H_x^{M+1}$, and its divergence belongs to $L_t^1 H_x^M$, hence to $L_t^1 H_x^{M-1}$. The pressure gradient belongs to $L_t^1 H_x^{M-1}$ by (53). Finally,

$$\Delta V_n \in L^2(I; H^{M-1}) \hookrightarrow L^1(I; H^{M-1})$$

because I has finite length. The complete temporal recurrence holds in $L^1(I; H^{M-1})$.

The upper velocity rows fix the pressure size and gauge. Its gradient meets the lower velocity topology, and the same finite rectangle retains the incompressibility equation that generated it.

The finite rectangle now retains the complete pressure-coupled response needed at the endpoint. Its upper velocity rows determine every normalized pressure row through (29); estimate (53) places that pressure in the common topology; and every term of (30) remains meaningful in $L^1(I; H^{M-1})$. Whole-terminal bounds and compatible traces can then carry the velocity, pressure, and temporal response to one endpoint, where the limiting recurrence rules out Dead.

12.2 Compatibility under restriction

A larger rectangle can belong to the same all-orders family only if lowering either cutoff retains the functions already present. Suppose $N' \leq N$ and $M' \leq M$. Deleting rows above N' leaves $V_0, \dots, V_{N'+1}$ and $Q_0, \dots, Q_{N'}$ unchanged, while the embeddings

$$H^{M+1} \hookrightarrow H^{M'+1}, \quad H^{M-1} \hookrightarrow H^{M'-1}, \quad H^M \hookrightarrow H^{M'}$$

view those same functions at the lower spatial order.

For example, the $(2, 3)$ rectangle contains u, u_t, u_{tt}, u_{ttt} and p, p_t, p_{tt} . Lowering the cutoffs to $(1, 2)$ discards u_{ttt} and p_{tt} , retains u, u_t, u_{tt} and p, p_t , and regards every retained function in the weaker Sobolev topology. The retained u_t is the same distributional derivative in both rectangles.

This equality on every shared derivative is compatibility. Once the bounds hold on the common terminal interval, these restriction identities assemble the finite rectangles into one all-orders object. Each coordinate retains its own constant indexed by finite derivative depth.

The same equality later forces agreement of the terminal traces. If two rectangles contain the row V_n , both use the same preterminal curve. When their adjacent estimates produce continuous representatives in a shared lower Sobolev topology, those representatives agree for every $t < T_*$ and hence at T_* . Compatibility is thus the reason a derivative cannot acquire one endpoint value in a strong rectangle and another value in a weaker one.

12.3 The terminal quantifier

Compatibility fixes the identity of the coordinates. Terminal control adds a second requirement: each fixed finite rectangle must remain bounded as its time interval approaches T_* .

Fix $T < T_*$, a temporal row n , and a spatial order q . The maximal solution is classical on the compact interval $[0, T]$, so

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^q} < \infty.$$

It follows that

$$\|V_n\|_{L^2(0, T; H^q)}^2 \leq T \sup_{0 \leq t \leq T} \|V_n(t)\|_{H^q}^2 < \infty.$$

Every sum defining $\mathcal{N}_{N, M}(u; T)$ contains only finitely many such terms. Hence local classicality gives

$$\forall T < T_* \quad \forall N, M < \infty, \quad \mathcal{N}_{N, M}(u; T) < \infty. \quad (54)$$

The bound obtained from compactness may depend on T . All of the proposed singular behavior can be carried by that dependence.

The scalar function

$$f(t) = (T_* - t)^{-1/2}$$

isolates the point. For each fixed $T < T_*$,

$$\int_0^T |f(t)|^2 dt = \log \frac{T_*}{T_* - T} < \infty.$$

On the complete interval,

$$\int_0^{T_*} |f(t)|^2 dt = \int_0^{T_*} \frac{dt}{T_* - t} = \infty.$$

The example is a scalar model of the quantifier failure. Finiteness on every strict subinterval coexists here with divergence on their union.

Local classicality says that $\mathcal{N}_{N,M}(u; T)$ is finite after a cutoff $T < T_*$ has been fixed; its finite value may diverge as the cutoff approaches the endpoint. The required statement begins with the rectangle directly on $I_* := (0, T_*)$. Write $\mathcal{N}_{N,M}(u; I_*)$ for the finite sum in (51) with every time norm taken on I_* . The whole-terminal statement is

$$\mathcal{N}_{N,M}(u; I_*) < \infty \quad \text{for every finite } N, M. \quad (55)$$

This direct datum-generated whole-terminal finiteness is the established theorem that opens Part III, supplies the rectangle family, completes its pressure rows, and assembles its shared derivatives.

Part III

Whole-terminal control

13 The datum-generated compatible family

Part II ended with the distinction between strict-subinterval regularity and the terminal object required by the proof. The next theorem supplies that object directly from the datum. Temporal and spatial depth are fixed independently. Because the equation generated every coordinate before either depth was chosen, all overlapping rectangles retain the same derivatives of the same history. The adjacent rows give terminal traces, and the compatible family reconstructs one common endpoint from which the equation can restart.

13.1 The energy-paid base of the producer

Before the positive rectangles are opened, the kinetic-energy row already places the first pressure-complete response in two negative Sobolev topologies. This is the datum-side base from which the mixed construction starts.

Proposition 13.1 (Energy-paid response faces). *Let $0 < T < T_*$, put $I_T = (0, T)$, and set*

$$U_0 := \|u_0\|_2, \quad \mathcal{E}_\nu(T) := U_0 \left(T + \frac{1}{2\nu} \right)^{1/2}. \quad (56)$$

With

$$\mathcal{Q}(u, p) := (u \cdot \nabla)u + \nabla p = \mathbb{P}\nabla \cdot (u \otimes u),$$

the one datum-generated history satisfies

$$\|u\|_{L^\infty(I_T; L^2)} \leq U_0, \quad \|u\|_{L^2(I_T; H^1)} \leq \mathcal{E}_\nu(T), \quad (57)$$

$$\|p\|_{L^{4/3}(I_T; L^2)} + \|\mathcal{Q}(u, p)\|_{L^{4/3}(I_T; H^{-1})} \leq CU_0^{1/2} \mathcal{E}_\nu(T)^{3/2}, \quad (58)$$

$$\|p\|_{L^2(I_T; H^{-1/2})} + \|\mathcal{Q}(u, p)\|_{L^2(I_T; H^{-3/2})} \leq CU_0 \mathcal{E}_\nu(T), \quad (59)$$

$$\|u_t\|_{L^{4/3}(I_T; H^{-1})} \leq CU_0^{1/2} \mathcal{E}_\nu(T)^{3/2} + \nu T^{1/4} \mathcal{E}_\nu(T), \quad (60)$$

$$\|u_t\|_{L^2(I_T; H^{-3/2})} \leq CU_0 \mathcal{E}_\nu(T) + \nu \mathcal{E}_\nu(T). \quad (61)$$

Every right-hand side is fixed by u_0, ν, T .

Proof. The energy identity gives

$$\sup_{0 \leq t \leq T} \|u(t)\|_2^2 \leq U_0^2, \quad \int_0^T \|\nabla u(t)\|_2^2 dt \leq \frac{U_0^2}{2\nu}.$$

Adding the L^2 part of the inhomogeneous H^1 norm proves (57).

At each time, the three-dimensional interpolation inequality gives

$$\|u\|_4^2 \leq C \|u\|_2^{1/2} \|\nabla u\|_2^{3/2}.$$

Since the Riesz transforms are bounded on L^2 , while divergence maps L^2 to H^{-1} , raising this estimate to the power $4/3$ and integrating proves (58).

For the second face, duality with $H^{1/2} \hookrightarrow L^3$ gives

$$\|u_i u_j\|_{H^{-1/2}} \leq C \|u\|_3^2 \leq C \|u\|_2 \|u\|_{H^1}.$$

The Riesz formula for pressure and one divergence now give (59) after squaring and integrating in time. Finally,

$$u_t = -\nu \mathbb{P}(-\Delta)u - \mathcal{Q}(u, p).$$

The diffusion map sends H^1 to H^{-1} , hence also to $H^{-3/2}$. Its $L^{4/3}$ -time norm is bounded by $T^{1/4}$ times its L^2 -time norm. Combining these facts with (58) and (59) proves the last two estimates. $\square$

These are the first faces of the temporal tower: energy pays the nonlinear response and u_t before any positive spatial row is requested. The datum-generated theorem below upgrades this same pressure-complete history to every finite positive adjacent pair on the whole proposed terminal interval. The recurrence does not switch histories during that upgrade.

Theorem 13.2 (Datum-generated whole-terminal compatible rectangles). *Let $\nu > 0$, let $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let (u, p) be the pressure-normalized maximal classical solution generated by this datum on $[0, T_*)$. If $T_* < \infty$, set $I_* = (0, T_*)$ and $V_n = \partial_t^n u$. Then, for every finite $N, M \in \mathbb{N}_0$, the same pressure-complete history produces a whole-terminal rectangle satisfying*

$$\begin{aligned} \mathcal{N}_{N,M}(u; I_*) &:= \sum_{n=0}^N \|V_n\|_{L^2(I_*; H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|V_{n+1}\|_{L^2(I_*; H^{M-1})}^2 < \infty. \end{aligned} \tag{62}$$

In adjacent-row coordinates, the output is, for every finite $n, m \in \mathbb{N}_0$,

$$V_n \in L^2(I_*; H^{m+1}), \quad V_{n+1} = \partial_t V_n \in L^2(I_*; H^{m-1}). \tag{63}$$

Together with the pressure topology in (53), these bounds place the actual finite rectangle

$$\mathcal{R}_{N,M}[u_0; T_*] := (V_0, \dots, V_N; V_{N+1}; Q_0, \dots, Q_N)|_{I_*}$$

in $\mathcal{R}_{N,M}(u_0; I_*)$. If $N' \leq N$ and $M' \leq M$, restriction satisfies

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0; T_*] = \mathcal{R}_{N',M'}[u_0; T_*]. \tag{64}$$

Every shared velocity or pressure coordinate in this identity is the same distributional derivative of the maximal solution.

Fix one finite pair (N, M) . Estimate (62) allows the time cutoff to reach T_* , so every coordinate of that rectangle belongs to its stated whole-terminal space. The restriction identity then says that the corresponding coordinate in any larger rectangle is literally the same derivative of (u, p) . The estimates arrive on the terminal interval already attached to one another.

13.2 Reading the estimate by rows or by rectangles

The theorem has two useful readings. The trace argument uses one adjacent temporal pair at a time; compatibility uses a finite rectangle. For fixed N and M , these are the same finitely many norms. The passage between them is worth making explicit because it changes no derivative depth and adds no infinite-order summation.

Fix N, M and write

$$F_{N,M}(T) := \mathcal{N}_{N,M}(u; T).$$

Every term in $F_{N,M}(T)$ is the integral of a nonnegative function over $(0, T)$, so $F_{N,M}$ is increasing. Monotone convergence gives

$$\begin{aligned} \lim_{T \uparrow T_*} F_{N,M}(T) &= \sum_{n=0}^N \|V_n\|_{L^2(I_*; H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|V_{n+1}\|_{L^2(I_*; H^{M-1})}^2. \end{aligned}$$

There are only finitely many summands. Applying (63) to each retained row makes the right side finite, and restriction to $(0, T)$ gives (62).

Conversely, fix n, m and use the rectangle with $(N, M) = (n, m)$. Estimate (62) bounds the increasing finite-rectangle norm, so its monotone limit is finite. The two nonnegative summands containing V_n and V_{n+1} give exactly (63). The temporal and spatial depths remain fixed throughout both directions.

14 Pressure completion on the terminal interval

The upper velocity rows controlled by Theorem 13.2 already determine every retained pressure row. Applying (53) on I_* gives

$$\|Q_n\|_{L^1(I_*; H^M)} \leq C_{n,M} \sum_{a=0}^n \|V_a\|_{L^2(I_*; H^{M+1})} \|V_{n-a}\|_{L^2(I_*; H^{M+1})} < \infty. \quad (65)$$

Thus the velocity rectangle itself completes pressure on the terminal interval.

The same product estimate controls the transport sum

$$\sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} \quad \text{in } L^1(I_*; H^{M-1}).$$

The viscous term belongs to $L^2(I_*; H^{M-1})$, and hence to $L^1(I_*; H^{M-1})$ because $T_* < \infty$. Thus every differentiated momentum row holds in one common whole-terminal topology. Because every larger rectangle uses the same Riesz-transform formula, shared velocity rows determine the same normalized pressure row under restriction.

15 Restriction and projective assembly

The coordinatewise restriction described in Part II becomes the bonding map of the whole-terminal finite family. It deletes higher rows, weakens the Sobolev topology of the rows that remain, and never changes the retained functions.

Order the finite pairs by

$$(N', M') \preceq (N, M) \iff N' \leq N \text{ and } M' \leq M.$$

For such a pair, define

$$\begin{aligned} \rho_{N', M'}^{N, M}(V_0, \dots, V_N; V_{N+1}; Q_0, \dots, Q_N) \\ := (V_0, \dots, V_{N'}; V_{N'+1}; Q_0, \dots, Q_{N'}), \end{aligned}$$

where the retained coordinates are carried through the continuous embeddings

$$H^{M+1} \hookrightarrow H^{M'+1}, \quad H^{M-1} \hookrightarrow H^{M'-1}, \quad H^M \hookrightarrow H^{M'}.$$

This gives a continuous map

$$\rho_{N', M'}^{N, M} : \mathcal{R}_{N, M}(u_0; I_*) \longrightarrow \mathcal{R}_{N', M'}(u_0; I_*).$$

When $N' < N$, the distinguished $V_{N'+1}$ in the smaller rectangle is an ordinary retained row of the larger one. When $N' = N$, it is the larger rectangle's distinguished lower-space derivative. In either case the target coordinate is the same $V_{N'+1}$.

The maps obey

$$\rho_{N, M}^{N, M} = \text{id}, \quad \rho_{N'', M''}^{N', M'} \circ \rho_{N', M'}^{N, M} = \rho_{N'', M''}^{N, M}$$

whenever $(N'', M'') \preceq (N', M') \preceq (N, M)$. Deleting rows in two stages is the same operation as deleting them at once, and lowering the Sobolev topology twice leaves the same coordinate as lowering it directly.

The identity and composition laws make the rectangles a projective family. For the actual solution, compatibility is the concrete identity (64): every smaller rectangle is obtained from every larger one by forgetting coordinates and weakening Sobolev topologies. No new velocity or pressure is chosen in that passage. The differentiated momentum equations fix the velocity coordinates, and (65) fixes the normalized pressure coordinates.

This is the all-orders intersection in its useful form. It is not one norm of an infinite array. It is the assertion that every finite rectangle has its own whole-terminal bound and that any derivative appearing in two rectangles is the same function in both. The first fact supplies depth; the second keeps identity while the depth increases.

Now fix a temporal row V_n . Every rectangle tall enough to contain it uses the same distributional derivative, and increasing M places that same function in stronger finite Sobolev spaces. Their constants may grow with N and M , while derivative depth is handled coordinatewise. The restriction identities preserve the fact that all of these bounds belong to one coordinate.

That identity becomes decisive at the terminal time. An H^k trace and an H^m trace of V_n , with $k > m$, can define one endpoint only because both arise from the same preterminal function. The projective assembly has already supplied that compatibility before any trace is taken.

16 What the compatible family gives at all orders

Continuation begins from the velocity itself. Fixing the zeroth temporal row and increasing the spatial depth retains the same u in every finite Sobolev space. For each fixed m , the theorem supplies the velocity coordinate of a rectangle whose spatial depth is at least m . Compatibility says that all of these coordinates are the same u . It follows that

$$u \in \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)).$$

Every fixed $L_t^2 H_x^m$ norm is finite, with a separate constant permitted at each spatial order.

The same argument applies to every temporal row. Fix $n, q \in \mathbb{N}_0$ and choose a rectangle deep enough that its n -th row is viewed in H^q . Then

$$V_n = \partial_t^n u \in L^2(I_*; H^q) \quad (n, q \in \mathbb{N}_0).$$

In particular, fix finite $r, q \in \mathbb{N}_0$. The rows $V_0, \dots, V_r$ are the successive distributional time derivatives of one velocity and all belong to $L^2(I_*; H^q)$. Thus

$$u \in H^r(I_*; H^q) \quad \text{for every finite } r, q, \quad u \in \bigcap_{r, q \in \mathbb{N}_0} H^r(I_*; H^q).$$

This is the all-orders velocity realization of the compatible finite family. Every finite time–space demand is retained separately, and the intersection is understood coordinatewise.

16.1 One finite coordinate gives one finite estimate

The intersection changes the estimate problem because the size of one finite rectangle can now be carried all the way to any readout contained in it. Fix $N, M \in \mathbb{N}_0$ and a row $0 \leq n \leq N$. Applying the quantitative adjacent-row trace estimate (49) to V_n gives

$$\begin{aligned} \sup_{0 \leq t \leq T_*} \|V_n(t)\|_{H^M}^2 &\leq \frac{1}{T_*} \|V_n\|_{L^2(I_*; H^M)}^2 \\ &\quad + 2 \|V_{n+1}\|_{L^2(I_*; H^{M-1})} \|V_n\|_{L^2(I_*; H^{M+1})} \\ &\leq (1 + T_*^{-1}) \mathcal{N}_{N,M}(u; I_*). \end{aligned} \tag{66}$$

The last line uses $H^{M+1} \hookrightarrow H^M$ and $2xy \leq x^2 + y^2$. No all-orders norm enters: the one rectangle on the right contains both factors that produce the middle trace on the left.

Taking $N = n$ and $M = m$ gives, for every fixed $n, m \in \mathbb{N}_0$,

$$V_n \in C([0, T_*]; H^m), \quad \mathcal{M}_{n,m} := \sup_{0 \leq t \leq T_*} \|V_n(t)\|_{H^m} \leq (1 + T_*^{-1})^{1/2} \mathcal{N}_{n,m}(u; I_*)^{1/2}. \tag{67}$$

The two-unit gap between the upper and lower rows is doing exact work here: their shifted pairing lands at the H^m norm whose endpoint value is being measured.

Now fix $k, n \in \mathbb{N}_0$, $1 \leq a \leq \infty$, and $2 \leq b \leq \infty$. Choose an integer m satisfying

$$m > k + \frac{3}{2} - \frac{3}{b}.$$

The spatial embedding (387), followed by (67), gives

$$\begin{aligned} \nabla^k \partial_t^n u &\in C([0, T_*]; L^b(\mathbb{R}^3)) \cap L^a(I_*; L^b(\mathbb{R}^3)), \\ \sup_{0 \leq t \leq T_*} \left\| \nabla^k \partial_t^n u(t) \right\|_{L^b} &\leq C_{m,k,b} (1 + T_*^{-1})^{1/2} \mathcal{N}_{n,m}(u; I_*)^{1/2}, \\ \left\| \nabla^k \partial_t^n u \right\|_{L^a(I_*; L^b)} &\leq C_{m,k,b} T_*^{1/a} (1 + T_*^{-1})^{1/2} \mathcal{N}_{n,m}(u; I_*)^{1/2}, \end{aligned} \quad (68)$$

where $T_*^{1/\infty} = 1$. This is the general projection rule. Once a finite estimate involving a prescribed number of derivatives and one of these Sobolev-accessible mixed norms has been named, one sufficiently high finite coordinate of the compatible family supplies it. Taking $a = b = \infty$ after choosing $m > k + \frac{3}{2}$ gives the pointwise version

$$\sup_{0 \leq t \leq T_*} \left\| \nabla^k \partial_t^n u(t) \right\|_{\infty} < \infty \quad (n, k \in \mathbb{N}_0).$$

Pressure is read from the same coordinates, but now its size can be retained rather than only its membership. Fix $n, s \in \mathbb{N}_0$ and put $r_s := \max\{s, 2\}$. The normalized identity

$$Q_n = R_i R_j \sum_{c=0}^n \binom{n}{c} V_{c,i} V_{n-c,j}$$

and the product estimate from Appendix A give

$$\begin{aligned} \sup_{0 \leq t \leq T_*} \|Q_n(t)\|_{H^s} &\leq C_s \sum_{c=0}^n \binom{n}{c} \mathcal{M}_{c,r_s} \mathcal{M}_{n-c,r_s} \\ &\leq C_s 2^n (1 + T_*^{-1}) \mathcal{N}_{n,r_s}(u; I_*). \end{aligned} \quad (69)$$

The common rectangle (n, r_s) contains every factor in the finite Leibniz sum; the binomial weights contribute exactly 2^n .

For fixed $n, k \in \mathbb{N}_0$, $1 \leq a \leq \infty$, and $2 \leq b \leq \infty$, choose an integer $s > k + \frac{3}{2} - \frac{3}{b}$. Spatial embedding and (69) then give

$$\begin{aligned} \nabla^k Q_n &\in C([0, T_*]; L^b(\mathbb{R}^3)) \cap L^a(I_*; L^b(\mathbb{R}^3)), \\ \sup_{0 \leq t \leq T_*} \left\| \nabla^k Q_n(t) \right\|_{L^b} &\leq C_{s,k,b} 2^n (1 + T_*^{-1}) \mathcal{N}_{n,r_s}(u; I_*), \\ \left\| \nabla^k Q_n \right\|_{L^a(I_*; L^b)} &\leq C_{s,k,b} 2^n T_*^{1/a} (1 + T_*^{-1}) \mathcal{N}_{n,r_s}(u; I_*). \end{aligned} \quad (70)$$

Equation (65) is the direct $L_t^1 H_x^M$ pressure completion. Equations (69) and (70) are its uniform-time readouts. Both come from the same velocity rectangle and the same normalized pressure formula.

The family is complete before the endpoint is taken. Its zeroth row now has a terminal value in every fixed Sobolev space. Compatibility must still show that the values obtained from different finite rectangles are one common endpoint, equal across every continuous Sobolev embedding. That is the next step.

Part IV

From finite projections to continuation

17 The compatible family as an estimate generator

Part III changed the estimate problem. Once a finite derivative and target norm have been named, equation (68) selects one sufficiently deep rectangle and supplies the required bound. The calculations below return to the quantities that made the terminal threat visible in Parts I and II: the classical continuation coefficient, the Ladyzhenskaya–Prodi–Serrin scale, vorticity and strain, normalized pressure, the critical height, and the high-frequency tail. Each comes from a finite projection of the same compatible family.

These bounds will exclude Blown and the critical escape scenario. They do not yet prove that traces obtained in different finite topologies are one terminal state, or that the normalized pressure–momentum response survives there. Those are the remaining Jump and Dead obligations. The finite readouts will lead directly to them.

On a proposed finite maximal interval, Theorem 13.2 at the rectangle $(N, M) = (0, 2)$ gives

$$u \in L^2(I_*; H^3).$$

The embedding $H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3)$ and Cauchy–Schwarz in time yield

$$L_* := \int_0^{T_*} \|\nabla u(t)\|_\infty dt \leq CT_*^{1/2} \|u\|_{L^2(I_*; H^3)} \leq CT_*^{1/2} \mathcal{N}_{0,2}(u; I_*)^{1/2} < \infty. \quad (71)$$

This is exactly the quantity whose divergence was forced by (16), and it is the coefficient in the high-order energy inequality (14). Substituting (71) into (15) gives, for each fixed integer $k \geq 3$,

$$\sup_{0 \leq t < T_*} \|u(t)\|_{H^k}^2 \leq \|u_0\|_{H^k}^2 \exp(2C_k L_*) \leq \|u_0\|_{H^k}^2 \exp\left(2C_k CT_*^{1/2} \mathcal{N}_{0,2}(u; I_*)^{1/2}\right) < \infty. \quad (72)$$

The constant may grow with k . The conclusion is one finite bound after one finite order has been chosen.

This pays the direct continuation coefficient isolated in Part I. It is finite on the complete proposed terminal interval, and (72) propagates the resulting control through every fixed Sobolev order.

17.1 The numerical size of the completed family

Theorem 13.2 has already supplied every compatible rectangle. We can now ask how much of that family must remain visible in the constants of the final estimates. Put

$$\mathfrak{B}_2[u_0, \nu; T_*] := \mathcal{N}_{0,2}(u; I_*) = \|u\|_{L^2(I_*; H^3)}^2 + \|u_t\|_{L^2(I_*; H^1)}^2. \quad (73)$$

This is the finite $(0, 2)$ output of the datum theorem. The calculation below does not use it to construct that theorem; it shows that, once the family exists, every higher numerical readout can be written through this one datum-generated value and finitely many Sobolev sizes of u_0 .

The equation itself opens this base value one step further. Let

$$A := -\Delta, \quad B(u, u) := \mathbb{P}\nabla \cdot (u \otimes u),$$

so that the projected momentum equation is $u_t + \nu Au + B(u, u) = 0$. For $0 < S < T_*$, put

$$X_\nu(S) := \int_0^S \left(\|u_t(t)\|_{H^1}^2 + \nu^2 \|Au(t)\|_{H^1}^2 \right) dt,$$

$$Z_0 := \left\| A^{1/2}u_0 \right\|_{H^1}^2 = \|\nabla u_0\|_2^2 + \|\Delta u_0\|_2^2, \quad J_1(S) := \int_0^S \|B(u(t), u(t))\|_{H^1}^2 dt.$$

Squaring the projected equation in H^1 leaves one cross term. The self-adjointness of A turns it into an endpoint difference:

$$\begin{aligned} J_1(S) &= X_\nu(S) + 2\nu \int_0^S \langle u_t, Au \rangle_{H^1} dt \\ &= X_\nu(S) + \nu \left(\left\| A^{1/2}u(S) \right\|_{H^1}^2 - Z_0 \right). \end{aligned} \quad (74)$$

In particular,

$$X_\nu(S) \leq \nu Z_0 + J_1(S). \quad (75)$$

The elementary Fourier inequality $(1 + |\xi|^2)^3 \leq 4(1 + |\xi|^4 + |\xi|^6)$, together with the kinetic-energy bound, now gives

$$\mathfrak{B}_2[u_0, \nu; T_*] \leq 4T_* \|u_0\|_2^2 + \kappa_\nu (\nu Z_0 + \mathfrak{J}_1[u_0, \nu; T_*]), \quad \kappa_\nu := \max\{1, 4\nu^{-2}\}, \quad (76)$$

where

$$\mathfrak{J}_1[u_0, \nu; T_*] := \int_0^{T_*} \|\mathbb{P}\nabla \cdot (u \otimes u)\|_{H^1}^2 dt.$$

This nonlinear action is evaluated on the unique history generated by the full datum. Equation (76) exposes its exact place in the base rectangle. The signed critical current gives a second description that is closer to the height mechanism from Part II. Define

$$\mathfrak{C}_{1/2}[u_0, \nu; T_*] := \sup_{0 \leq S < T_*} \int_0^S \mathcal{P}_{1/2}(t) dt. \quad (77)$$

The value at $S = 0$ makes $\mathfrak{C}_{1/2} \geq 0$. The whole-terminal theorem makes it finite; the direct L^1 calculation appears in (124). Thus $\mathfrak{C}_{1/2}$ is a scalar value generated by the same datum and history, not an additional hypothesis.

The exact critical balance can now be read numerically. For every $0 < S < T_*$,

$$H(S) + \nu \int_0^S \left\| \Lambda^{3/2}u(t) \right\|_2^2 dt = H(0) + \int_0^S \mathcal{P}_{1/2}(t) dt.$$

Since $H(S) \geq 0$, monotone convergence gives

$$\int_0^{T_*} \left\| \Lambda^{3/2}u(t) \right\|_2^2 dt \leq \mathfrak{A}_{1/2}[u_0, \nu; T_*] := \frac{H(0) + \mathfrak{C}_{1/2}[u_0, \nu; T_*]}{\nu}, \quad H(0) = \frac{1}{2} \left\| \Lambda^{1/2}u_0 \right\|_2^2. \quad (78)$$

The base rectangle supplies the same coefficient without passing through the signed current:

$$\int_0^{T_*} \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \|u\|_{L^2(I_*; H^3)}^2 \leq \mathfrak{B}_2[u_0, \nu; T_*]. \quad (79)$$

The kinetic-energy payment sharpens this direct inclusion. Fourier log-convexity at each time and Hölder's inequality in time give

$$\begin{aligned} \int_0^{T_*} \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt &\leq \left(\int_0^{T_*} \left\| \Lambda u(t) \right\|_2^2 dt \right)^{3/4} \left(\int_0^{T_*} \left\| \Lambda^3 u(t) \right\|_2^2 dt \right)^{1/4} \\ &\leq \left(\frac{\|u_0\|_2^2}{2\nu} \right)^{3/4} \mathfrak{B}_2[u_0, \nu; T_*]^{1/4}. \end{aligned}$$

This prefix bound is already paid by the same base rectangle. On every strict subinterval, the critical balance $\mathcal{P}_{1/2} = H' + 2\nu E_{3/2}$ and Cauchy–Schwarz give

$$\begin{aligned} \int_0^S |H'(t)| dt &\leq \|u_t\|_{L^2(0,S; H^1)} \|u\|_{L^2(0,S; H^1)} \leq \frac{1}{2} \mathfrak{B}_2[u_0, \nu; T_*], \\ 2\nu \int_0^S E_{3/2}(t) dt &\leq \nu \mathfrak{B}_2[u_0, \nu; T_*]. \end{aligned}$$

Taking the supremum in S gives

$$\mathfrak{C}_{1/2}[u_0, \nu; T_*] \leq \|\mathcal{P}_{1/2}\|_{L^1(I_*)} \leq \left(\frac{1}{2} + \nu \right) \mathfrak{B}_2[u_0, \nu; T_*]. \quad (80)$$

The strongest current-free coefficient obtained from these two payments is

$$\mathfrak{A}_{1/2}^{\text{prod}}[u_0, \nu; T_*] := \min \left\{ \mathfrak{B}_2[u_0, \nu; T_*], \left(\frac{\|u_0\|_2^2}{2\nu} \right)^{3/4} \mathfrak{B}_2[u_0, \nu; T_*]^{1/4} \right\}. \quad (81)$$

Then

$$\int_0^{T_*} \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \mathfrak{A}_{1/2}^{\text{prod}}[u_0, \nu; T_*]. \quad (82)$$

Thus every spatial row is driven directly by the single $(0, 2)$ rectangle; the signed prefix remains an optional sharper readout, not a parameter needed by the final estimates. For $0 < T \leq T_*$, the same symbols with T in the final slot denote the definitions above with $(0, T)$ in place of I_* .

Proposition 17.1 (Datum-only calibrations of the critical coefficient). *Let C_{crit} be the constant in (36), and put*

$$q_0 := \left\| \Lambda^{1/2} u_0 \right\|_2, \quad \vartheta := \frac{C_{\text{crit}} q_0}{\nu}.$$

If $\vartheta < 1$, then every finite horizon satisfies

$$\mathfrak{C}_{1/2}[u_0, \nu; T] \leq \frac{\vartheta}{1 - \vartheta} H(0), \quad \mathfrak{A}_{1/2}[u_0, \nu; T] \leq \frac{H(0)}{\nu(1 - \vartheta)}. \quad (83)$$

For arbitrary datum, let C_{ens} be the constant in (10), set

$$E_0 := \|u_0\|_2^2, \quad D_1 := \|\Lambda u_0\|_2^2, \quad \Theta(T) := 1 - 2C_{\text{ens}} \nu^{-3} D_1^2 T,$$

and suppose $\Theta(T) > 0$. Then

$$\sup_{0 \leq t \leq T} \|\Lambda u(t)\|_2^2 \leq D_1 \Theta(T)^{-1/2}, \quad (84)$$

$$\int_0^T \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \frac{\sqrt{E_0 D_1/2}}{\nu \Theta(T)^{1/4}}, \quad (85)$$

$$\mathfrak{C}_{1/2}[u_0, \nu; T] \leq \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \right) \sqrt{E_0 D_1} \Theta(T)^{-1/4}. \quad (86)$$

Proof. When $\vartheta < 1$, estimate (36) and a first-crossing argument give

$$H(S) + \nu(1 - \vartheta) \int_0^S \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq H(0).$$

Writing the signed prefix as

$$\left[H(S) - H(0) + \nu(1 - \vartheta) \int_0^S \left\| \Lambda^{3/2} u \right\|_2^2 dt \right] + \nu \vartheta \int_0^S \left\| \Lambda^{3/2} u \right\|_2^2 dt$$

gives (83).

For the second statement, put $Y = \|\Lambda u\|_2^2$ and $Z = \|\Lambda^2 u\|_2^2$. Scalar comparison in (10) gives

$$Y(t) \leq D_1 (1 - 2C_{\text{ens}} \nu^{-3} D_1^2 t)^{-1/2}.$$

Integrating the same inequality and inserting this comparison yields

$$\int_0^T Z(t) dt \leq \frac{D_1}{\nu \Theta(T)^{1/2}}.$$

Fourier interpolation, Cauchy–Schwarz in time, and the kinetic-energy identity now give

$$\int_0^T \left\| \Lambda^{3/2} u \right\|_2^2 dt \leq \left(\int_0^T Y dt \right)^{1/2} \left(\int_0^T Z dt \right)^{1/2} \leq \frac{\sqrt{E_0 D_1/2}}{\nu \Theta(T)^{1/4}}.$$

Finally,

$$H(S) \leq \frac{1}{2} \|u(S)\|_2 \|\Lambda u(S)\|_2 \leq \frac{1}{2} \sqrt{E_0 D_1} \Theta(T)^{-1/4}.$$

Insert the last two estimates into the exact critical balance and take the supremum over $S \leq T$ to obtain (86). $\square$

The preceding estimate uses the first differentiated row and is strongest near the local enstrophy scale. A higher datum row offers a different trade. It can be very large, but it enters with a fractional exponent that decreases as the chosen order rises. This produces a closed datum condition in which the spatial order, the horizon, and the viscosity remain visible separately.

Proposition 17.2 (A finite-order datum window). *Fix an integer $\ell \geq 3$. Let C_ℓ^{Lip} be a constant for which the H^ℓ transport estimate gives*

$$\|u(t)\|_{H^\ell} \leq \|u_0\|_{H^\ell} \exp\left(C_\ell^{\text{Lip}} \mathcal{L}(t)\right), \quad \mathcal{L}(t) := \int_0^t \|\nabla u(s)\|_\infty ds. \quad (87)$$

Put

$$\theta_\ell := \frac{3}{2(\ell-1)}, \quad \lambda_\ell := C_\ell^{\text{Lip}} \theta_\ell, \quad (88)$$

and choose G_ℓ so that

$$\|\nabla f\|_\infty \leq G_\ell \|\Lambda f\|_2^{1-\theta_\ell} \|f\|_{H^\ell}^{\theta_\ell}. \quad (89)$$

For $T > 0$, define the closed datum value

$$\Xi_\ell[u_0, \nu; T] := \lambda_\ell G_\ell \|u_0\|_{H^\ell}^{\theta_\ell} T^{(1+\theta_\ell)/2} \left(\frac{\|u_0\|_2^2}{2\nu} \right)^{(1-\theta_\ell)/2}. \quad (90)$$

If $\Xi_\ell[u_0, \nu; T] < 1$, then the solution continues through T and

$$\mathcal{L}(T) \leq \frac{1}{\lambda_\ell} \log \frac{1}{1 - \Xi_\ell[u_0, \nu; T]}. \quad (91)$$

Let C_1^{Lip} be a constant in the first-order estimate

$$Y'(t) + 2\nu Z(t) \leq 2C_1^{\text{Lip}} \|\nabla u(t)\|_\infty Y(t), \quad Y := \|\Lambda u\|_2^2, \quad Z := \|\Lambda^2 u\|_2^2. \quad (92)$$

With $E_0 = \|u_0\|_2^2$ and $D_1 = \|\Lambda u_0\|_2^2$, the same datum window gives

$$\begin{aligned} \mathfrak{C}_{1/2}[u_0, \nu; T] &\leq \widehat{\mathfrak{C}}_{1/2}^{(\ell)}[u_0, \nu; T] \\ &:= \sqrt{E_0 D_1} (1 - \Xi_\ell[u_0, \nu; T])^{-C_1^{\text{Lip}}/\lambda_\ell} - H(0). \end{aligned} \quad (93)$$

$$\mathfrak{A}_{1/2}[u_0, \nu; T] \leq \widehat{\mathfrak{A}}_{1/2}^{(\ell)}[u_0, \nu; T] := \frac{\sqrt{E_0 D_1}}{\nu} (1 - \Xi_\ell[u_0, \nu; T])^{-C_1^{\text{Lip}}/\lambda_\ell}. \quad (94)$$

Every quantity on the right is calculated directly from u_0, ν, T, ℓ , and the fixed analytic constants.

Proof. Fourier inversion gives

$$\|\nabla f\|_\infty \leq C \int_{\mathbb{R}^3} |\xi| |\widehat{f}(\xi)| d\xi.$$

Split the integral at a radius $R > 0$. Cauchy–Schwarz on the two pieces gives

$$\|\nabla f\|_\infty \leq C_\ell \left(R^{3/2} \|\Lambda f\|_2 + R^{5/2-\ell} \|f\|_{H^\ell} \right).$$

If $\|\Lambda f\|_2 = 0$, then $f = 0$ in $H^\ell(\mathbb{R}^3)$, and the estimate is immediate. Otherwise, balance the two terms by choosing $R^{\ell-1} = \|f\|_{H^\ell} / \|\Lambda f\|_2$. This proves (89).

Let $T_{\max}$ be the maximal classical lifespan from u_0 , and fix

$$0 < \tau < \min\{T, T_{\max}\}.$$

Apply the Fourier inequality on $[0, \tau]$, then insert (87). Since $\lambda_\ell = C_\ell^{\text{Lip}} \theta_\ell$, throughout this interval

$$-\frac{d}{dt} e^{-\lambda_\ell \mathcal{L}(t)} \leq \lambda_\ell G_\ell \|u_0\|_{H^\ell}^{\theta_\ell} \|\Lambda u(t)\|_2^{1-\theta_\ell}.$$

The kinetic-energy identity and Hölder's inequality give

$$\int_0^\tau \|\Lambda u(t)\|_2^{1-\theta_\ell} dt \leq \tau^{(1+\theta_\ell)/2} \left(\frac{\|u_0\|_2^2}{2\nu} \right)^{(1-\theta_\ell)/2} \leq T^{(1+\theta_\ell)/2} \left(\frac{\|u_0\|_2^2}{2\nu} \right)^{(1-\theta_\ell)/2}.$$

Integration now yields

$$e^{-\lambda_\ell \mathcal{L}(\tau)} \geq 1 - \Xi_\ell[u_0, \nu; T],$$

and hence the right side of (91) bounds $\mathcal{L}(\tau)$, uniformly in every such τ . If $T_{\max} \leq T$, this bound and (87) keep the H^ℓ norm bounded as $\tau \uparrow T_{\max}$. The continuation estimate (15) would then extend the classical solution past $T_{\max}$, a contradiction. Thus $T < T_{\max}$; letting $\tau \uparrow T$ proves (91).

It remains to retain the signed critical value rather than only the Lipschitz clock. Equation (92) gives

$$\sup_{0 \leq t \leq T} Y(t) \leq D_1 e^{2C_1^{\text{Lip}} \mathcal{L}(T)}, \quad 2\nu \int_0^T Z(t) dt \leq D_1 e^{2C_1^{\text{Lip}} \mathcal{L}(T)}.$$

Energy also gives $\int_0^T Y(t) dt \leq E_0/(2\nu)$. Spatial interpolation on each time slice, followed by Cauchy–Schwarz in time, now gives

$$\nu \int_0^T \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \frac{1}{2} \sqrt{E_0 D_1} e^{C_1^{\text{Lip}} \mathcal{L}(T)}.$$

At every $S \leq T$,

$$H(S) \leq \frac{1}{2} \|u(S)\|_2 \|\Lambda u(S)\|_2 \leq \frac{1}{2} \sqrt{E_0 D_1} e^{C_1^{\text{Lip}} \mathcal{L}(T)}.$$

Insert these two bounds into the exact critical balance and take the supremum over S . The balance subtracts $H(0)$, producing the last term in (93). Notice that

$$2H(0) = \left\| \Lambda^{1/2} u_0 \right\|_2^2 \leq \sqrt{E_0 D_1},$$

so the displayed majorant is nonnegative. Equation (91) converts the last exponential into the power in (93), and (94) follows from its definition. $\square$

Fix an integer $m \geq 1$. Differentiating the equation m times in space, pairing with the same derivative of u , and summing with the multinomial weights keeps the cancellation of the undifferentiated transport term and of pressure. If k derivatives land on the transporting velocity, Gagliardo–Nirenberg interpolation places that factor between $\nabla u \in L^3$ and $\nabla^m u \in L^6$; the remaining differentiated factor uses the complementary interpolation exponent. Hölder’s inequality then gives the complete-row estimate

$$|\mathcal{P}_m(t)| \leq K_m \left\| \Lambda^{3/2} u(t) \right\|_2 \left\| \Lambda^{m+1} u(t) \right\|_2 \left\| \Lambda^m u(t) \right\|_2, \quad m \geq 1, \quad (95)$$

where K_m depends only on the derivative order and the fixed Sobolev constants. This is the spatially differentiated VPI row: every product term is retained before the common estimate is taken.

Set

$$X_m(t) := \left\| \Lambda^m u(t) \right\|_2^2, \quad Y_m(t) := \left\| \Lambda^{m+1} u(t) \right\|_2^2, \quad D_m(u_0) := \left\| \Lambda^m u_0 \right\|_2^2.$$

Young’s inequality in (95) gives

$$X'_m(t) + \nu Y_m(t) \leq \frac{K_m^2}{\nu} \left\| \Lambda^{3/2} u(t) \right\|_2^2 X_m(t).$$

The producer coefficient bound (82) controls the complete coefficient. With

$$\Gamma_m[u_0, \nu; T_*] := \frac{K_m^2}{\nu} \mathfrak{A}_{1/2}^{\text{prod}}[u_0, \nu; T_*] \quad (96)$$

Gronwall and one subsequent integration give the explicit spatial bounds

$$\begin{aligned} \sup_{0 \leq t < T_*} \|\Lambda^m u(t)\|_2^2 &\leq D_m(u_0) e^{\Gamma_m}, \\ \nu \int_0^{T_*} \|\Lambda^{m+1} u(t)\|_2^2 dt &\leq D_m(u_0) e^{\Gamma_m}. \end{aligned} \quad (97)$$

The dependence on the derivative order is carried by D_m and K_m ; the viscosity appears both in the exponent and in the dissipative factor.

It will be useful to place the inhomogeneous norms into one notation. Define

$$\mathfrak{S}_0 := \|u_0\|_2, \quad \mathfrak{S}_m^2 := 2^{m-1} \left(\|u_0\|_2^2 + D_m(u_0) e^{\Gamma_m} \right) \quad (m \geq 1). \quad (98)$$

Since $(1+r)^m \leq 2^{m-1}(1+r^m)$ for $r \geq 0$,

$$\sup_{0 \leq t < T_*} \|u(t)\|_{H^m} \leq \mathfrak{S}_m \quad (m \in \mathbb{N}_0). \quad (99)$$

The integrated companion begins with the exact energy row,

$$\mathfrak{D}_0^2 := T_* \|u_0\|_2^2 + \frac{\|u_0\|_2^2}{2\nu}, \quad \|u\|_{L^2(I_*, H^1)} \leq \mathfrak{D}_0. \quad (100)$$

For $m \geq 1$, set

$$\mathfrak{D}_m^2 := 2^m \left[T_* \|u_0\|_2^2 + \frac{D_m(u_0)}{\nu} e^{\Gamma_m} \right], \quad \|u\|_{L^2(I_*; H^{m+1})} \leq \mathfrak{D}_m. \quad (101)$$

The temporal rows now inherit an entirely finite recursion. Put $\sigma_r := \max\{2, r+1\}$. Enlarge the order-dependent constants from Appendix A, if needed, so that for every integer $r \geq 0$,

$$\begin{aligned} \|\mathbb{P} \nabla \cdot (f \otimes g)\|_{H^r} + \|R_i R_j (f_i g_j)\|_{H^r} \\ \leq A_r \|f\|_{H^{\sigma_r}} \|g\|_{H^{\sigma_r}}. \end{aligned} \quad (102)$$

At $r = 1$, this product estimate and (99) close the nonlinear graph action:

$$\mathfrak{J}_1[u_0, \nu; T_*] \leq A_1^2 \int_0^{T_*} \|u(t)\|_{H^2}^4 dt \leq A_1^2 T_* \mathfrak{S}_2^4. \quad (103)$$

There is a sharper estimate that uses the differentiated spatial envelopes instead of the uniform H^2 norm. Write

$$a(t) := \|\nabla u(t)\|_2, \quad b(t) := \|\nabla^2 u(t)\|_2.$$

Sobolev interpolation and boundedness of $\mathbb{P}$ give a constant G_1 , depending only on the fixed three-dimensional inequalities, such that

$$\|B(u, u)\|_{H^1}^2 \leq G_1 (a^3 b + a b^3). \quad (104)$$

Indeed,

$$\|u \cdot \nabla u\|_2 \leq C a^{3/2} b^{1/2}, \quad \|\nabla(u \cdot \nabla u)\|_2 \leq C a^{1/2} b^{3/2}.$$

The two time integrals in (104) retain one supremum and one copy of the energy–dissipation pairing:

$$\begin{aligned} \int_0^{T_*} a^3 b \, dt &\leq \sup_{t < T_*} a(t)^2 \left(\int_0^{T_*} a^2 \, dt \right)^{1/2} \left(\int_0^{T_*} b^2 \, dt \right)^{1/2}, \\ \int_0^{T_*} a b^3 \, dt &\leq \sup_{t < T_*} b(t)^2 \left(\int_0^{T_*} a^2 \, dt \right)^{1/2} \left(\int_0^{T_*} b^2 \, dt \right)^{1/2}. \end{aligned}$$

The kinetic-energy identity and (97) at $m = 1, 2$ now yield

$$\mathfrak{J}_1[u_0, \nu; T_*] \leq \mathfrak{J}_1^\sharp[u_0, \nu; T_*] := \frac{G_1}{\sqrt{2}\nu} \sqrt{\|u_0\|_2^2 D_1(u_0)} e^{\Gamma_1/2} (D_1(u_0) e^{\Gamma_1} + D_2(u_0) e^{\Gamma_2}). \quad (105)$$

This version has no explicit factor of T_* . Its horizon dependence is carried by the effective coefficient inside Γ_1 and Γ_2 . Together, (103) and (105) give

$$\mathfrak{J}_1[u_0, \nu; T_*] \leq \mathfrak{J}_1^{\text{fin}} := \min \left\{ A_1^2 T_* \mathfrak{S}_2^4, \mathfrak{J}_1^\sharp[u_0, \nu; T_*] \right\}. \quad (106)$$

Substitution into (76) gives the analytic feedback estimate

$$\begin{aligned} \mathfrak{B}_2[u_0, \nu; T_*] &\leq \overline{\mathfrak{B}}_2[u_0, \nu; T_*], \\ \overline{\mathfrak{B}}_2[u_0, \nu; T_*] &:= 4T_* \|u_0\|_2^2 + \kappa_\nu \left(\nu Z_0 + \mathfrak{J}_1^{\text{fin}} \right). \end{aligned} \quad (107)$$

Here $\mathfrak{S}_2$ is the explicit expression (98), with exponent (96). The right side of (107) depends on u_0, ν, T_* , the fixed analytic constants, and the single producer coefficient $\mathfrak{A}_{1/2}^{\text{prod}}$. By (79), the current-free choice $\mathfrak{A}_{1/2}^{\text{prod}} \leq \mathfrak{B}_2$ reduces every higher readout to the one base output of the datum producer. Thus $\mathfrak{B}_2$ is the sole solution-dependent input; no higher rectangle value remains on the right. This form exposes the nonlinear feedback but does not yet remove the base output itself. After global continuation has been proved, Section 22 removes that final dependence by a residual certificate computed from the datum.

Starting from $\mathfrak{V}_{0,r} := \mathfrak{S}_r$, define

$$\mathfrak{V}_{n+1,r} := \nu \mathfrak{V}_{n,r+2} + A_r \sum_{a=0}^n \binom{n}{a} \mathfrak{V}_{a,\sigma_r} \mathfrak{V}_{n-a,\sigma_r}, \quad n, r \in \mathbb{N}_0, \quad (108)$$

and

$$\mathfrak{Q}_{n,r} := A_r \sum_{a=0}^n \binom{n}{a} \mathfrak{V}_{a,\sigma_r} \mathfrak{V}_{n-a,\sigma_r}. \quad (109)$$

The projected momentum recurrence and the normalized Riesz formula give, by induction in n ,

$$\sup_{0 \leq t < T_*} \|V_n(t)\|_{H^r} \leq \mathfrak{V}_{n,r}, \quad \sup_{0 \leq t < T_*} \|Q_n(t)\|_{H^r} \leq \mathfrak{Q}_{n,r}. \quad (110)$$

Every entry on the right of (108) has lower temporal order, while the two additional spatial derivatives pay the Laplacian and the differentiated quadratic term. The binomial sum retains all temporal distributions of the original transport product.

The lowest rectangle can keep its natural negative row. Define

$$\mathfrak{V}_{n+1,-1} := \nu \mathfrak{V}_{n,1} + A_{-1} \sum_{a=0}^n \binom{n}{a} \mathfrak{V}_{a,1} \mathfrak{V}_{n-a,1}, \quad (111)$$

where A_{-1} is the constant in

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^{-1}} \leq A_{-1} \|f\|_{H^1} \|g\|_{H^1}.$$

The projected recurrence gives $\sup_{t < T_*} \|V_{n+1}(t)\|_{H^{-1}} \leq \mathfrak{V}_{n+1,-1}$.

For $M \in \mathbb{N}_0$, set

$$\mathfrak{R}_{N,M}[u_0, \nu; T_*] := T_* \sum_{n=0}^N (\mathfrak{V}_{n,M+1}^2 + \mathfrak{V}_{n+1,M-1}^2). \quad (112)$$

For $M = 0$, (111) controls the lower row. For $M \geq 1$, the nonnegative recursion controls it. In either case,

$$\mathcal{N}_{N,M}(u; I_*) \leq \mathfrak{R}_{N,M}[u_0, \nu; T_*] < \infty. \quad (113)$$

Once N and M are chosen, the right side is a finite recursive expression in $\mathfrak{A}_{1/2}^{\text{prod}}[u_0, \nu; T_*]$, ν , T_* , the named analytic constants through the required orders, and the corresponding finite Sobolev norms of u_0 . The velocity rectangle norm (N, M) uses datum derivatives only through order $M + 2N + 1$. If the retained pressure row $Q_N \in H^M$ is included, the current $\sigma_M = \max\{2, M + 1\}$ recursion also uses order $\sigma_M + 2N$: this is $M + 2N + 1$ for $M \geq 1$ and $2N + 2$ for $M = 0$. No higher rectangle value remains on the right.

17.2 Classical continuation quantities as finite projections

The Ladyzhenskaya–Prodi–Serrin scale is the first direct example. Select the zeroth temporal row in $L^2(I_*; H^2)$. Since $H^2(\mathbb{R}^3) \hookrightarrow L^\infty(\mathbb{R}^3)$,

$$u \in L^2(I_*; L^\infty), \quad \frac{2}{2} + \frac{3}{\infty} = 1.$$

One finite coordinate has reached the scaling line. More generally, if $b > 3$ and $1 \leq a \leq \infty$ satisfy

$$\frac{2}{a} + \frac{3}{b} \leq 1,$$

equation (68) with $n = k = 0$ gives

$$u \in L^a(I_*; L^b).$$

More precisely, for any integer $m > \frac{3}{2} - \frac{3}{b}$,

$$\|u\|_{L^a(I_*; L^b)} \leq C_{m,b} T_*^{1/a} \mathfrak{S}_m. \quad (114)$$

Thus every classical Ladyzhenskaya–Prodi–Serrin pair with $b > 3$ is a finite projection of the same compatible family.

The same calculation gives the final mixed-jet estimates at arbitrary finite orders. If $m > k + \frac{3}{2} - \frac{3}{b}$, then

$$\begin{aligned} \sup_{0 \leq t < T_*} \left\| \nabla^k \partial_t^n u(t) \right\|_{L^b} &\leq C_{m,k,b} \mathfrak{V}_{n,m}, \\ \left\| \nabla^k \partial_t^n u \right\|_{L^a(I_*; L^b)} &\leq C_{m,k,b} T_*^{1/a} \mathfrak{V}_{n,m}. \end{aligned} \quad (115)$$

If $s \in \mathbb{N}_0$ satisfies $s > k + \frac{3}{2} - \frac{3}{b}$, the normalized pressure rows obey

$$\begin{aligned} \sup_{0 \leq t < T_*} \left\| \nabla^k Q_n(t) \right\|_{L^b} &\leq C_{s,k,b} \mathfrak{Q}_{n,s}, \\ \left\| \nabla^k Q_n \right\|_{L^a(I_*; L^b)} &\leq C_{s,k,b} T_*^{1/a} \mathfrak{Q}_{n,s}. \end{aligned} \quad (116)$$

Here and below $T_*^{1/\infty} = 1$.

The same $L^2(I_*; H^3)$ coordinate controls the stretching quantities from Part I. Put

$$\omega := \nabla \times u, \quad S := \frac{1}{2} (\nabla u + \nabla u^\top).$$

Spatial embedding gives

$$\omega, S, \nabla u \in L^2(I_*; L^\infty).$$

Since $T_* < \infty$, Cauchy–Schwarz in time gives

$$\int_0^{T_*} (\|\omega(t)\|_\infty + \|S(t)\|_\infty + \|\nabla u(t)\|_\infty) dt \leq CT_*^{1/2} \mathfrak{B}_2[u_0, \nu; T_*]^{1/2} \leq CT_*^{1/2} \overline{\mathfrak{B}}_2[u_0, \nu; T_*]^{1/2} < \infty. \quad (117)$$

The vorticity and strain actions are not new assumptions at the end of the proof. They are low spatial projections of the rectangle that already produced (71).

Pressure is another projection of the same object. The normalized identity

$$p = R_i R_j (u_i u_j)$$

and the pressure projection at the end of Part III give $p \in C([0, T_*]; H^m)$ for every fixed m . Choosing $m = 3$ and using (386) gives

$$\int_0^{T_*} \|\nabla p(t)\|_\infty dt \leq C \|p\|_{L^1(I_*; H^3)} \leq C \|u\|_{L^2(I_*; H^4)}^2 \leq C \mathfrak{R}_{0,3}[u_0, \nu; T_*] < \infty. \quad (118)$$

The same argument applies to any fixed temporal pressure row after a finite spatial margin is added.

The critical endpoint is a second direct projection. Equation (67) with $n = 0$ and $m = 1$ gives

$$u \in C([0, T_*]; H^1).$$

Since $H^1 \hookrightarrow L^6$ and $H^1 \subset L^2$, interpolation gives

$$\|u(t)\|_3 \leq \|u(t)\|_2^{1/2} \|u(t)\|_6^{1/2} \leq C \|u(t)\|_{H^1}.$$

Thus

$$\sup_{0 \leq t < T_*} \|u(t)\|_3 \leq C \mathfrak{S}_1 < \infty, \quad (119)$$

which is incompatible with the necessary L^3 limsup behavior in the Escauriaza–Seregin–Šverák theorem^[10]. It also excludes the full terminal divergence (20) when u_0 belongs to the compactly supported subclass used for Seregin’s full terminal theorem^[11].

The H^1 trace that bounds the L^3 endpoint also controls the critical height that generated the response tower in Part II. Split frequency at $|\xi| = 1$:

$$\begin{aligned} 2H(t) &= \int_{|\xi| \leq 1} |\xi| |\widehat{u}(\xi, t)|^2 d\xi + \int_{|\xi| > 1} |\xi| |\widehat{u}(\xi, t)|^2 d\xi \\ &\leq \|u(t)\|_2^2 + \|\nabla u(t)\|_2^2 \leq \|u(t)\|_{H^1}^2. \end{aligned}$$

Hence

$$H_{\max} := \sup_{0 \leq t < T_*} H(t) \leq \frac{1}{2} \mathfrak{S}_1^2 < \infty. \quad (120)$$

Let N_{pass} denote the number of completed passages in the dyadic ladder from Part II. If $N_{\text{pass}} \geq 1$, the last endpoint level reached is

$$L_{N_{\text{pass}}+1} = 2^{N_{\text{pass}}} L_1,$$

and hence

$$2^{N_{\text{pass}}} L_1 \leq H_{\max}.$$

Thus $N_{\text{pass}} = 0$ when $H_{\max} < L_1$, while

$$N_{\text{pass}} \leq \left\lfloor \log_2 \frac{H_{\max}}{L_1} \right\rfloor \quad (H_{\max} \geq L_1).$$

The height bound does not say that the solution never rises. It says that the infinite terminal ladder demanded by (24) cannot be completed. For the compactly supported subclass used in Part I, this also excludes the necessary divergence in Seregin's critical Sobolev endpoint theorem^[12]. Both Seregin comparisons concern the narrower compactly supported data class, while the compatible-family argument uses the stated Schwartz data.

17.3 The first rectangle seen through the critical height

The height bound ends the level ladder by bounding H . The same $(0, 2)$ rectangle also resolves the mechanism inside each attempted passage: the signed response and the next half-integer dissipation from Part II. To expose that critical block on the complete terminal interval, retain both adjacent rows:

$$u \in L^2(I_*; H^3), \quad u_t \in L^2(I_*; H^1).$$

In particular, $u \in L^2 H^{5/2}$ and $u_t \in L^2 H^{1/2}$. Lemma 12.1 at $m = 3/2$ gives

$$u \in C([0, T_*]; H^{3/2}).$$

This pair also controls the variation of the next critical energy. Let S_R be a smooth self-adjoint Fourier cutoff, equal to one on $|\xi| \leq R$ and supported in $|\xi| \leq 2R$, and put $u_R = S_R u$. For classical times,

$$\begin{aligned} \frac{d}{dt} \frac{1}{2} \left\| \Lambda^{3/2} u_R(t) \right\|_2^2 &= \text{Re}(\Lambda^{3/2} S_R u_t, \Lambda^{3/2} S_R u)_2 \\ &= \text{Re}(\Lambda^{1/2} S_R u_t, \Lambda^{5/2} S_R u)_2. \end{aligned}$$

Integrate from r to t . The two cutoff factors converge in $L^2(I_*; L^2)$, so their product converges in $L^1(I_*)$. The $H^{3/2}$ trace handles the endpoint energies. Passing $R \rightarrow \infty$ gives

$$E_{3/2}(t) - E_{3/2}(r) = \int_r^t \operatorname{Re}(\Lambda^{1/2} u_\tau(\tau), \Lambda^{5/2} u(\tau))_2 d\tau.$$

The integrand is absolutely integrable because

$$\int_0^{T_*} |(\Lambda^{1/2} u_t, \Lambda^{5/2} u)_2| dt \leq \|u_t\|_{L^2(I_*; H^{1/2})} \|u\|_{L^2(I_*; H^{5/2})} \leq \frac{1}{2} \mathcal{N}_{0,2}(u; I_*).$$

Moreover,

$$\int_0^{T_*} E_{3/2}(t) dt \leq \frac{1}{2} \|u\|_{L^2(I_*; H^3)}^2 \leq \frac{1}{2} \mathfrak{B}_2[u_0, \nu; T_*] \leq \frac{1}{2} \overline{\mathfrak{B}}_2[u_0, \nu; T_*], \quad (121)$$

while (66) at the middle order 2 gives

$$\sup_{0 \leq t \leq T_*} E_{3/2}(t) \leq \frac{1}{2} \mathfrak{S}_2^2. \quad (122)$$

Thus

$$E_{3/2} \in W^{1,1}(I_*) \cap L^1(I_*) \subset C([0, T_*]) \cap L^\infty(I_*).$$

A second cutoff calculation at order 1/2 gives

$$H'(t) = \operatorname{Re}(\Lambda^{1/2} u_t(t), \Lambda^{1/2} u(t))_2$$

in distributions. Both factors are square-integrable in time, and the zeroth spatial row gives $H \in L^1(I_*)$. Thus

$$H \in W^{1,1}(I_*) \hookrightarrow C([0, T_*]), \quad \|H'\|_{L^1(I_*)} \leq \frac{1}{2} \mathfrak{B}_2[u_0, \nu; T_*] \leq \frac{1}{2} \overline{\mathfrak{B}}_2[u_0, \nu; T_*]. \quad (123)$$

The exact critical balance (32) now yields

$$\mathcal{P}_{1/2} = H' + 2\nu E_{3/2} \in L^1(I_*), \quad \|\mathcal{P}_{1/2}\|_{L^1(I_*)} \leq \left(\frac{1}{2} + \nu\right) \mathfrak{B}_2[u_0, \nu; T_*] \leq \left(\frac{1}{2} + \nu\right) \overline{\mathfrak{B}}_2[u_0, \nu; T_*]. \quad (124)$$

The complete first critical block is

$$H \in W^{1,1}(I_*), \quad E_{3/2} \in L^\infty(I_*) \cap L^1(I_*), \quad \mathcal{P}_{1/2} \in L^1(I_*). \quad (125)$$

Thus (125) records finite critical variation, integrable next dissipation, and the complete signed response on the terminal interval.

The same spatial-row estimate controls every fixed height readout. For $j \in \mathbb{N}_0$, the complete row gives $u \in L^2(I_*; H^{j+1})$. Since $|\xi|^{2j+1} \leq (1 + |\xi|^2)^{j+1}$,

$$\begin{aligned} \int_0^{T_*} E_{j+1/2}(t) dt &= \frac{1}{2} \int_0^{T_*} \left\| \Lambda^{j+1/2} u(t) \right\|_2^2 dt \\ &\leq \frac{1}{2} \|u\|_{L^2(I_*; H^{j+1})}^2 \leq \frac{1}{2} \mathfrak{D}_j^2 < \infty. \end{aligned}$$

By Lemma 9.1, this is the spatial energy exposed by the j -th differentiated critical-height row. Temporal ascent and spatial depth remain different coordinates, but every fixed spatial readout they expose is controlled on the same terminal interval.

17.4 Uniform tightness of the critical tail

The finite height bound stops the level ladder. Part II's outward scale migration leaves a second question: can the activity responsible for the terminal approach retreat through an unbounded collection of higher frequencies? Part I identified vorticity as the local stretching diagnostic. When that activity is distributed across the shell index, its complete dyadic L^∞ sum is the relevant quantity. The first estimate controls this sum in time; the second returns to H and makes its critical tail vanish uniformly.

For the inhomogeneous decomposition from Appendix A, Bernstein's inequality gives, for $j \geq 0$,

$$\|P_j \omega(t)\|_\infty \leq C 2^{3j/2} \|P_j \omega(t)\|_2 = C 2^{-j/2} (2^{2j} \|P_j \omega(t)\|_2).$$

Cauchy–Schwarz in j , followed by the dyadic Sobolev equivalence, gives

$$\sum_{j \geq 0} \|P_j \omega(t)\|_\infty \leq C \|\omega(t)\|_{H^2} \leq C \|u(t)\|_{H^3}.$$

The low block $S_{1/2} \omega$ satisfies the same estimate by the low-frequency Bernstein bound. Integration over the finite interval yields

$$\begin{aligned} \int_0^{T_*} \left(\|S_{1/2} \omega(t)\|_\infty + \sum_{j \geq 0} \|P_j \omega(t)\|_\infty \right) dt &\leq C T_*^{1/2} \|u\|_{L^2(I_*; H^3)} \\ &\leq C T_*^{1/2} \mathfrak{B}_2[u_0, \nu; T_*]^{1/2} \\ &\leq C T_*^{1/2} \overline{\mathfrak{B}}_2[u_0, \nu; T_*]^{1/2} < \infty. \end{aligned}$$

This is how the intersection controls a summation over scales: a fixed shell weight is dominated by one sufficiently high finite Sobolev coordinate. More generally, fix $\sigma \in \mathbb{R}$ and $k, n \in \mathbb{N}_0$, and choose an integer $M \geq 1$ satisfying

$$M > \sigma + k + \frac{3}{2}.$$

Bernstein and Cauchy–Schwarz in the shell index give

$$\left\| S_{1/2} \nabla^k \partial_t^n u(t) \right\|_\infty + \sum_{j \geq 0} 2^{\sigma j} \left\| P_j \nabla^k \partial_t^n u(t) \right\|_\infty \leq C_{\sigma, k, M} \|\partial_t^n u(t)\|_{H^M}.$$

Selecting the corresponding finite rectangle and integrating in time yields

$$\int_0^{T_*} \left(\left\| S_{1/2} \nabla^k \partial_t^n u(t) \right\|_\infty + \sum_{j \geq 0} 2^{\sigma j} \left\| P_j \nabla^k \partial_t^n u(t) \right\|_\infty \right) dt \leq C_{\sigma, k, M} T_* \mathfrak{V}_{n, M} < \infty.$$

Thus every fixed dyadic weight, spatial derivative, and temporal row is supplied by one sufficiently high finite coordinate.

Fix an integer $m \geq 1$. For $K \geq 1$, define

$$H_{>K}(t) := \frac{1}{2} \int_{|\xi| > K} |\xi| |\widehat{u}(t, \xi)|^2 d\xi.$$

On $|\xi| > K$, $|\xi| \leq K^{1-2m}|\xi|^{2m}$, so

$$\sup_{0 \leq t < T_*} H_{>K}(t) \leq \frac{1}{2} K^{1-2m} \sup_{0 \leq t < T_*} \|u(t)\|_{H^m}^2 \leq \frac{1}{2} K^{1-2m} D_m(u_0) e^{\Gamma_m} \longrightarrow 0 \quad (K \rightarrow \infty). \quad (126)$$

The convergence in (126) is uniform in time along the proposed terminal approach. Consequently, any critical height carried beyond frequency K vanishes uniformly as $K \rightarrow \infty$. This is the scale-level conclusion behind the failure of the Zeno scenario: the whole-terminal estimate removes both its unbounded height sequence and its ultraviolet residue.

Thus the critical-height scenario from Part II cannot occur. It required a rising scale-invariant norm, shorter level passages, and energy pushed toward higher frequencies. Whole-terminal control gives the opposite conclusion: every selected finite coordinate is bounded, every fixed critical tail vanishes uniformly, and the sequence of higher level passages cannot continue through infinitely many levels.

This excludes Blown at every fixed finite coordinate, but it has not yet made the separately obtained traces into one state. A list of finite limits is not enough: compatibility must identify one velocity jet across all Sobolev topologies, and the limiting rows must still satisfy the normalized pressure and momentum laws. The next two sections pay those remaining obligations.

18 Every temporal row reaches one endpoint

The projection argument in (67) has carried each temporal row to the endpoint in every fixed Sobolev space. The readouts above used those bounds one target at a time. The new point is identity: the H^m value and the H^k value obtained from different finite rectangles must belong to one endpoint jet of the original history.

Write $V_n^{*,m} = V_n(T_*) \in H^m$. If $k > m$, the H^k -continuous and H^m -continuous representatives both equal the classical row on I_* . After the embedding $H^k \hookrightarrow H^m$, uniqueness of the continuous representative gives

$$V_n^{*,k} = V_n^{*,m} \quad \text{in } H^m.$$

We can now drop the topology from the notation and set

$$V_n^* := \lim_{t \uparrow T_*} V_n(t) \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (127)$$

The trace retains the size of the rectangle that produced it. For every $n, m \in \mathbb{N}_0$,

$$\|V_n^*\|_{H^m} \leq \mathfrak{V}_{n,m}, \quad \|V_n(t) - V_n^*\|_{H^m} \leq (T_* - t) \mathfrak{V}_{n+1,m}. \quad (128)$$

The first estimate follows by passing the uniform row bound (110) to the trace. For the second, the Banach-valued fundamental theorem of calculus gives

$$V_n^* - V_n(t) = \int_t^{T_*} V_{n+1}(\tau) d\tau \quad \text{in } H^m,$$

and the uniform bound for the next temporal row gives the stated Lipschitz modulus. For $n = 0$, define

$$u_* := V_0^* = \lim_{t \uparrow T_*} u(t) \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (129)$$

Continuity of divergence from H^{m+1} to H^m passes $\nabla \cdot u = 0$ to this limit. The H^0 trace and the energy identity give finite kinetic energy. In particular,

$$u_* \in H_\sigma^3(\mathbb{R}^3),$$

so the endpoint is admissible initial data for the local construction in Proposition 1.1.

The zeroth trace is enough to start a right-hand solution. The higher traces answer a different question: whether that solution meets the old one without a corner in time. The equation will settle this by showing that the left and right jets are generated from the same u_* .

Every mixed velocity coordinate reaches the endpoint as well. For a spatial multi-index α , choose $m \geq |\alpha|$. Continuity of $\partial_x^\alpha : H^m \rightarrow H^{m-|\alpha|}$ gives

$$\partial_x^\alpha V_n(t) \longrightarrow \partial_x^\alpha V_n^* \quad \text{in every fixed admissible Sobolev topology.}$$

Sobolev embedding converts sufficiently high source orders into convergence in every fixed C_b^j topology. The remaining endpoint work belongs to the normalized pressure and the differentiated momentum equation.

19 The pressure-complete endpoint jet

The traces in (127) carry the nonlinear terms to T_* , one fixed target topology at a time. Fix $q \in \mathbb{N}_0$ and choose an integer $\ell \geq q + 3$. Sobolev multiplication gives

$$\|fg\|_{H^{q+1}} \leq C_{q,\ell} \|f\|_{H^\ell} \|g\|_{H^\ell}.$$

If $f(t) \rightarrow f_*$ and $g(t) \rightarrow g_*$ in H^ℓ , then

$$\begin{aligned} \|f(t)g(t) - f_*g_*\|_{H^{q+1}} &\leq C_{q,\ell} \|f(t) - f_*\|_{H^\ell} \|g(t)\|_{H^\ell} \\ &\quad + C_{q,\ell} \|f_*\|_{H^\ell} \|g(t) - g_*\|_{H^\ell} \longrightarrow 0. \end{aligned}$$

The convergent factor $g(t)$ is bounded near T_* . After one spatial derivative,

$$(f(t) \cdot \nabla)g(t) \longrightarrow (f_* \cdot \nabla)g_* \quad \text{in } H^q.$$

Riesz transforms preserve this convergence in every finite Sobolev space.

For each $n \in \mathbb{N}_0$, define the normalized terminal pressure row by

$$Q_n^* := R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^* V_{n-a,j}^*. \quad (130)$$

Every factor belongs to H^∞ , so $Q_n^* \in H^\infty$. If $s \in \mathbb{N}_0$ and $r_s = \max\{s, 2\}$, the same product calculation as (69) gives the endpoint bound

$$\|Q_n^*\|_{H^s} \leq \mathfrak{Q}_{n,s}. \quad (131)$$

The pressure approaches this value with an explicit modulus as well. With $\sigma_s = \max\{2, s + 1\}$, put

$$\begin{aligned} \mathfrak{L}_{n,s}^Q &:= A_s \sum_{a=0}^n \binom{n}{a} (\mathfrak{V}_{a+1,\sigma_s} \mathfrak{V}_{n-a,\sigma_s} \\ &\quad + \mathfrak{V}_{a,\sigma_s} \mathfrak{V}_{n-a+1,\sigma_s}). \end{aligned} \quad (132)$$

Adding and subtracting $V_a^* V_{n-a}(t)$ in each product and using (128) gives

$$\|Q_n(t) - Q_n^*\|_{H^s} \leq (T_* - t) \mathfrak{L}_{n,s}^Q. \quad (133)$$

Proposition 19.1 (Endpoint jet reconstruction). *For every $n \in \mathbb{N}_0$, the left velocity traces satisfy*

$$V_{n+1}^* = \nu \Delta V_n^* - \sum_{a=0}^n \binom{n}{a} (V_a^* \cdot \nabla) V_{n-a}^* - \nabla Q_n^*. \quad (134)$$

For every finite q ,

$$\lim_{t \uparrow T_*} (\|Q_n(t) - Q_n^*\|_{H^q} + \|V_{n+1}(t) - V_{n+1}^*\|_{H^q}) = 0. \quad (135)$$

The family $(V_n^*, Q_n^*)_{n \geq 0}$ is the unique normalized terminal jet rooted at u_* .

Proof. Fix n, q and choose $\ell \geq q + 3$. At every $t < T_*$,

$$Q_n(t) = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}(t) V_{n-a,j}(t).$$

Equation (67) gives $V_a(t) \rightarrow V_a^*$ in H^ℓ for every $a \leq n$. The product estimate above, applied term by term and followed by the Riesz transforms, gives

$$Q_n(t) \rightarrow Q_n^* \quad \text{in } H^{q+1}.$$

In particular, $\nabla Q_n(t) \rightarrow \nabla Q_n^*$ in H^q .

The velocity recurrence on I_* is

$$V_{n+1}(t) = \nu \Delta V_n(t) - \sum_{a=0}^n \binom{n}{a} (V_a(t) \cdot \nabla) V_{n-a}(t) - \nabla Q_n(t).$$

The trace of V_n in H^{q+2} handles diffusion. The traces in H^ℓ handle every transport product, and the pressure convergence handles the final term. The right side converges in H^q to the right side of (134). At the same time, the adjacent trace for row $n + 1$ gives $V_{n+1}(t) \rightarrow V_{n+1}^*$ in H^q . Uniqueness of limits proves (134) and (135).

Finally, $V_0^* = u_*$ fixes Q_0^* through (130), and (134) then fixes V_1^* . Those rows fix Q_1^* , which fixes V_2^* , and induction continues through every finite order. This proves uniqueness of the normalized jet. Spatial differentiation of the convergent identities recovers every mixed endpoint coordinate. $\square$

The three endpoint failures from Part I have now been separated and paid. The finite projections exclude Blown. Equality of the traces across Sobolev embeddings excludes Jump. The limiting pressure and momentum recurrences exclude Dead. What remains is not another estimate. The recovered state has to start the same equation on the right and meet the old history without a corner in time.

The terminal rows still satisfy the same incompressible, pressure-normalized momentum recurrence as the preterminal solution. Nothing new is chosen for pressure or for the higher time derivatives. A right-hand solution starting from u_* generates its jet by this same recurrence, so the two jets can now be compared row by row.

20 The exact local restart

Because $u_* \in H_\sigma^\infty$, the local theorem applies at T_* and produces a unique pressure-normalized solution for a time

$$\begin{aligned} \delta &\geq \min \{1, c_\nu(1 + \|u_*\|_{H^3})^{-2}\} \\ &\geq \min \{1, c_\nu(1 + \mathfrak{S}_3)^{-2}\} > 0. \end{aligned} \tag{136}$$

Persistence retains every higher spatial order on that same interval, while the pressure normalization and differentiated equation determine the complete right-hand jet from u_* . Proposition 1.1 states this construction, and Appendix C derives its lifespan, pressure, persistence, and uniqueness in the strong class.

Theorem 20.1 (Continuation from the compatible intersection). *Let (u, p) be a pressure-normalized maximal classical solution on $[0, T_*)$, where $T_* < \infty$. If its datum-generated rows satisfy (63) for every finite n, m , then there are $\delta > 0$ and a unique pressure-normalized classical solution $(\tilde{u}, \tilde{p})$ on $[0, T_* + \delta]$ whose restriction to $[0, T_*)$ is (u, p) . The continuation time can be chosen to satisfy (136).*

Proof. Apply Proposition 1.1 at $t_0 = T_*$ with the datum u_* constructed in (129). It gives a unique normalized solution (w, q) on $[T_*, T_* + \delta]$, where

$$\delta = \delta(\nu, \|u_*\|_{H^3}) > 0, \quad w(T_*) = u_*.$$

Estimate (408), followed by (99) at $m = 3$, gives (136). Because $u_* \in H_\sigma^\infty$, persistence keeps every finite spatial Sobolev order on this one interval. The pressure formula and momentum recurrence then give

$$w, q \in \bigcap_{r, m \in \mathbb{N}_0} C^r([T_*, T_* + \delta]; H^m).$$

Write the right-hand jet as

$$W_n^+ := \partial_t^n w(T_*), \quad P_n^+ := \partial_t^n q(T_*).$$

The local solution uses the same pressure normalization, so its initial jet obeys

$$\begin{aligned} P_n^+ &= R_i R_j \sum_{a=0}^n \binom{n}{a} W_{a,i}^+ W_{n-a,j}^+, \\ W_{n+1}^+ &= \nu \Delta W_n^+ - \sum_{a=0}^n \binom{n}{a} (W_a^+ \cdot \nabla) W_{n-a}^+ - \nabla P_n^+. \end{aligned}$$

The equality $W_0^+ = w(T_*) = u_* = V_0^*$ starts an induction. If $W_a^+ = V_a^*$ for $0 \leq a \leq n$, then the two pressure formulas give $P_n^+ = Q_n^*$. The two velocity recurrences then give $W_{n+1}^+ = V_{n+1}^*$. Thus

$$W_n^+ = V_n^*, \quad P_n^+ = Q_n^* \quad (n \in \mathbb{N}_0). \tag{137}$$

Continuous spatial differentiation matches every mixed derivative at the joining time.

The left side already has the boundary regularity needed for gluing. Part III gave

$$u \in \bigcap_{r, m \in \mathbb{N}_0} H^r(I_*; H^m).$$

For each finite a, m , apply the time-Sobolev embedding (400) with source order $a + 1$. This gives

$$u \in \bigcap_{a, m \in \mathbb{N}_0} C^a([0, T_*]; H^m), \quad \partial_t^n u(T_*) = V_n^*.$$

The pressure formula, the temporal Leibniz rule, and product continuity give the analogous left extension for p , with $\partial_t^n p(T_*) = Q_n^*$. Thus both halves are smooth curves into every fixed H^m , and (137) identifies all of their one-sided derivatives.

Define

$$(\tilde{u}, \tilde{p})(t) := \begin{cases} (u, p)(t), & 0 \leq t < T_*, \\ (w, q)(t), & T_* \leq t \leq T_* + \delta. \end{cases}$$

For a fixed m , equality at order zero makes the piecewise H^m -valued curve continuous. If it is C^k , equality of the $(k + 1)$ -st one-sided derivatives makes the difference quotient of its k -th derivative agree from both sides. Induction in k gives

$$\tilde{u}, \tilde{p} \in \bigcap_{r, m \in \mathbb{N}_0} C^r([0, T_* + \delta]; H^m).$$

Taking m above each spatial embedding threshold gives

$$\tilde{u}, \tilde{p} \in C^\infty(\mathbb{R}^3 \times [0, T_* + \delta]).$$

The joined pair solves Navier–Stokes on each side of T_* . Every term is continuous there in sufficiently high finite Sobolev spaces, so the equation also holds at the joining time. Divergence and the Riesz-transform pressure normalization pass in the same way. The joined pair is a classical normalized solution on a strictly longer interval. Its left restriction is the original history by construction, and local uniqueness fixes the right-hand continuation issued from u_* . $\square$

21 Global conclusion

Proof of Theorem 1.2. Fix $\nu > 0$ and $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let (u, p) be the unique pressure-normalized maximal classical solution supplied by Proposition 1.1, with lifespan $[0, T_*)$.

No finite value of T_* can be maximal. At such an endpoint, Theorem 13.2 places every finite pressure-complete rectangle on $(0, T_*)$. The projection rule gives one terminal value for every temporal row in every finite Sobolev space, compatibility identifies those values as one H^∞ endpoint jet, and Proposition 19.1 shows that the jet still obeys the normalized momentum and pressure recurrences. Theorem 20.1 then extends the original solution to a strictly longer interval. Hence

$$T_* = \infty.$$

Every finite interval now lies inside this one maximal solution. Persistence of all spatial orders, followed by the velocity and pressure recurrences, gives

$$u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)), \quad p(t) = R_i R_j (u_i(t) u_j(t)).$$

The force is zero, $\nabla \cdot u = 0$, and $u(0) = u_0$. Finally, the energy identity (5) holds for every $t \geq 0$:

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2.$$

This is the uniform kinetic-energy bound (4). □

22 The final finite-horizon estimates

The contradiction argument has ended, so the constants can now be stated on an arbitrary physical horizon. Fix $T > 0$, write $I_T = (0, T)$, and begin with the one quantity through which all of the preceding finite recursions run:

$$\mathfrak{B}_2[u_0, \nu; T] := \|u\|_{L^2(I_T; H^3)}^2 + \|u_t\|_{L^2(I_T; H^1)}^2. \quad (138)$$

The graph identity (74) on $[0, T]$ gives

$$\mathfrak{B}_2[u_0, \nu; T] \leq 4T \|u_0\|_2^2 + \kappa_\nu (\nu Z_0 + \mathfrak{J}_1[u_0, \nu; T]). \quad (139)$$

This value belongs to the unique global history generated by the datum. Writing it down quantifies that history, but a final estimate must also let a reader obtain an upper bound without knowing the history in advance. The equation supplies that last step. A proposed history can be inserted into Navier–Stokes, its defect can be measured, and one scalar comparison then decides whether the proposal encloses the exact solution on the chosen horizon.

For each requested finite output, fix once and for all validated rational upper bounds for every analytic constant used through the requested orders. We write these as $\overline{K}_q$, $\overline{A}_r$, $\overline{C}_{s,k,b}$, $\overline{C}_\ell^{\text{Lip}}$, $\overline{G}_\ell$, $\overline{C}_1^{\text{Lip}}$, $\overline{C}_{\text{quad}}$, and $\overline{C}_{\text{lin}}(\nu)$. Only finitely many are needed for a finite request, and the Fourier, convolution, embedding, and heat estimates used above give computable rational upper enclosures for them. In every expression carrying the superscript *alg*, the corresponding barred constants are used. For the restart put

$$\underline{c}_\nu := (8\overline{C}_{\text{quad}}\overline{C}_{\text{lin}}(\nu)^2)^{-2} \leq c_\nu. \quad (140)$$

The unbarred constants remain in the classical estimates.

Proposition 22.1 (Residual certificate and terminating datum search). *Fix an integer $s \geq 3$. Enlarge the order constant K_s from (95), if needed, and fix a proved computable rational value for it so that, for divergence-free fields v, w ,*

$$\begin{aligned} |\langle B(v, w) + B(w, v), w \rangle_{H^s}| &\leq K_s \|v\|_{H^{s+1}} \|w\|_{H^s}^2, \\ |\langle B(w, w), w \rangle_{H^s}| &\leq K_s \|w\|_{H^s}^3. \end{aligned} \quad (141)$$

Let

$$v \in C^1([0, T]; H_\sigma^s) \cap C([0, T]; H_\sigma^{s+2})$$

be any trial history and define its full equation residual by

$$\text{Res}_\nu[v] := v_t - \nu \Delta v + B(v, v). \quad (142)$$

Put

$$\begin{aligned} A_{s,v}(t) &:= K_s \int_0^t \|v(\tau)\|_{H^{s+1}} d\tau, \\ d_{s,v} &:= \|u_0 - v(0)\|_{H^s} + \int_0^T e^{-A_{s,v}(t)} \|\text{Res}_\nu[v](t)\|_{H^s} dt, \\ J_{s,v}(t) &:= K_s \int_0^t e^{A_{s,v}(\tau)} d\tau, \quad \Theta_{s,v} := 1 - d_{s,v} J_{s,v}(T). \end{aligned} \quad (143)$$

If $\Theta_{s,v} > 0$, then

$$\|u(t) - v(t)\|_{H^s} \leq \mathcal{E}_{s,v}(t) := \frac{e^{A_{s,v}(t)} d_{s,v}}{1 - d_{s,v} J_{s,v}(t)} \quad (0 \leq t \leq T). \quad (144)$$

There is a fixed countable family of such divergence-free trial histories $\mathcal{D}_{s,T} = \{v_{s,T}^{(q)} : q \in \mathbb{N}\}$, independent of the solution, for which the accepted set

$$\mathcal{Q}_s[u_0, \nu; T] := \left\{ q \in \mathbb{N} : \Theta_{s,v_{s,T}^{(q)}} > 0 \right\} \quad (145)$$

is nonempty. Its canonical classical selector is

$$q_s^{\text{cl}}[u_0, \nu; T] := \min \mathcal{Q}_s[u_0, \nu; T]. \quad (146)$$

This selector and the majorant below are canonical relative to the once-fixed effective enumeration $\mathcal{D}_{s,T}$ and the selected rational value of K_s . Those fixed selector choices are suppressed in the notation. Define

$$\mathfrak{M}_s[u_0, \nu; T] := \sup_{0 \leq t \leq T} \left(\|v_{s,T}^{(q_s^{\text{cl}})}(t)\|_{H^s} + \mathcal{E}_{s,v_{s,T}^{(q_s^{\text{cl}})}}(t) \right). \quad (147)$$

Then

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^s} \leq \mathfrak{M}_s[u_0, \nu; T] < \infty. \quad (148)$$

Every quantity on the right of (147) is evaluated on a finite trial history and the given datum. If u_0 has an effective H^s Cauchy name and $\nu > 0$ and $T > 0$ are computable reals, a fixed dovetailed validated search returns an accepted index $q_s^{\text{alg}}[u_0, \nu; T]$ and a rational upper certificate $\overline{\mathfrak{M}}_s^{\text{alg}}[u_0, \nu; T]$ in finite time, with

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^s} \leq \overline{\mathfrak{M}}_s^{\text{alg}}[u_0, \nu; T]. \quad (149)$$

The algorithmic selector need not equal the canonical classical selector.

Proof. Let $w = u - v$. Subtracting the residual equation from Navier–Stokes gives

$$w_t - \nu \Delta w + B(v, w) + B(w, v) + B(w, w) = -\text{Res}_\nu[v].$$

Pair this equation with w in H^s . The viscous term is nonnegative, and (141) gives, after regularizing the norm at $w = 0$,

$$D^+ \|w\|_{H^s} \leq K_s \|v\|_{H^{s+1}} \|w\|_{H^s} + K_s \|w\|_{H^s}^2 + \|\text{Res}_\nu[v]\|_{H^s}. \quad (150)$$

Set $y = e^{-A_{s,v}} \|w\|_{H^s}$. Integration of (150) gives

$$y(t) \leq d_{s,v} + K_s \int_0^t e^{A_{s,v}(\tau)} y(\tau)^2 d\tau.$$

Bihari's inequality now yields

$$y(t) \leq \frac{d_{s,v}}{1 - d_{s,v} J_{s,v}(t)}$$

as long as the denominator is positive. The acceptance condition keeps it positive through T , proving (144).

It remains to construct the search family and prove that the search stops. In Fourier space, fix a uniform effective enumeration of real vector fields whose transforms are finite rational combinations of computable C_c^∞ bumps supported in rational boxes away from the origin. Apply the solenoidal multiplier

$$\widehat{\mathbb{P}f}(\xi) = \left(I - \frac{\xi \otimes \xi}{|\xi|^2} \right) \widehat{f}(\xi).$$

The resulting fields are divergence-free and belong to every Sobolev space. They are dense in $H_\sigma^m(\mathbb{R}^3)$ for each finite m : smooth compact Fourier support away from zero is dense in the weighted Fourier L^2 space, and $\mathbb{P}$ is bounded and fixes divergence-free fields. Rational time polynomials with coefficients in this spatial family form $\mathcal{D}_{s,T}$. To see density in the two-norm path space, first apply smooth spatial Fourier cutoffs to a path and its time derivative; the cutoffs converge in $C^1 H^s \cap C H^{s+2}$ and place both in every higher Sobolev space. Banach-valued polynomial approximation in time, followed by rational approximation of its finitely many spatial coefficients, then makes $\mathcal{D}_{s,T}$ dense in

$$C^1([0, T]; H_\sigma^s) \cap C([0, T]; H_\sigma^{s+2}).$$

The global solution belongs to this space. Choose a sequence $v_j \in \mathcal{D}_{s,T}$ converging to u in its defining norm. Since $\text{Res}_\nu[u] = 0$, the product estimate gives

$$\|\text{Res}_\nu[v_j]\|_{C([0,T];H^s)} \longrightarrow 0, \quad \|u_0 - v_j(0)\|_{H^s} \longrightarrow 0.$$

The values $A_{s,v_j}(T)$, and hence $J_{s,v_j}(T)$, stay bounded. Thus $d_{s,v_j} J_{s,v_j}(T) \rightarrow 0$, so some finite candidate is accepted.

For the effective statement, a finite candidate, its nonlinear Fourier convolution, and its full residual have compact computable Fourier support. Validated quadrature thus encloses every norm and time integral in (143); the effective H^s name of u_0 encloses the initial mismatch, while the computable ν, T , and K_s enclose the remaining operations. At stage L , run these enclosures for every candidate $q \leq L$ to width at most 2^{-L} , and accept the first one whose resulting interval for $\Theta_{s,v_{s,T}^{(q)}}$ has positive lower endpoint. The candidates and enclosure depths are being dovetailed, so no zero or negative candidate can stall the search. Some candidate has a strictly positive margin; at a finite stage its enclosure exposes that margin and supplies q_s^{alg} . At that accepting stage let

$$\underline{\Theta}_s > 0, \quad \overline{V}_s \geq \sup_{0 \leq t \leq T} \left\| v_{s,T}^{(q_s^{\text{alg}})}(t) \right\|_{H^s}, \quad \overline{d}_s \geq d_{s,v_{s,T}^{(q_s^{\text{alg}})}}, \quad \overline{e}_s \geq e_{s,v_{s,T}^{(q_s^{\text{alg}})}(T)}^A$$

be rational validated bounds from the same enclosure schedule. Since $A_{s,v}$ and $J_{s,v}$ are increasing,

$$\mathcal{E}_{s,v_{s,T}^{(q_s^{\text{alg}})}}(t) \leq \frac{\overline{e}_s \overline{d}_s}{\underline{\Theta}_s} \quad (0 \leq t \leq T).$$

Thus the search outputs the rational number

$$\overline{\mathfrak{M}}_s^{\text{alg}} := \overline{V}_s + \frac{\overline{e}_s \overline{d}_s}{\underline{\Theta}_s}, \tag{151}$$

which proves (149). This separates the canonical mathematical data functional from the terminating validated certificate. $\square$

The order $s = 3$ pays the base rectangle directly. The projected equation, (102), and (148) give

$$\|u_t(t)\|_{H^1} \leq \nu \mathfrak{M}_3[u_0, \nu; T] + A_1 \mathfrak{M}_3[u_0, \nu; T]^2.$$

Consequently,

$$\begin{aligned} \mathfrak{B}_2[u_0, \nu; T] &\leq \mathfrak{B}_2^{\text{cert}}[u_0, \nu; T], \\ \mathfrak{B}_2^{\text{cert}}[u_0, \nu; T] &:= T \left\{ \mathfrak{M}_3[u_0, \nu; T]^2 + (\nu \mathfrak{M}_3[u_0, \nu; T] + A_1 \mathfrak{M}_3[u_0, \nu; T]^2)^2 \right\}. \end{aligned} \quad (152)$$

For effective inputs, the terminating version is obtained from (149):

$$\mathfrak{B}_2[u_0, \nu; T] \leq \overline{\mathfrak{B}}_2^{\text{alg}}[u_0, \nu; T], \quad \overline{\mathfrak{B}}_2^{\text{alg}} \geq T \left\{ (\overline{\mathfrak{M}}_3^{\text{alg}})^2 + \left(\nu \overline{\mathfrak{M}}_3^{\text{alg}} + \overline{A}_1 (\overline{\mathfrak{M}}_3^{\text{alg}})^2 \right)^2 \right\}, \quad (153)$$

where the right member is a validated rational upper enclosure of the displayed computable-real expression. The canonical value $\mathfrak{B}_2^{\text{cert}}$ is a datum-to-number map for every Schwartz datum; $\overline{\mathfrak{B}}_2^{\text{alg}}$ is the independently checkable finite output for effective data. Both constructions are downstream from global continuation, whose existence guarantees acceptance on every finite horizon. Every monotone certified recursion below also has its terminating algorithmic version after replacing $\mathfrak{B}_2^{\text{cert}}$ by $\overline{\mathfrak{B}}_2^{\text{alg}}$ and every analytic constant it uses by the corresponding validated rational upper bound fixed above. Its restart output uses $\underline{c}_\nu$.

On the same horizon define

$$\mathfrak{C}_{1/2}[u_0, \nu; T] := \sup_{0 \leq S \leq T} \int_0^S \mathcal{P}_{1/2}(t) dt, \quad \mathfrak{A}_{1/2}[u_0, \nu; T] := \frac{H(0) + \mathfrak{C}_{1/2}[u_0, \nu; T]}{\nu}. \quad (154)$$

The exact critical balance gives

$$\int_0^T \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \mathfrak{A}_{1/2}[u_0, \nu; T]. \quad (155)$$

The base rectangle gives the sharper history-resolved coefficient

$$\mathfrak{A}_{1/2}^{\text{prod}}[u_0, \nu; T] := \min \left\{ \mathfrak{B}_2[u_0, \nu; T], \left(\frac{\|u_0\|_2^2}{2\nu} \right)^{3/4} \mathfrak{B}_2[u_0, \nu; T]^{1/4} \right\}, \quad (156)$$

while the certificate supplies the data-resolved coefficient

$$\mathfrak{A}_{1/2}^{\text{cert}}[u_0, \nu; T] := \min \left\{ \mathfrak{B}_2^{\text{cert}}[u_0, \nu; T], \left(\frac{\|u_0\|_2^2}{2\nu} \right)^{3/4} \mathfrak{B}_2^{\text{cert}}[u_0, \nu; T]^{1/4} \right\}. \quad (157)$$

Monotonicity and (152) give

$$\int_0^T \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \mathfrak{A}_{1/2}^{\text{prod}}[u_0, \nu; T] \leq \mathfrak{A}_{1/2}^{\text{cert}}[u_0, \nu; T]. \quad (158)$$

Repeat the finite definitions (96)–(112) with T in place of T_* . Using $\mathfrak{A}_{1/2}^{\text{prod}}$ gives the unadorned quantities

$$\Gamma_m(T), \quad \mathfrak{S}_m(T), \quad \mathfrak{D}_m(T), \quad \mathfrak{V}_{n,m}(T), \quad \mathfrak{Q}_{n,m}(T), \quad \mathfrak{R}_{N,M}(T).$$

Using $\mathfrak{A}_{1/2}^{\text{cert}}$ gives

$$\Gamma_m^{\text{cert}}(T), \quad \mathfrak{S}_m^{\text{cert}}(T), \quad \mathfrak{D}_m^{\text{cert}}(T), \quad \mathfrak{V}_{n,m}^{\text{cert}}(T), \quad \mathfrak{Q}_{n,m}^{\text{cert}}(T), \quad \mathfrak{R}_{N,M}^{\text{cert}}(T).$$

Every operation in these definitions is nonnegative and increasing, so each unadorned value is bounded by its certified counterpart.

The $r = 1$ product row also retains the two analytic feedback bounds. Put

$$\begin{aligned} \mathfrak{J}_1^\sharp(T) &:= \frac{G_1}{\sqrt{2}\nu} \sqrt{\|u_0\|_2^2 D_1(u_0) e^{\Gamma_1(T)/2}} \left(D_1(u_0) e^{\Gamma_1(T)} + D_2(u_0) e^{\Gamma_2(T)} \right), \\ \mathfrak{J}_1^{\text{fin}}(T) &:= \min \left\{ A_1^2 T \mathfrak{S}_2(T)^4, \mathfrak{J}_1^\sharp(T) \right\}. \end{aligned} \quad (159)$$

Then

$$\begin{aligned} \mathfrak{B}_2[u_0, \nu; T] &\leq \overline{\mathfrak{B}}_2[u_0, \nu; T], \\ \overline{\mathfrak{B}}_2[u_0, \nu; T] &:= 4T \|u_0\|_2^2 + \kappa_\nu \left(\nu Z_0 + \mathfrak{J}_1^{\text{fin}}(T) \right). \end{aligned} \quad (160)$$

This feedback form can be sharper once the history has been evaluated. Equation (152) is the unconditional datum-resolved majorant used in the final statement.

Whenever one finite order $\ell \geq 3$ satisfies $\Xi_\ell[u_0, \nu; T] < 1$, Proposition 17.2 also gives the closed short-horizon values

$$\mathfrak{C}_{1/2}[u_0, \nu; T] \leq \widehat{\mathfrak{C}}_{1/2}^{(\ell)}[u_0, \nu; T], \quad \mathfrak{A}_{1/2}[u_0, \nu; T] \leq \widehat{\mathfrak{A}}_{1/2}^{(\ell)}[u_0, \nu; T]. \quad (161)$$

Repeating the same recursions with $\widehat{\mathfrak{A}}_{1/2}^{(\ell)}$ gives hatted quantities. On that window the strongest displayed estimate is the minimum of the certified and hatted majorants.

Corollary 22.2 (Complete quantitative output). *For every $N, M, n, m, k \in \mathbb{N}_0$, every $1 \leq a \leq \infty$, and every $2 \leq b \leq \infty$, the global solution satisfies the following datum-resolved estimates on $[0, T]$.*

Every finite rectangle and every uniform jet row obey

$$\mathcal{N}_{N,M}(u; I_T) \leq \mathfrak{R}_{N,M}(T) \leq \mathfrak{R}_{N,M}^{\text{cert}}(T), \quad (162)$$

$$\sup_{0 \leq t \leq T} \|\partial_t^n u(t)\|_{H^m} \leq \mathfrak{V}_{n,m}(T) \leq \mathfrak{V}_{n,m}^{\text{cert}}(T), \quad (163)$$

$$\sup_{0 \leq t \leq T} \|\partial_t^n p(t)\|_{H^m} \leq \mathfrak{Q}_{n,m}(T) \leq \mathfrak{Q}_{n,m}^{\text{cert}}(T). \quad (164)$$

If $s \in \mathbb{N}_0$ satisfies $s > k + \frac{3}{2} - \frac{3}{b}$, then

$$\left\| \nabla^k \partial_t^n u \right\|_{L^a(I_T; L^b)} \leq C_{s,k,b} T^{1/a} \mathfrak{V}_{n,s}^{\text{cert}}(T), \quad (165)$$

$$\left\| \nabla^k \partial_t^n p \right\|_{L^a(I_T; L^b)} \leq C_{s,k,b} T^{1/a} \mathfrak{Q}_{n,s}^{\text{cert}}(T), \quad (166)$$

where $T^{1/\infty} = 1$. The same right sides without the factor $T^{1/a}$ control the corresponding $L_t^\infty L_x^b$ norms.

For $j \in \mathbb{N}_0$, $m > j + \frac{1}{2}$, and $K \geq 1$, define

$$E_{j+1/2, >K}(t) := \frac{1}{2} \int_{|\xi| > K} |\xi|^{2j+1} |\widehat{u}(t, \xi)|^2 d\xi. \quad (167)$$

The complete critical-height ladder and every one of its frequency tails satisfy

$$\sup_{0 \leq t \leq T} E_{j+1/2}(t) \leq \frac{1}{2} \mathfrak{S}_{j+1}^{\text{cert}}(T)^2, \quad (168)$$

$$\int_0^T E_{j+1/2}(t) dt \leq \frac{1}{2} \mathfrak{D}_j^{\text{cert}}(T)^2, \quad (169)$$

$$\sup_{0 \leq t \leq T} E_{j+1/2, > K}(t) \leq \frac{1}{2} K^{2j+1-2m} D_m(u_0) e^{\Gamma_m^{\text{cert}}(T)}. \quad (170)$$

For the original critical row,

$$\begin{aligned} \|H'\|_{L^1(0,T)} &\leq \frac{1}{2} \mathfrak{B}_2^{\text{cert}}[u_0, \nu; T], \\ \|\mathcal{P}_{1/2}\|_{L^1(0,T)} &\leq \left(\frac{1}{2} + \nu\right) \mathfrak{B}_2^{\text{cert}}[u_0, \nu; T]. \end{aligned} \quad (171)$$

The endpoint at the chosen horizon is quantitative. For $0 \leq t \leq T$,

$$\|\partial_t^n u(t) - \partial_t^n u(T)\|_{H^m} \leq (T-t) \mathfrak{V}_{n+1,m}^{\text{cert}}(T), \quad (172)$$

$$\|\partial_t^n p(t) - \partial_t^n p(T)\|_{H^m} \leq (T-t) (\mathfrak{L}_{n,m}^Q)^{\text{cert}}(T), \quad (173)$$

where $(\mathfrak{L}_{n,m}^Q)^{\text{cert}}(T)$ is (132) evaluated with the certified row values. Restarting the solution from $u(T)$ has the explicit lower lifespan

$$\delta_T \geq \min \left\{ 1, c_\nu (1 + \mathfrak{S}_3^{\text{cert}}(T))^{-2} \right\}. \quad (174)$$

The derivative ledger remains finite and exact. The certificate itself reads the datum in H^3 . The velocity row (n, m) additionally uses datum order $m + 2n$, while the pressure row uses order $\sigma_m + 2n$. The rectangle (N, M) uses order $M + 2N + 1$; retaining Q_N raises this, when needed, to

$$\max\{M + 2N + 1, \sigma_M + 2N\}.$$

The velocity endpoint modulus uses order $m + 2n + 2$, and its pressure counterpart uses order $\sigma_m + 2n + 2$. The uniform bound for $E_{j+1/2}$ uses datum order $j + 1$, while its time integral uses order j . The order- m frequency tail and the restart use datum orders m and 3. In particular, the displayed mixed velocity estimate at embedding order s reads depth $\max\{3, s + 2n\}$, while its pressure counterpart reads depth $\max\{3, \sigma_s + 2n\}$. Thus every certified datum depth is $\max\{3, d\}$, with d the order just listed. The hatted route reads depth $\max\{\ell, d\}$. Numerical evaluation requires an effective Sobolev name through that finite depth together with computable names for ν and T . It is also relative to the fixed trial-history enumeration, selector constant K_s , and the validated rational analytic constants specified above. Temporal order, spatial order, viscosity, horizon, and datum depth all remain visible.

If $\Xi_\ell[u_0, \nu; T] < 1$, the recursive jet, height, tail, and endpoint upper bounds may be replaced by the smaller of the certified value and its hatted counterpart. In particular,

$$\mathcal{N}_{N,M}(u; I_T) \leq \min \left\{ \mathfrak{R}_{N,M}^{\text{cert}}(T), \widehat{\mathfrak{R}}_{N,M}^{(\ell)}(T) \right\}. \quad (175)$$

The critical-action pair uses

$$\mathfrak{B}_{2,\ell}^{\text{best}}(T) := \min \left\{ \mathfrak{B}_2^{\text{cert}}[u_0, \nu; T], \widehat{\mathfrak{R}}_{0,2}^{(\ell)}(T) \right\}, \quad (176)$$

so that

$$\|H'\|_{L^1(0,T)} \leq \frac{1}{2} \mathfrak{B}_{2,\ell}^{\text{best}}(T), \quad \|\mathcal{P}_{1/2}\|_{L^1(0,T)} \leq \left(\frac{1}{2} + \nu\right) \mathfrak{B}_{2,\ell}^{\text{best}}(T). \quad (177)$$

The restart estimate is a lower bound, so its two versions combine in the opposite direction:

$$\begin{aligned} \delta_T &\geq \max \left\{ \min \left\{ 1, c_\nu (1 + \mathfrak{S}_3^{\text{cert}}(T))^{-2} \right\}, \min \left\{ 1, c_\nu (1 + \widehat{\mathfrak{S}}_3^{(\ell)}(T))^{-2} \right\} \right\} \\ &= \min \left\{ 1, c_\nu \left(1 + \min \left\{ \mathfrak{S}_3^{\text{cert}}(T), \widehat{\mathfrak{S}}_3^{(\ell)}(T) \right\} \right)^{-2} \right\}. \end{aligned} \quad (178)$$

For effective inputs, replacing c_ν by $\underline{c}_\nu$ and the upper envelopes by their validated rational enclosures gives a terminating rational lower bound for the same restart time.

Proof. Equation (152) and monotonicity of $x \mapsto \min\{x, cx^{1/4}\}$ give (158). Substituting the certified coefficient into (95)–(113) gives (162)–(164). Sobolev embedding gives (165)–(166). The comparisons

$$|\xi|^{2j+1} \leq (1 + |\xi|^2)^{j+1}, \quad |\xi|^{2j+1} \mathbf{1}_{\{|\xi| > K\}} \leq K^{2j+1-2m} |\xi|^{2m},$$

together with the certified spatial envelopes, give (168)–(170). The variation and response estimates give (171). Integrating the next velocity and pressure rows gives the two endpoint moduli, and the local construction in Appendix C gives (174). Under $\Xi_\ell[u_0, \nu; T] < 1$, the alternative coefficient (161) enters the same nonnegative recursions. Taking the smaller upper output proves (175) and (177); monotonic decrease of the restart map in its H^3 input proves (178). $\square$

Part V

Independent uses of the mechanism

23 What must survive when the equation changes

The three-dimensional proof does more than produce a terminal bound. The datum supplies finite spatial rows of one history; the differentiated equation generates its temporal and pressure rows; restriction keeps the overlaps equal; the intersection produces one endpoint; and uniqueness lets that endpoint restart the evolution. A genuinely different three-dimensional producer can preserve those later steps while changing the estimate that begins them.

Axisymmetric flow without swirl provides the first test. The physical vorticity is stretched, but the relative vorticity ω^θ/r obeys a scalar parabolic law whose maximum principle controls the stretching coefficient on every finite horizon. That datum-side coefficient produces every spatial row, after which the complete mixed jet, pressure jet, compatible rectangles, endpoint, and restart follow from the same finite mechanism used in the main proof. The result is classical at the level of global smoothness; the work below strengthens it into a quantitative realization of the complete proof engine.

The later tests change the equation itself. In viscous–resistive MHD, the producer must control a coupled velocity and magnetic field while pressure keeps the difference of their stresses. Critical hyperdissipation replaces the adjacent integer Sobolev rows by a fractional Gelfand triple. The Navier–Stokes–Voigt equation moves elliptic regularization onto the time derivative, with the striking consequence that temporal height no longer increases datum depth. Each problem is carried from its own datum estimate through the complete mixed and pressure jets, a common endpoint, and an explicit same-history restart. The comparison is made by the proofs, not by an analogy among their names.

24 Three-dimensional axisymmetric flow without swirl

The next problem restores three-dimensional vortex stretching. It does not remove the nonlinearity or pressure, and it permits arbitrarily large smooth data. Instead, axial symmetry exposes the precise part of vorticity that the stretching cannot amplify: the angular vorticity divided by its distance from the axis. This scalar becomes the producer for the complete mixed jet.

Write cylindrical coordinates as (r, θ, z) . A smooth axisymmetric velocity without swirl has the form

$$u = u^r(r, z, t)e_r + u^z(r, z, t)e_z, \quad \omega := \nabla \times u = \omega^\theta(r, z, t)e_\theta. \quad (179)$$

Let $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$ have this form. Rotational invariance and local uniqueness preserve the class on the maximal classical interval. Smoothness at the axis gives $\omega^\theta = O(r)$, so

$$\eta := \frac{\omega^\theta}{r} \quad (180)$$

extends smoothly across $r = 0$. Write $T_{\max}^{\text{axi}}$ for the maximal time in this invariant class.

Theorem 24.1 (Quantitative axisymmetric no-swirl jet). *Fix $p > 3$. Every Schwartz, divergence-free, axisymmetric datum without swirl generates a unique global smooth no-swirl solution. For every finite*

$T > 0$, every $n, r, N, M \in \mathbb{N}_0$, and every $q, m \in \mathbb{N}$ with $q, m \geq 3$, the solution obeys the datum-side spatial, Lipschitz, dissipation, velocity-row, pressure-row, rectangle, endpoint, frequency-tail, and restart estimates (194)–(196), (200), (204), (206), and (207)–(210).

Proof. Directly taking the curl of Navier–Stokes gives

$$\partial_t \eta + u^r \partial_r \eta + u^z \partial_z \eta = \nu \left(\partial_r^2 + \frac{3}{r} \partial_r + \partial_z^2 \right) \eta, \quad (181)$$

$$\partial_t \omega^\theta + u^r \partial_r \omega^\theta + u^z \partial_z \omega^\theta - \nu \left(\Delta - \frac{1}{r^2} \right) \omega^\theta = \frac{u^r}{r} \omega^\theta. \quad (182)$$

These are the classical no-swirl equations. [18] [19]

The extra radial term in (181) has the good axis sign. Multiplying by $|\eta|^{a-2} \eta$, integrating with the three-dimensional measure $r dr d\theta dz$, and integrating at $r = 0$ produces the boundary contribution

$$-\frac{4\pi\nu}{a} \int_{\mathbb{R}} |\eta(0, z, t)|^a dz, \quad (183)$$

in addition to the nonpositive bulk dissipation. Approximation at $a = 1$ and the maximum principle then give

$$\|\eta(t)\|_{L^a(\mathbb{R}^3)} \leq \|\eta_0\|_{L^a(\mathbb{R}^3)} \quad (1 \leq a \leq \infty). \quad (184)$$

This scalar producer remains viscous, and the radial diffusion records the geometry of the axis.

To see what (184) buys, fix $p > 3$. The axisymmetric Biot–Savart kernel satisfies

$$\left| \frac{u^r(x, t)}{r} \right| \leq C \int_{\mathbb{R}^3} \frac{|\eta(y, t)|}{|x - y|^2} dy. \quad (185)$$

[18] [19] Split the integral at a radius $R > 0$. Hölder’s inequality near the origin and Cauchy–Schwarz outside it give

$$\|u^r(t)/r\|_\infty \leq C_p \left(R^{1-3/p} \|\eta(t)\|_p + R^{-1/2} \|\eta(t)\|_2 \right). \quad (186)$$

Optimizing in R and using (184) yields

$$\|u^r(t)/r\|_\infty \leq K_p[u_0], \quad K_p[u_0] := C_p \|\eta_0\|_2^{\alpha_p} \|\eta_0\|_p^{1-\alpha_p}, \quad (187)$$

where

$$\alpha_p := \frac{2(p-3)}{3(p-2)}. \quad (188)$$

The exponent is forced by balancing the powers $R^{1-3/p}$ and $R^{-1/2}$. The maximum principle applied to (182) now gives

$$\|\omega(t)\|_\infty \leq W_0 e^{K_p[u_0]t}, \quad W_0 := \|\omega_0\|_\infty. \quad (189)$$

Thus the genuine three-dimensional stretching coefficient is finite on every finite horizon and is expressed entirely by the datum.

24.1 The spatial producer at every order

Fix an integer $q \geq 3$. A dyadic decomposition proves the logarithmic gradient estimate

$$\|\nabla u\|_\infty \leq C_q [1 + \|u_0\|_2 + \|\omega\|_\infty \log(e + \|u\|_{H^q})]. \quad (190)$$

Indeed, the negative dyadic shells are bounded by the energy norm $\|u(t)\|_2 \leq \|u_0\|_2$. Each of the first J nonnegative shells is bounded by the frequency-localized Biot–Savart multiplier times $\|\omega\|_\infty$. The high shells satisfy

$$\sum_{j>J} \|P_j \nabla u\|_\infty \leq C_q 2^{-J(q-5/2)} \|u\|_{H^q}.$$

Choosing J from $e + \|u\|_{H^q}$ gives (190). This is the usual Beale–Kato–Majda logarithm, derived here at the exact order used below. ^[17]

The complete H^q energy row gives

$$\frac{d}{dt} \|u(t)\|_{H^q} \leq C_q \|\nabla u(t)\|_\infty \|u(t)\|_{H^q}. \quad (191)$$

Put

$$\begin{aligned} L_0 &:= 1 + \|u_0\|_2, & \Psi_K(T) &:= \begin{cases} (e^{KT} - 1)/K, & K > 0, \\ T, & K = 0, \end{cases} \\ A_p(T) &:= W_0 \Psi_{K_p[u_0]}(T), & Y_{q,0} &:= \log(e + \|u_0\|_{H^q}). \end{aligned} \quad (192)$$

For $Y_q(t) := \log(e + \|u(t)\|_{H^q})$, equations (189)–(191) give

$$Y_q'(t) \leq C_q \left(L_0 + W_0 e^{K_p[u_0]t} Y_q(t) \right).$$

Gronwall's inequality yields the completely datum-side quantities

$$\bar{Y}_q(T) := (Y_{q,0} + C_q L_0 T) e^{C_q A_p(T)}, \quad \mathbf{S}_q^{\text{axi}}(T) := e^{\bar{Y}_q(T)}, \quad (193)$$

and the bounds

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^q} \leq \mathbf{S}_q^{\text{axi}}(T), \quad (194)$$

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq \mathbf{J}_q^{\text{axi}}(T) := C_q [L_0 T + A_p(T) \bar{Y}_q(T)]. \quad (195)$$

No norm of the evolved solution remains on either right-hand side.

The dissipative companion is also explicit. The squared high-order energy row and (195) give

$$\int_0^T \|u(t)\|_{H^{q+1}}^2 dt \leq \mathbf{D}_q^{\text{axi}}(T)^2, \quad (196)$$

where, after enlarging the norm-equivalence constant C_q , one may take

$$\mathbf{D}_q^{\text{axi}}(T)^2 := C_q \left[T \|u_0\|_2^2 + \frac{\|u_0\|_{H^q}^2 + 2C_q \mathbf{S}_q^{\text{axi}}(T)^2 \mathbf{J}_q^{\text{axi}}(T)}{\nu} \right]. \quad (197)$$

24.2 The axisymmetric mixed and pressure jets

The producer has now paid all spatial orders. The rest of the construction returns to the original three-dimensional equation. Put $V_n := \partial_t^n u$, set $\sigma_r := \max\{2, r+1\}$, and define

$$\mathbf{U}_{0,r}^{\text{axi}}(T) := \mathbf{S}_{\max\{3,r\}}^{\text{axi}}(T). \quad (198)$$

With the product constants from (102), define

$$\mathbf{U}_{n+1,r}^{\text{axi}}(T) := \nu \mathbf{U}_{n,r+2}^{\text{axi}}(T) + A_r \sum_{a=0}^n \binom{n}{a} \mathbf{U}_{a,\sigma_r}^{\text{axi}}(T) \mathbf{U}_{n-a,\sigma_r}^{\text{axi}}(T). \quad (199)$$

The projected momentum recurrence gives

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^r} \leq \mathbf{U}_{n,r}^{\text{axi}}(T). \quad (200)$$

For the natural negative row, put

$$\mathbf{U}_{n+1,-1}^{\text{axi}}(T) := \nu \mathbf{U}_{n,1}^{\text{axi}}(T) + A_{-1} \sum_{a=0}^n \binom{n}{a} \mathbf{U}_{a,1}^{\text{axi}}(T) \mathbf{U}_{n-a,1}^{\text{axi}}(T). \quad (201)$$

The normalized pressure rows are

$$Q_n := \partial_t^n p = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (202)$$

Accordingly, they obey

$$\mathbf{Q}_{n,r}^{\text{axi}}(T) := A_r \sum_{a=0}^n \binom{n}{a} \mathbf{U}_{a,\sigma_r}^{\text{axi}}(T) \mathbf{U}_{n-a,\sigma_r}^{\text{axi}}(T), \quad (203)$$

and hence

$$\sup_{0 \leq t \leq T} \|\partial_t^n p(t)\|_{H^r} \leq \mathbf{Q}_{n,r}^{\text{axi}}(T). \quad (204)$$

For every $N, M \in \mathbb{N}_0$, define

$$\mathfrak{R}_{N,M}^{\text{axi}}(T) := T \sum_{n=0}^N \left[(\mathbf{U}_{n,M+1}^{\text{axi}}(T))^2 + (\mathbf{U}_{n+1,M-1}^{\text{axi}}(T))^2 \right], \quad (205)$$

using (201) when $M = 0$. Measure the actual axisymmetric history by

$$\mathcal{N}_{N,M}^{\text{axi}}(u; T) := \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2.$$

It satisfies

$$\mathcal{N}_{N,M}^{\text{axi}}(u; T) \leq \mathfrak{R}_{N,M}^{\text{axi}}(T). \quad (206)$$

Expanding this finite recursion uses datum derivatives through order at most $\max\{3, M+2N+1\}$, together with η_0 in the two displayed Lebesgue spaces.

Every rectangle is a restriction of the same axisymmetric velocity and normalized pressure. At a chosen horizon, the endpoint moduli are

$$\|V_n(t) - V_n(T)\|_{H^m} \leq (T - t)U_{n+1,m}^{\text{axi}}(T), \quad (207)$$

$$\|Q_n(t) - Q_n(T)\|_{H^r} \leq (T - t)Q_{n+1,r}^{\text{axi}}(T). \quad (208)$$

The critical frequency tail has the explicit form

$$\sup_{0 \leq t \leq T} H_{>K}(t) \leq \frac{1}{2} K^{1-2m} S_m^{\text{axi}}(T)^2, \quad m \geq 3, K \geq 1. \quad (209)$$

For every $0 < T < T_{\max}^{\text{axi}}$, the local H^3 construction and (194) give a same-history restart of length

$$\delta_T^{\text{axi}} \geq \min \left\{ 1, c_\nu (1 + S_3^{\text{axi}}(T))^{-2} \right\}. \quad (210)$$

If $T_{\max}^{\text{axi}}$ were finite, apply the estimates first on $[0, \tau]$, $\tau < T_{\max}^{\text{axi}}$, with the right sides evaluated at $T_{\max}^{\text{axi}}$, and then let $\tau \uparrow T_{\max}^{\text{axi}}$. The compatible rectangles give one H^∞ endpoint and the complete pressure jet. At that endpoint the limiting rows obey

$$V_{n+1,*} = \nu \Delta V_{n,*} - \sum_{a=0}^n \binom{n}{a} \mathbb{P}(V_{a,*} \cdot \nabla V_{n-a,*}),$$

together with the normalized limit of (202). The local theorem restarts it for the positive time supplied by (210) with $T = T_{\max}^{\text{axi}}$. The adjacent-row trace proof, the normalized pressure recurrence, and the local H^3 construction now apply directly to these axisymmetric rectangles. The right local jet agrees inductively with the left endpoint jet, the joined curve is smooth through $T_{\max}^{\text{axi}}$, and uniqueness thereafter preserves axial symmetry and zero swirl. Finite maximality is impossible. This recovers global smoothness for every Schwartz axisymmetric no-swirl datum in the stated class, now with an explicit mixed jet, pressure jet, frequency tail, endpoint modulus, and restart time. The global regularity of this class is classical; the result here is its complete realization through the paper's quantitative producer–intersection mechanism. $\square$

25 A coupled three-dimensional fluid and magnetic field

The previous application still evolved one vector field. A stronger test is to let a second divergence-free field enter the transport itself. In Alfvénic units, viscous–resistive incompressible magnetohydrodynamics on $\mathbb{R}^3$ reads

$$\partial_t u - \nu \Delta u + (u \cdot \nabla)u - (b \cdot \nabla)b + \nabla \Pi = 0, \quad (211)$$

$$\partial_t b - \eta \Delta b + (u \cdot \nabla)b - (b \cdot \nabla)u = 0, \quad (212)$$

$$\nabla \cdot u = \nabla \cdot b = 0. \quad (213)$$

Here $\nu, \eta > 0$, and $\Pi = p + |b|^2/2$ is total pressure. The magnetic field can transfer energy to the velocity and take it back. The producer must control that exchange as one coupled state; the endpoint must carry two compatible jets; and the pressure must retain the difference between their quadratic stresses.

Let $\dot{\Delta}_j$ be homogeneous Littlewood–Paley blocks and write

$$\|f\|_{\dot{B}_{2,1}^s} := \sum_{j \in \mathbb{Z}} 2^{js} \left\| \dot{\Delta}_j f \right\|_2. \quad (214)$$

For a time norm with a tilde, the time norm is taken inside this dyadic sum. Schwartz data satisfy the usual homogeneous low-frequency normalization. The critical small-data theory in these spaces is classical. [15]

Theorem 25.1 (Quantitative coupled MHD jet). *Let $u_0, b_0 \in \mathcal{S}(\mathbb{R}^3)$ be divergence free and put*

$$\mu := \min\{\nu, \eta\}, \quad A_0 := \|u_0\|_{\dot{B}_{2,1}^{1/2}} + \|b_0\|_{\dot{B}_{2,1}^{1/2}}. \quad (215)$$

There are fixed dyadic constants $c_h, C_B > 0$ such that, if

$$A_0 \leq \varepsilon_{\text{MHD}} \mu, \quad \varepsilon_{\text{MHD}} := \frac{c_h}{4C_B}, \quad (216)$$

then (211)–(213) has a unique global smooth solution. Every finite spatial and temporal order obeys the explicit coupled producer, velocity, magnetic, total-pressure, mechanical-pressure, rectangle, endpoint, tail, datum-depth, and restart estimates (222)–(277) below.

Proof. The first cancellation already shows why (u, b) must be kept together. Pair (211) with u , pair (212) with b , and add. Incompressibility removes the two terms transported by u , while integration by parts gives

$$-\langle b \cdot \nabla b, u \rangle - \langle b \cdot \nabla u, b \rangle = 0.$$

Thus

$$\frac{1}{2} \frac{d}{dt} \left(\|u\|_2^2 + \|b\|_2^2 \right) + \nu \|\nabla u\|_2^2 + \eta \|\nabla b\|_2^2 = 0. \quad (217)$$

The energy identity alone does not control the Lipschitz action needed by the high Sobolev rows. The critical Besov producer does. For a finite horizon define

$$\mathcal{A}(T) := \|u\|_{\tilde{L}^\infty(0,T;\dot{B}_{2,1}^{1/2})} + \|b\|_{\tilde{L}^\infty(0,T;\dot{B}_{2,1}^{1/2})}, \quad (218)$$

$$\mathcal{D}(T) := \nu \|u\|_{L^1(0,T;\dot{B}_{2,1}^{5/2})} + \eta \|b\|_{L^1(0,T;\dot{B}_{2,1}^{5/2})}. \quad (219)$$

The product estimate that bears the weight is

$$\|f \cdot \nabla g\|_{\dot{B}_{2,1}^{1/2}} \leq C_B \|f\|_{\dot{B}_{2,1}^{1/2}} \|g\|_{\dot{B}_{2,1}^{5/2}}. \quad (220)$$

To see the derivative count, decompose

$$f \cdot \nabla g = T_f(\nabla g) + T_{\nabla g} f + R(f, \nabla g).$$

The three paraproduct bounds, together with $\dot{B}_{2,1}^{3/2} \hookrightarrow L^\infty$, give (220); the endpoint summation is paid by the Littlewood–Paley ℓ^1 index. [20]

Apply the dyadic heat estimate to the projected momentum equation and the induction equation. Summing over j and using (220) gives

$$\mathcal{A}(T) + c_h \mathcal{D}(T) \leq A_0 + \frac{C_B}{\mu} \mathcal{A}(T) \mathcal{D}(T). \quad (221)$$

On an interval where $\mathcal{A}(T) \leq 2A_0$, the smallness assumption (216) absorbs the last term and improves the bound. Continuity closes the bootstrap:

$$\mathcal{A}(T) + \frac{c_h}{2} \mathcal{D}(T) \leq A_0 \quad (T < T_{\max}). \quad (222)$$

If C_E is the embedding constant in $\|\nabla f\|_\infty \leq C_E \|f\|_{\dot{B}_{2,1}^{5/2}}$, then

$$\int_0^T (\|\nabla u(t)\|_\infty + \|\nabla b(t)\|_\infty) dt \leq J_0 := \frac{2C_E A_0}{c_h \mu}. \quad (223)$$

The right side no longer depends on the horizon.

Fix an integer $q \geq 3$ and set

$$Z_q(t) := \|u(t)\|_{H^q}^2 + \|b(t)\|_{H^q}^2, \quad L(t) := \|\nabla u(t)\|_\infty + \|\nabla b(t)\|_\infty. \quad (224)$$

After ∂^γ , $|\gamma| \leq q$, the leading u -transport terms vanish. The two leading b -terms cancel against each other just as they did in (217). Every remaining term is a commutator, and

$$\|[\partial^\gamma, f \cdot \nabla]g\|_2 \leq C_q (\|\nabla f\|_\infty \|g\|_{H^q} + \|f\|_{H^q} \|\nabla g\|_\infty). \quad (225)$$

Summing the two equations gives

$$\frac{1}{2} Z'_q(t) + \nu \|\nabla u\|_{H^q}^2 + \eta \|\nabla b\|_{H^q}^2 \leq C_q L(t) Z_q(t). \quad (226)$$

Define

$$\mathcal{S}_q^{\text{MHD}} := Z_q(0)^{1/2} e^{C_q J_0}. \quad (227)$$

After enlarging C_q to absorb the harmless factor from (226), one obtains

$$\sup_{0 \leq t \leq T} Z_q(t)^{1/2} \leq \mathcal{S}_q^{\text{MHD}}. \quad (228)$$

The integrated companion is

$$(\mathcal{D}_q^{\text{MHD}}(T))^2 := T(\mathcal{S}_q^{\text{MHD}})^2 + \frac{\frac{1}{2} Z_q(0) + C_q J_0 (\mathcal{S}_q^{\text{MHD}})^2}{\mu}, \quad (229)$$

for which

$$\|u\|_{L^2(0,T;H^{q+1})}^2 + \|b\|_{L^2(0,T;H^{q+1})}^2 \leq (\mathcal{D}_q^{\text{MHD}}(T))^2. \quad (230)$$

The bound (228), first at one $q > 5/2$ and then at every finite q , prevents finite-time loss of the classical coupled solution.

25.1 The coupled mixed jet

Write

$$V_n := \partial_t^n u, \quad W_n := \partial_t^n b. \quad (231)$$

Differentiating the projected equations gives

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} \mathbb{P}[(V_a \cdot \nabla) V_{n-a} - (W_a \cdot \nabla) W_{n-a}], \quad (232)$$

$$W_{n+1} = \eta \Delta W_n - \sum_{a=0}^n \binom{n}{a} [(V_a \cdot \nabla) W_{n-a} - (W_a \cdot \nabla) V_{n-a}]. \quad (233)$$

For $r \geq 0$, let $\sigma_r := \max\{2, r + 1\}$ and fix A_r so that

$$\|\mathbb{P}(f \cdot \nabla g)\|_{H^r} + \|f \cdot \nabla g\|_{H^r} \leq A_r \|f\|_{H^{\sigma_r}} \|g\|_{H^{\sigma_r}}. \quad (234)$$

Set

$$\mathbf{U}_{0,r}^{\text{MHD}} = \mathbf{B}_{0,r}^{\text{MHD}} := \mathbf{S}_{\max\{3,r\}}^{\text{MHD}}. \quad (235)$$

Define the two finite recursions

$$\mathbf{U}_{n+1,r}^{\text{MHD}} := \nu \mathbf{U}_{n,r+2}^{\text{MHD}} + A_r \sum_{a=0}^n \binom{n}{a} (\mathbf{U}_{a,\sigma_r}^{\text{MHD}} \mathbf{U}_{n-a,\sigma_r}^{\text{MHD}} + \mathbf{B}_{a,\sigma_r}^{\text{MHD}} \mathbf{B}_{n-a,\sigma_r}^{\text{MHD}}), \quad (236)$$

$$\mathbf{B}_{n+1,r}^{\text{MHD}} := \eta \mathbf{B}_{n,r+2}^{\text{MHD}} + A_r \sum_{a=0}^n \binom{n}{a} (\mathbf{U}_{a,\sigma_r}^{\text{MHD}} \mathbf{B}_{n-a,\sigma_r}^{\text{MHD}} + \mathbf{B}_{a,\sigma_r}^{\text{MHD}} \mathbf{U}_{n-a,\sigma_r}^{\text{MHD}}). \quad (237)$$

They give

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^r} \leq \mathbf{U}_{n,r}^{\text{MHD}}, \quad (238)$$

$$\sup_{0 \leq t \leq T} \|W_n(t)\|_{H^r} \leq \mathbf{B}_{n,r}^{\text{MHD}}. \quad (239)$$

At the natural negative row, fix A_{-1} so that

$$\|\mathbb{P}(f \cdot \nabla g)\|_{H^{-1}} + \|f \cdot \nabla g\|_{H^{-1}} \leq A_{-1} \|f\|_{H^1} \|g\|_{H^1}, \quad (240)$$

for the divergence-free factors used here, because $f \cdot \nabla g = \nabla \cdot (f \otimes g)$, $\|f \otimes g\|_2 \leq \|f\|_4 \|g\|_4$, and $\mathbb{P}$ is bounded on H^{-1} . Put

$$\mathbf{U}_{0,-1}^{\text{MHD}} = \mathbf{B}_{0,-1}^{\text{MHD}} := \mathbf{S}_3^{\text{MHD}} \quad (241)$$

and define

$$\mathbf{U}_{n+1,-1}^{\text{MHD}} := \nu \mathbf{U}_{n,1}^{\text{MHD}} + A_{-1} \sum_{a=0}^n \binom{n}{a} (\mathbf{U}_{a,1}^{\text{MHD}} \mathbf{U}_{n-a,1}^{\text{MHD}} + \mathbf{B}_{a,1}^{\text{MHD}} \mathbf{B}_{n-a,1}^{\text{MHD}}), \quad (242)$$

$$\mathbf{B}_{n+1,-1}^{\text{MHD}} := \eta \mathbf{B}_{n,1}^{\text{MHD}} + A_{-1} \sum_{a=0}^n \binom{n}{a} (\mathbf{U}_{a,1}^{\text{MHD}} \mathbf{B}_{n-a,1}^{\text{MHD}} + \mathbf{B}_{a,1}^{\text{MHD}} \mathbf{U}_{n-a,1}^{\text{MHD}}). \quad (243)$$

Equations (232)–(240) give

$$\sup_{0 \leq t \leq T} (\|V_n(t)\|_{H^{-1}} + \|W_n(t)\|_{H^{-1}}) \leq \mathbf{U}_{n,-1}^{\text{MHD}} + \mathbf{B}_{n,-1}^{\text{MHD}}. \quad (244)$$

This supplies the lower side of the $M = 0$ rectangle with the projected transport term retained.

25.2 Both pressures and every finite rectangle

The normalized total pressure is

$$\Pi = R_i R_j (u_i u_j - b_i b_j). \quad (245)$$

For $Q_n := \partial_t^n \Pi$,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} [(V_a)_i (V_{n-a})_j - (W_a)_i (W_{n-a})_j]. \quad (246)$$

Let

$$\tau_0 := 1, \quad \tau_1 := 2, \quad \tau_r := r \quad (r \geq 2), \quad (247)$$

and fix P_r so that

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|f_i g_j\|_{H^r} \leq P_r \|f\|_{H^{\tau_r}} \|g\|_{H^{\tau_r}}. \quad (248)$$

Then

$$\mathbf{Q}_{n,r}^{\text{MHD}} := P_r \sum_{a=0}^n \binom{n}{a} (\mathbf{U}_{a,\tau_r}^{\text{MHD}} \mathbf{U}_{n-a,\tau_r}^{\text{MHD}} + \mathbf{B}_{a,\tau_r}^{\text{MHD}} \mathbf{B}_{n-a,\tau_r}^{\text{MHD}}) \quad (249)$$

obeys

$$\sup_{0 \leq t \leq T} \|Q_n(t)\|_{H^r} \leq \mathbf{Q}_{n,r}^{\text{MHD}}. \quad (250)$$

The mechanical pressure is not discarded. Since $p = \Pi - |b|^2/2$, its row is

$$\partial_t^n p = Q_n - \frac{1}{2} \sum_{a=0}^n \binom{n}{a} W_a \cdot W_{n-a}, \quad (251)$$

Define its explicit companion envelope by

$$\mathbf{P}_{n,r}^{\text{mech}}(T) := \mathbf{Q}_{n,r}^{\text{MHD}} + \frac{P_r}{2} \sum_{a=0}^n \binom{n}{a} \mathbf{B}_{a,\tau_r}^{\text{MHD}} \mathbf{B}_{n-a,\tau_r}^{\text{MHD}}. \quad (252)$$

Then

$$\sup_{0 \leq t \leq T} \|\partial_t^n p(t)\|_{H^r} \leq \mathbf{P}_{n,r}^{\text{mech}}(T). \quad (253)$$

For $N, M \in \mathbb{N}_0$, define

$$\begin{aligned} \mathcal{N}_{N,M}^{\text{MHD}}(u, b; T) := & \sum_{n=0}^N \left[\|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|W_n\|_{L^2(0,T;H^{M+1})}^2 \right. \\ & \left. + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 + \|W_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right]. \end{aligned} \quad (254)$$

The uniform rows give the datum-side enclosure

$$\mathcal{N}_{N,M}^{\text{MHD}}(u, b; T) \leq \mathfrak{R}_{N,M}^{\text{MHD}}(T), \quad (255)$$

$$\begin{aligned} \mathfrak{R}_{N,M}^{\text{MHD}}(T) := & T \sum_{n=0}^N \left[(\mathbf{U}_{n,M+1}^{\text{MHD}})^2 + (\mathbf{B}_{n,M+1}^{\text{MHD}})^2 \right. \\ & \left. + (\mathbf{U}_{n+1,M-1}^{\text{MHD}})^2 + (\mathbf{B}_{n+1,M-1}^{\text{MHD}})^2 \right]. \end{aligned} \quad (256)$$

The negative rows just defined are used when $M = 0$. The pressure companion is

$$\sum_{n=0}^N \left[\|Q_n\|_{L^2(0,T;H^M)}^2 + \|\partial_t^n p\|_{L^2(0,T;H^M)}^2 \right] \leq T \sum_{n=0}^N \left[(\mathbf{Q}_{n,M}^{\text{MHD}})^2 + (\mathbf{P}_{n,M}^{\text{mech}}(T))^2 \right]. \quad (257)$$

Every rectangle is a restriction of the same coupled history, so all overlaps agree without choosing either field again.

The velocity and magnetic row (n, r) reads the data through no order above

$$d_{n,r}^{u,b} := \max\{3, r + 2n\}. \quad (258)$$

The total or mechanical pressure row reads no higher than

$$d_{n,r}^p := \max\{3, \tau_r + 2n\}, \quad (259)$$

and the rectangle (N, M) reads no higher than

$$d_{N,M}^{\text{rect}} := \max\{3, M + 2N + 1\}. \quad (260)$$

25.3 Coupled endpoint and restart

For every $n, m \in \mathbb{N}_0$, Banach-valued integration gives

$$\|V_n(t) - V_n(T)\|_{H^m} \leq (T - t) \mathbf{U}_{n+1,m}^{\text{MHD}}, \quad (261)$$

$$\|W_n(t) - W_n(T)\|_{H^m} \leq (T - t) \mathbf{B}_{n+1,m}^{\text{MHD}}, \quad (262)$$

$$\|Q_n(t) - Q_n(T)\|_{H^m} \leq (T - t) \mathbf{Q}_{n+1,m}^{\text{MHD}}, \quad (263)$$

$$\|\partial_t^n p(t) - \partial_t^n p(T)\|_{H^m} \leq (T - t) \mathbf{P}_{n+1,m}^{\text{mech}}(T). \quad (264)$$

The two field moduli read the data through no order above

$$\max\{3, m + 2n + 2\}. \quad (265)$$

The datum depth of both pressure moduli is at most

$$\max\{3, \tau_m + 2n + 2\}. \quad (266)$$

For every temporal row define the coupled critical height and its high-frequency tail by

$$\mathcal{H}_n^{\text{MHD}}(t) := \frac{1}{2} \int_{\mathbb{R}^3} |\xi| \left(|\widehat{V}_n(\xi, t)|^2 + |\widehat{W}_n(\xi, t)|^2 \right) d\xi, \quad (267)$$

$$\mathcal{H}_{n,>K}^{\text{MHD}}(t) := \frac{1}{2} \int_{|\xi|>K} |\xi| \left(|\widehat{V}_n|^2 + |\widehat{W}_n|^2 \right) d\xi. \quad (268)$$

Thus $\mathcal{H}_0^{\text{MHD}}$ is the base coupled height. The row envelopes give

$$\sup_{0 \leq t \leq T} \mathcal{H}_n^{\text{MHD}}(t) \leq \frac{1}{2} [(\mathbf{U}_{n,1}^{\text{MHD}})^2 + (\mathbf{B}_{n,1}^{\text{MHD}})^2], \quad (269)$$

$$\sup_{0 \leq t \leq T} \mathcal{H}_{n,>K}^{\text{MHD}}(t) \leq \frac{1}{2} K^{1-2m} [(\mathbf{U}_{n,m}^{\text{MHD}})^2 + (\mathbf{B}_{n,m}^{\text{MHD}})^2], \quad m \geq 3, \ K \geq 1. \quad (270)$$

It remains to cross a hypothetical finite terminal time with the coupled state. Use the linear constant C_{lin} from (404) and define

$$C_{\text{sys}}(\nu, \eta) := \max\{C_{\text{lin}}(\nu), C_{\text{lin}}(\eta)\}. \quad (271)$$

Let $C_{\text{quad}}^{\text{MHD}}$ be a fixed dimension-dependent constant for the sum of the four H^3 -based quadratic estimates in (211) and (212), and set

$$c_{\nu,\eta}^{\text{MHD}} := [8C_{\text{quad}}^{\text{MHD}} C_{\text{sys}}(\nu, \eta)^2]^{-2}. \quad (272)$$

For data $(a, c) \in H_\sigma^3 \times H_\sigma^3$, define the paired mild map

$$\begin{aligned}\Phi_u(w, h)(t) &:= e^{\nu t \Delta} a - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P}[(w \cdot \nabla)w - (h \cdot \nabla)h](s) ds, \\ \Phi_b(w, h)(t) &:= e^{\eta t \Delta} c - \int_0^t e^{\eta(t-s)\Delta} [(w \cdot \nabla)h - (h \cdot \nabla)w](s) ds.\end{aligned}\tag{273}$$

Equip $X_\delta \times X_\delta$, with X_δ from Appendix C, with the sum norm, and write $A = \|a\|_{H^3} + \|c\|_{H^3}$. The two linear estimates and the four quadratic products give

$$\|(\Phi_u, \Phi_b)(w, h)\|_{X_\delta \times X_\delta} \leq C_{\text{sys}} A + C_{\text{sys}} C_{\text{quad}}^{\text{MHD}} \delta^{1/2} \|(w, h)\|_{X_\delta \times X_\delta}^2,\tag{274}$$

$$\begin{aligned}\|(\Phi_u, \Phi_b)(w, h) - (\Phi_u, \Phi_b)(v, k)\|_{X_\delta \times X_\delta} &\leq C_{\text{sys}} C_{\text{quad}}^{\text{MHD}} \delta^{1/2} \left(\|(w, h)\|_{X_\delta \times X_\delta} + \|(v, k)\|_{X_\delta \times X_\delta} \right) \\ &\quad \cdot \|(w - v, h - k)\|_{X_\delta \times X_\delta}.\end{aligned}\tag{275}$$

Taking $R = 2C_{\text{sys}}A$ and $4C_{\text{sys}}C_{\text{quad}}^{\text{MHD}}\delta^{1/2}R \leq 1$ makes the radius- R ball invariant and the map contractive. Banach's theorem gives

$$\delta(\nu, \eta, a, c) \geq \min \left\{ 1, c_{\nu, \eta}^{\text{MHD}} (1 + \|a\|_{H^3} + \|c\|_{H^3})^{-2} \right\}.\tag{276}$$

At the common endpoint, $\|u(T)\|_{H^3} + \|b(T)\|_{H^3} \leq \sqrt{2}S_3^{\text{MHD}}$. Hence the same-history restart obeys

$$\delta_T^{\text{MHD}} \geq \min \left\{ 1, c_{\nu, \eta}^{\text{MHD}} (1 + \sqrt{2}S_3^{\text{MHD}})^{-2} \right\}.\tag{277}$$

Suppose instead that the maximal smooth MHD time T_{max} were finite. Apply the estimates above on $[0, \tau]$, $\tau < T_{\text{max}}$. Their uniform right sides are independent of τ , and every integrated right side is bounded by the same expression with T_{max} in place of τ . Monotone convergence gives every coupled rectangle on $(0, T_{\text{max}})$. The endpoint moduli then produce one pair $(u_*, b_*) \in H_\sigma^\infty \times H_\sigma^\infty$ and both normalized pressure jets at T_{max} . In particular, $\|u_*\|_{H^3} + \|b_*\|_{H^3} \leq \sqrt{2}S_3^{\text{MHD}}$, so (277) continues the pair beyond T_{max} . Local uniqueness identifies the new branch with the old one on their overlap. Passing to the limit in (232) and (233) identifies every right row with its left endpoint row. The joined coupled history is smooth, contradicting finite maximality. This proves the theorem under the stated critical datum condition. $\square$

26 Critical hyperdissipation in three dimensions

The axisymmetric argument changed the geometry seen by the stretching term while leaving the Laplacian untouched. We now keep unrestricted three-dimensional transport and change the dissipation instead. Consider

$$\partial_t u + (u \cdot \nabla)u + \nabla p + \nu \Lambda^{2\alpha} u = 0, \quad \nabla \cdot u = 0, \quad \Lambda := (-\Delta)^{1/2},\tag{278}$$

on $\mathbb{R}^3$, with $\alpha \geq 5/4$. At the borderline value, the basic energy dissipation carries exactly the spatial integrability needed to pay the differentiated transport term. Once that payment is made, the rest of the proof can be read from the equation: the fractional operator changes the width of each spatial rectangle, but it does not change compatibility, pressure normalization, endpoint formation, or same-history restart. The classical global regularity range begins at this exponent. ^[14]

Theorem 26.1 (Quantitative hyperdissipative jet). *Let $\nu > 0$, $\alpha \geq 5/4$, and $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$. Equation (278) has a unique global smooth solution with normalized pressure $p = R_i R_j (u_i u_j)$. For every finite horizon T , every finite temporal order, and every nonnegative real spatial order, its spatial rows, mixed velocity and pressure rows, compatible fractional rectangles, endpoint moduli, frequency tails, datum-depth ledger, and restart lifespan obey the explicit estimates (294)–(334) below.*

Proof. Put $M_0 := \|u_0\|_2$. Pairing (278) with u gives

$$\sup_{t \geq 0} \|u(t)\|_2 \leq M_0, \quad \int_0^T \|\Lambda^\alpha u(t)\|_2^2 dt \leq \frac{M_0^2}{2\nu}. \quad (279)$$

The transport term has vanished from this row, but it returns as soon as a spatial derivative is applied. The exponent $5/4$ appears when we ask whether the one integral in (279) is enough to absorb that return at every order.

Choose the finite exponent

$$q_\alpha := \begin{cases} 6/(3-2\alpha), & 5/4 \leq \alpha < 3/2, \\ 6, & \alpha \geq 3/2, \end{cases} \quad \vartheta_\alpha := \frac{3}{\alpha} \left(\frac{1}{2} - \frac{1}{q_\alpha} \right). \quad (280)$$

Then

$$\frac{3}{q_\alpha} \leq \alpha - 1, \quad 0 < \vartheta_\alpha \leq 1. \quad (281)$$

The Gagliardo–Nirenberg inequality and (279) first give

$$\|u(t)\|_{q_\alpha} \leq G_\alpha M_0^{1-\vartheta_\alpha} \|\Lambda^\alpha u(t)\|_2^{\vartheta_\alpha}. \quad (282)$$

Squaring, applying Hölder in time with exponents $1/\vartheta_\alpha$ and $1/(1-\vartheta_\alpha)$, and then using (279) yield the datum-side action

$$\int_0^T \|u(t)\|_{q_\alpha}^2 dt \leq \mathcal{A}_\alpha(M_0, \nu, T), \quad (283)$$

$$\mathcal{A}_\alpha(M_0, \nu, T) := G_\alpha^2 M_0^2 (2\nu)^{-\vartheta_\alpha} T^{1-\vartheta_\alpha}. \quad (284)$$

Fix $s \geq \alpha - 1$ and set $r = s + 1 - \alpha$. Moving α derivatives from the transport term onto the tested velocity gives

$$|\langle \Lambda^s \mathbb{P}(u \cdot \nabla u), \Lambda^s u \rangle| \leq \|\Lambda^r(u \otimes u)\|_2 \|\Lambda^{s+\alpha} u\|_2. \quad (285)$$

If $p_\alpha = 2q_\alpha/(q_\alpha - 2)$, fractional Leibniz and homogeneous Sobolev embedding give

$$\begin{aligned} \|\Lambda^r(u \otimes u)\|_2 &\leq C_{\alpha,s} \|u\|_{q_\alpha} \|\Lambda^r u\|_{p_\alpha} \\ &\leq C_{\alpha,s} \|u\|_{q_\alpha} \left\| \Lambda^{r+3/q_\alpha} u \right\|_2 \leq C_{\alpha,s} \|u\|_{q_\alpha} \|u\|_{H^s}. \end{aligned} \quad (286)$$

The last inequality is exactly $r + 3/q_\alpha \leq s$, which is the first relation in (281). Define

$$X_s(t) := M_0^2 + \|\Lambda^s u(t)\|_2^2, \quad Y_s(t) := \|\Lambda^{s+\alpha} u(t)\|_2. \quad (287)$$

Young's inequality now turns (285)–(286) into

$$X'_s(t) + \nu Y_s(t)^2 \leq \frac{C_{\alpha,s}}{\nu} \|u(t)\|_{q_\alpha}^2 X_s(t). \quad (288)$$

Set

$$\Gamma_{\alpha,s}(T) := \frac{C_{\alpha,s}}{\nu} \mathcal{A}_\alpha(M_0, \nu, T). \quad (289)$$

Gronwall's inequality, followed once by the integrating-factor form of (288), gives

$$\sup_{0 \leq t \leq T} X_s(t) \leq X_s(0) e^{\Gamma_{\alpha,s}(T)}, \quad (290)$$

$$\int_0^T Y_s(t)^2 dt \leq \frac{X_s(0)}{\nu} e^{\Gamma_{\alpha,s}(T)}. \quad (291)$$

At $\alpha = 5/4$, one has $q_\alpha = 12$ and $\vartheta_\alpha = 1$. The derivative count in (286) becomes

$$s + 1 - \alpha + \frac{3}{q_\alpha} = s,$$

and the exponent reduces to

$$\Gamma_{5/4,s}(T) = \frac{C_s G_{5/4}^2 M_0^2}{2\nu^2}. \quad (292)$$

No derivative and no time integrability remain unpaid at the critical exponent.

For any requested real $r \geq 0$, put

$$\bar{s}_r := \max\{r, \alpha - 1\}, \quad \kappa_{r,\alpha} := \sup_{\rho \geq 0} \frac{(1 + \rho^2)^r}{1 + \rho^{2\bar{s}_r}} < \infty, \quad \mathcal{S}_r^\alpha(T) := \kappa_{r,\alpha}^{1/2} X_{\bar{s}_r}(0)^{1/2} e^{\Gamma_{\alpha,\bar{s}_r}(T)/2}. \quad (293)$$

Indeed,

$$\|u(t)\|_{H^r}^2 \leq \kappa_{r,\alpha} \left(\|u(t)\|_2^2 + \|\Lambda^{\bar{s}_r} u(t)\|_2^2 \right) \leq \kappa_{r,\alpha} X_{\bar{s}_r}(t).$$

Thus

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^r} \leq \mathcal{S}_r^\alpha(T). \quad (294)$$

For the integrated row, let

$$s_r^- := \max\{\alpha - 1, r - \alpha\} \quad (295)$$

and define

$$(\mathcal{D}_{0,r}^\alpha(T))^2 := C_{\alpha,r} \left[T M_0^2 + \frac{X_{s_r^-}(0)}{\nu} e^{\Gamma_{\alpha,s_r^-}(T)} \right]. \quad (296)$$

Since $s_r^- + \alpha \geq r$, equations (279) and (291) give

$$\|u\|_{L^2(0,T;H^r)} \leq \mathcal{D}_{0,r}^\alpha(T). \quad (297)$$

Choosing one $s > \max\{5/2, \alpha - 1\}$ in (290) prevents a finite-time loss of the local classical solution. This already proves global smoothness. We keep going because the stronger conclusion is the complete quantitative jet.

26.1 The fractional mixed and pressure jets

Let $V_n := \partial_t^n u$ and $Q_n := \partial_t^n p$. Differentiating (278) and then applying $\mathbb{P}$ gives

$$V_{n+1} = -\nu \Lambda^{2\alpha} V_n - \sum_{a=0}^n \binom{n}{a} \mathbb{P}(V_a \cdot \nabla V_{n-a}). \quad (298)$$

Taking divergence before projection fixes every pressure row:

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (299)$$

For real $r \geq 0$, let $\sigma_r := \max\{2, r+1\}$, and enlarge the fixed constant A_r so that

$$\|\mathbb{P}(f \cdot \nabla g)\|_{H^r} + \|R_i R_j (f_i g_j)\|_{H^r} \leq A_r \|f\|_{H^{\sigma_r}} \|g\|_{H^{\sigma_r}}. \quad (300)$$

Starting from

$$\mathbf{V}_{0,r}^\alpha(T) := \mathbf{S}_r^\alpha(T), \quad (301)$$

define

$$\mathbf{V}_{n+1,r}^\alpha(T) := \nu \mathbf{V}_{n,r+2\alpha}^\alpha(T) + A_r \sum_{a=0}^n \binom{n}{a} \mathbf{V}_{a,\sigma_r}^\alpha(T) \mathbf{V}_{n-a,\sigma_r}^\alpha(T), \quad (302)$$

and

$$\mathbf{Q}_{n,r}^\alpha(T) := A_r \sum_{a=0}^n \binom{n}{a} \mathbf{V}_{a,\sigma_r}^\alpha(T) \mathbf{V}_{n-a,\sigma_r}^\alpha(T). \quad (303)$$

Induction through (298) and (299) proves

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^r} \leq \mathbf{V}_{n,r}^\alpha(T), \quad (304)$$

$$\sup_{0 \leq t \leq T} \|Q_n(t)\|_{H^r} \leq \mathbf{Q}_{n,r}^\alpha(T). \quad (305)$$

The integrated recursion begins with (296). Define

$$\mathbf{D}_{n+1,r}^\alpha(T) := \nu \mathbf{D}_{n,r+2\alpha}^\alpha(T) + A_r \sum_{a=0}^n \binom{n}{a} \mathbf{V}_{a,\sigma_r}^\alpha(T) \mathbf{D}_{n-a,\sigma_r}^\alpha(T). \quad (306)$$

The same product estimate, with one factor in L_t^∞ and the other in L_t^2 , gives

$$\|V_n\|_{L^2(0,T;H^r)} \leq \mathbf{D}_{n,r}^\alpha(T). \quad (307)$$

The corresponding pressure row is

$$\|Q_n\|_{L^2(0,T;H^r)} \leq \mathbf{Q}_{n,r}^{\alpha,(2)}(T) := A_r \sum_{a=0}^n \binom{n}{a} \mathbf{V}_{a,\sigma_r}^\alpha(T) \mathbf{D}_{n-a,\sigma_r}^\alpha(T). \quad (308)$$

26.2 Fractional rectangles and their common endpoint

The dissipation selects the Gelfand triple

$$H_\sigma^{M+\alpha}(\mathbb{R}^3) \hookrightarrow H_\sigma^M(\mathbb{R}^3) \hookrightarrow H_\sigma^{M-\alpha}(\mathbb{R}^3), \quad M \geq 0. \quad (309)$$

For $N \in \mathbb{N}_0$ and real $M \geq 0$, define

$$\mathcal{N}_{N,M}^\alpha(u; T) := \sum_{n=0}^N \left[\|V_n\|_{L^2(0,T;H^{M+\alpha})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-\alpha})}^2 \right]. \quad (310)$$

Write $x_+ := \max\{x, 0\}$. Since $H^0 \hookrightarrow H^r$ for $r < 0$, (307) gives

$$\mathcal{N}_{N,M}^\alpha(u; T) \leq \mathfrak{R}_{N,M}^\alpha(T), \quad (311)$$

$$\mathfrak{R}_{N,M}^\alpha(T) := \sum_{n=0}^N \left[(\mathsf{D}_{n,M+\alpha}^\alpha(T))^2 + (\mathsf{D}_{n+1,(M-\alpha)_+}^\alpha(T))^2 \right]. \quad (312)$$

The pressure also fits the upper side of the rectangle. The real-order product theorem gives

$$\|fg\|_{H^M} \leq C_{\alpha,M} \|f\|_{H^{M+\alpha}} \|g\|_{H^{M+\alpha}}, \quad (313)$$

because $M < 2(M + \alpha) - 3/2$. Hence

$$\|Q_n\|_{L^1(0,T;H^M)} \leq C_{\alpha,M} \sum_{a=0}^n \binom{n}{a} \mathsf{D}_{a,M+\alpha}^\alpha(T) \mathsf{D}_{n-a,M+\alpha}^\alpha(T). \quad (314)$$

The differentiated transport term lies in $L^1(0,T;H^{M-\alpha})$ because

$$\|\mathbb{P}(V_a \cdot \nabla V_b)\|_{H^{M-\alpha}} \leq C_{\alpha,M} \|V_a\|_{H^{M+\alpha}} \|V_b\|_{H^{M+\alpha}}. \quad (315)$$

The remaining lower-row terms obey

$$\|\Lambda^{2\alpha} V_n\|_{H^{M-\alpha}} \leq \|V_n\|_{H^{M+\alpha}}, \quad \|\nabla Q_n\|_{H^{M-\alpha}} \leq \|Q_n\|_{H^M}. \quad (316)$$

Thus every term in the differentiated momentum equation occupies the lower member of (309).

On the Fourier side the pivot factorization is exact:

$$\langle \xi \rangle^{2M} = \langle \xi \rangle^{M-\alpha} \langle \xi \rangle^{M+\alpha}. \quad (317)$$

Apply this identity first to smooth temporal cutoffs and then pass by density. For $0 \leq s < t \leq T$, the adjacent-row identity gives

$$\left| \|V_n(t)\|_{H^M}^2 - \|V_n(s)\|_{H^M}^2 \right| \leq 2 \|V_n\|_{L^2(s,t;H^{M+\alpha})} \|V_{n+1}\|_{L^2(s,t;H^{M-\alpha})}. \quad (318)$$

It follows that $V_n \in C([0,T];H^M)$ for each finite pair (n, M) . Every rectangle is cut from the same velocity and the same Riesz-normalized pressure, so deleting higher rows or lowering the Sobolev order leaves its overlap unchanged. Their intersection is one mixed jet.

The uniform next row sharpens the trace to the endpoint moduli

$$\|V_n(t) - V_n(T)\|_{H^r} \leq (T-t)V_{n+1,r}^\alpha(T), \quad (319)$$

$$\|Q_n(t) - Q_n(T)\|_{H^r} \leq (T-t)Q_{n+1,r}^\alpha(T). \quad (320)$$

If $s > k + 3/2 - 3/b$, $2 \leq b \leq \infty$, and $1 \leq a \leq \infty$, Sobolev embedding also gives

$$\|\nabla^k V_n\|_{L^a(0,T;L^b)} \leq C_{s,k,b} T^{1/a} V_{n,s}^\alpha(T), \quad (321)$$

$$\|\nabla^k Q_n\|_{L^a(0,T;L^b)} \leq C_{s,k,b} T^{1/a} Q_{n,s}^\alpha(T), \quad (322)$$

where $T^{1/\infty} = 1$. Define the sharp high-frequency multiplier by

$$\widehat{P_{>K} f}(\xi) := \mathbf{1}_{\{|\xi| > K\}} \widehat{f}(\xi). \quad (323)$$

For $K \geq 1$ and $s > k$,

$$\sup_{0 \leq t \leq T} \|P_{>K} \nabla^k V_n(t)\|_2 \leq K^{k-s} V_{n,s}^\alpha(T), \quad (324)$$

$$\sup_{0 \leq t \leq T} \|P_{>K} \nabla^k Q_n(t)\|_2 \leq K^{k-s} Q_{n,s}^\alpha(T). \quad (325)$$

The datum order stays finite at every requested coordinate. In real-order notation the velocity row (n, r) reads u_0 no higher than

$$d_{n,r}^{u,\alpha} := \max\{\alpha - 1, r + 2\alpha n\}, \quad (326)$$

and the pressure row no higher than

$$d_{n,r}^{p,\alpha} := \max\{\alpha - 1, \sigma_r + 2\alpha n\}. \quad (327)$$

The rectangle (N, M) reads no higher than

$$d_{N,M}^{\text{rect},\alpha} := \max\{\alpha - 1, M + \alpha + 2\alpha N, (M - \alpha)_+ + 2\alpha(N + 1)\}. \quad (328)$$

Taking ceilings converts these values into a classical integer-derivative ledger.

26.3 The fractional restart

Fix $s_R > 3/2$, $s_R \geq \alpha - 1$, and put $v = u(T)$, $M_R = \|v\|_{H^{s_R}}$. The fractional heat semigroup satisfies

$$\left\| \Lambda e^{-\nu t \Lambda^{2\alpha}} F \right\|_{H^{s_R}} \leq c_\alpha (\nu t)^{-1/(2\alpha)} \|F\|_{H^{s_R}}, \quad c_\alpha = (2\alpha e)^{-1/(2\alpha)}. \quad (329)$$

The operator in the mild equation consequently satisfies

$$\left\| e^{-\nu t \Lambda^{2\alpha}} \mathbb{P} \nabla \cdot F \right\|_{H^{s_R}} \leq C_{\mathbb{P}} c_\alpha (\nu t)^{-1/(2\alpha)} \|F\|_{H^{s_R}}. \quad (330)$$

Since H^{s_R} is an algebra, the mild bilinear map on $C([0, \delta]; H^{s_R})$ has norm at most

$$K_{\alpha,s_R} \nu^{-1/(2\alpha)} \delta^{1-1/(2\alpha)}, \quad K_{\alpha,s_R} := \frac{C_{s_R} c_\alpha}{1 - 1/(2\alpha)}. \quad (331)$$

Here C_{s_R} includes $C_{\mathbb{P}}$ and the H^{s_R} -algebra constant. The ball of radius $2M_R$ is invariant and contractive when

$$8K_{\alpha,s_R}\nu^{-1/(2\alpha)}\delta^{1-1/(2\alpha)}M_R \leq 1. \quad (332)$$

For $M_R > 0$, the resulting lifespan is

$$\delta_T^\alpha \geq c_{\alpha,s_R}\nu^{1/(2\alpha-1)}M_R^{-2\alpha/(2\alpha-1)}, \quad c_{\alpha,s_R} := (8K_{\alpha,s_R})^{-2\alpha/(2\alpha-1)}. \quad (333)$$

Using $M_R \leq S_{s_R}^\alpha(T)$ makes the restart entirely datum-side:

$$\delta_T^\alpha \geq c_{\alpha,s_R}\nu^{1/(2\alpha-1)}(S_{s_R}^\alpha(T))^{-2\alpha/(2\alpha-1)}. \quad (334)$$

For zero endpoint data the zero solution is global. Otherwise the local solution from v agrees with the old history wherever both exist. To make the terminal passage explicit, suppose a maximal time $T_{\max} < \infty$ existed and apply every estimate above on $[0, \tau]$, $\tau < T_{\max}$. Each right side is bounded by the same nondecreasing envelope evaluated at $T_{\max}$. Monotone convergence supplies every fractional rectangle on $(0, T_{\max})$, while (317) and (318) give one common H^∞ endpoint. The uniform next-row bounds identify the complete left jet there. Inserting that endpoint into (334) gives a positive interval crossing $T_{\max}$, and uniqueness glues it to the same solution. Finite maximality is impossible. At $\alpha = 5/4$, the lower bound has the concrete form

$$\delta_T^{5/4} \geq c_{5/4,s_R}\nu^{2/3}(S_{s_R}^{5/4}(T))^{-5/3}.$$

This completes the hyperdissipative application. $\square$

27 Elliptic regularization of the time derivative

The last change acts in a different place. Hyperdissipation strengthens the operator applied to u ; the Navier–Stokes–Voigt equation places an elliptic operator on u_t . Work on the three-dimensional torus $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with mean-zero divergence-free fields, and set

$$A := -\mathbb{P}\Delta, \quad G_\ell := (I + \ell^2 A)^{-1}, \quad B(v, w) := \mathbb{P}\nabla \cdot (v \otimes w). \quad (335)$$

For $\ell, \nu > 0$, the projected equation is

$$(I + \ell^2 A)u_t + \nu Au + B(u, u) = 0, \quad u(0) = u_0. \quad (336)$$

The inverse G_ℓ cancels the two derivatives in the viscous term and gains one derivative over transport. Temporal differentiation now costs no additional spatial order. That change is visible not just in existence, but in the exact datum ledger of the entire mixed jet. Global well-posedness and long-time dynamics for this model are classical. ^[16]

Theorem 27.1 (Quantitative Navier–Stokes–Voigt jet). *Fix $\ell, \nu > 0$. Every mean-zero $u_0 \in H_\sigma^\infty(\mathbb{T}^3)$ generates a unique global smooth solution of (336). For every $T > 0$, every $n, r, N, M \in \mathbb{N}_0$, every $m \in \mathbb{N}$ with $m \geq 2$, and every $K \geq 1$, the spatial producer, mixed velocity rows, pressure rows, compatible rectangles, filtered critical height, frequency tails, endpoint moduli, restart lifespan, and time analyticity radius obey (347)–(384) below. The datum order of a velocity or pressure row is independent of its temporal height.*

Proof. Write

$$|v|_s := \left\| A^{s/2} v \right\|_2 \quad (s \in \mathbb{R}), \quad E_q(t) := |u(t)|_{q-1}^2 + \ell^2 |u(t)|_q^2 \quad (q \geq 1). \quad (337)$$

Throughout this section the mean-zero space $H_\sigma^q(\mathbb{T}^3)$ is equipped with the A -scale norm $|\cdot|_q$. Poincaré's inequality makes it equivalent to the usual inhomogeneous Sobolev norm, while this choice keeps every displayed constant tied to the equation's own Fourier multiplier. Pairing (336) with u gives the exact modified energy identity

$$E_1(t) + 2\nu \int_0^t |u(s)|_1^2 ds = E_1(0). \quad (338)$$

Put $K_1 := E_1(0)$. At the next order, pairing with Au gives

$$\frac{1}{2} E_2'(t) + \nu |u|_2^2 = -\langle B(u, u), Au \rangle.$$

The three-dimensional interpolation estimate

$$|\langle B(u, u), Au \rangle| \leq C \|u\|_6 \|\nabla u\|_3 |u|_2 \leq C |u|_1^{3/2} |u|_2^{3/2} \quad (339)$$

and Young's inequality yield

$$E_2'(t) + \nu |u|_2^2 \leq C_2 \nu^{-3} \ell^{-6} K_1^3. \quad (340)$$

Define

$$K_2(T) := E_2(0) + C_2 \nu^{-3} \ell^{-6} K_1^3 T. \quad (341)$$

For every integer $q \geq 3$, pair the equation with $A^{q-1}u$. The periodic Sobolev algebra estimate gives

$$|B(u, u)|_{q-2} \leq C_q |u|_{q-1}^2. \quad (342)$$

Hence

$$\begin{aligned} |\langle B(u, u), A^{q-1}u \rangle| &\leq C_q |u|_{q-1}^2 |u|_q \\ &\leq \frac{\nu}{2} |u|_q^2 + C_q \nu^{-1} |u|_{q-1}^4. \end{aligned} \quad (343)$$

Starting from (341), define recursively

$$K_q(T) := E_q(0) + C_q \nu^{-1} \ell^{-4} \int_0^T K_{q-1}(t)^2 dt \quad (q \geq 3). \quad (344)$$

Induction gives

$$\sup_{0 \leq t \leq T} E_q(t) \leq K_q(T), \quad (345)$$

$$\nu \int_0^T |u(t)|_q^2 dt \leq K_q(T) \quad (q \geq 2), \quad (346)$$

while (338) gives $\int_0^T |u|_1^2 \leq K_1/(2\nu)$. In particular,

$$\sup_{0 \leq t \leq T} |u(t)|_q \leq \ell^{-1} K_q(T)^{1/2}. \quad (347)$$

Every K_q is a computable nondecreasing polynomial in T , with

$$\deg K_q \leq 2^{q-1} - 1 \quad (q \geq 2). \quad (348)$$

Its datum dependence enters only through $E_j(0) = |u_0|_{j-1}^2 + \ell^2 |u_0|_j^2$, $j \leq q$.

The base phase space already explains globality. The Fourier multipliers satisfy

$$\|G_\ell A v\|_{H^1} \leq \ell^{-2} \|v\|_{H^1}, \quad \|G_\ell B(v, w)\|_{H^1} \leq C \ell^{-2} \|v\|_{H^1} \|w\|_{H^1}. \quad (349)$$

Thus (336) is an ordinary differential equation on H_σ^1 . Its local solution can end only if the H^1 norm becomes unbounded, and (338) prevents that. The higher polynomials then propagate every smooth datum order globally.

27.1 A mixed jet with no temporal derivative loss

Let $V_n := \partial_t^n u$. Applying G_ℓ after differentiating (336) gives

$$V_{n+1} = -\nu G_\ell A V_n - G_\ell \sum_{a=0}^n \binom{n}{a} B(V_a, V_{n-a}). \quad (350)$$

For $r \in \mathbb{N}_0$, put $\sigma_r := \max\{2, r-1\}$. The same multipliers and Sobolev multiplication give

$$|G_\ell A v|_r \leq \ell^{-2} |v|_r, \quad (351)$$

$$|G_\ell B(v, w)|_r \leq C_r \ell^{-2} |v|_{\sigma_r} |w|_{\sigma_r}. \quad (352)$$

Define the spatial base values

$$S_0^V(T) := K_1^{1/2}, \quad S_r^V(T) := \ell^{-1} K_r(T)^{1/2} \quad (r \geq 1), \quad (353)$$

and set $U_{0,r}^V := S_r^V$. The row recursion is

$$U_{n+1,r}^V(T) := \ell^{-2} \left[\nu U_{n,r}^V(T) + C_r \sum_{a=0}^n \binom{n}{a} U_{a,\sigma_r}^V(T) U_{n-a,\sigma_r}^V(T) \right]. \quad (354)$$

It follows that

$$\sup_{0 \leq t \leq T} |V_n(t)|_r \leq U_{n,r}^V(T). \quad (355)$$

Unlike the Laplacian and fractional recursions, the right side of (354) never asks for a higher spatial order of V_n . Consequently,

$$U_{n,r}^V(T) \text{ reads } u_0 \text{ only through } H^{\max\{2,r\}}, \quad (356)$$

independently of n .

The integrated rows are just as explicit. Put

$$\begin{aligned} D_{0,0}^V(T) &:= T^{1/2} S_0^V(T), \\ D_{0,1}^V(T) &:= \left(\frac{K_1}{2\nu} \right)^{1/2}, \\ D_{0,r}^V(T) &:= \left(\frac{K_r(T)}{\nu} \right)^{1/2} \quad (r \geq 2). \end{aligned} \quad (357)$$

Define

$$D_{n+1,r}^V(T) := \ell^{-2} \left[\nu D_{n,r}^V(T) + C_r \sum_{a=0}^n \binom{n}{a} U_{a,\sigma_r}^V(T) D_{n-a,\sigma_r}^V(T) \right]. \quad (358)$$

Taking one factor in L_t^∞ and the other in L_t^2 gives

$$\|V_n\|_{L^2(0,T;H^r)} \leq 2^{r+/2} D_{n,r}^V(T), \quad r_+ := \max\{r, 0\}. \quad (359)$$

For the lower side of the $M = 0$ rectangle, set

$$\sigma_{-1} := 2, \quad U_{0,-1}^V := S_0^V, \quad D_{0,-1}^V := T^{1/2} S_0^V, \quad (360)$$

and extend (354) and (358) to $r = -1$ using

$$|G_\ell A v|_{-1} \leq \ell^{-2} |v|_{-1}, \quad |G_\ell B(v, w)|_{-1} \leq C \ell^{-2} |v|_2 |w|_2. \quad (361)$$

27.2 Pressure, rectangles, and the filtered critical height

Restoring pressure gives

$$(I - \ell^2 \Delta) u_t - \nu \Delta u + (u \cdot \nabla) u + \nabla p = 0. \quad (362)$$

The filtered time derivative remains divergence free, so

$$Q_n := \partial_t^n p = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (363)$$

With $\tau_r := \max\{2, r\}$, define

$$P_{n,r}^V(T) := C_r \sum_{a=0}^n \binom{n}{a} U_{a,\tau_r}^V(T) U_{n-a,\tau_r}^V(T), \quad (364)$$

$$\Pi_{n,r}^V(T) := C_r \sum_{a=0}^n \binom{n}{a} U_{a,\tau_r}^V(T) D_{n-a,\tau_r}^V(T). \quad (365)$$

Then

$$\sup_{0 \leq t \leq T} \|Q_n(t)\|_{H^r} \leq P_{n,r}^V(T), \quad (366)$$

$$\|Q_n\|_{L^2(0,T;H^r)} \leq \Pi_{n,r}^V(T). \quad (367)$$

The pressure row reads no datum order above $\max\{2, r\}$, again independently of n .

For $N, M \in \mathbb{N}_0$, let

$$\mathcal{N}_{N,M}^V(u; T) := \sum_{n=0}^N \left[\|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right]. \quad (368)$$

Equations (359)–(361) give

$$\mathcal{N}_{N,M}^V(u; T) \leq \mathfrak{R}_{N,M}^V(T), \quad (369)$$

$$\mathfrak{R}_{N,M}^V(T) := \sum_{n=0}^N \left[2^{M+1} (D_{n,M+1}^V(T))^2 + 2^{(M-1)+} (D_{n+1,M-1}^V(T))^2 \right]. \quad (370)$$

The complete rectangle reads u_0 only through

$$H^{\max\{2, M+1\}}, \quad (371)$$

independently of N . Compatibility is exact because every row and every pressure is a restriction of one periodic solution.

The natural critical quantity for this equation includes its time filter:

$$\mathcal{H}^V(t) := \frac{1}{2} \left(|u(t)|_{1/2}^2 + \ell^2 |u(t)|_{3/2}^2 \right). \quad (372)$$

Poincaré monotonicity and the first two energy envelopes give

$$\sup_{0 \leq t \leq T} \mathcal{H}^V(t) \leq \frac{1}{2} (\ell^{-2} K_1 + K_2(T)). \quad (373)$$

Define the ordinary kinetic high-frequency tail by

$$H_{>K}^V(t) := \frac{1}{2} \sum_{|k| > K} |k| |\hat{u}(k, t)|^2. \quad (374)$$

Then

$$\sup_{0 \leq t \leq T} H_{>K}^V(t) \leq \frac{1}{2} K^{1-2m} (S_m^V(T))^2, \quad m \in \mathbb{N}, \quad m \geq 2, \quad K \geq 1. \quad (375)$$

27.3 Endpoint, restart, and time analyticity

The uniform next row gives

$$|V_n(T) - V_n(t)|_r \leq (T - t) U_{n+1,r}^V(T), \quad (376)$$

$$\|Q_n(T) - Q_n(t)\|_{H^r} \leq (T - t) P_{n+1,r}^V(T). \quad (377)$$

Every finite rectangle reaches the same H^∞ endpoint and the same normalized pressure jet.

On an H^1 ball of radius R , the vector field in (349) has Lipschitz constant at most $C\ell^{-2}(\nu + R)$. Picard iteration, with the torus and Poincaré constants absorbed into c_V, C_V , gives

$$\delta(w) \geq c_V \min \left\{ 1, \frac{\ell^2}{\nu + 1 + \|w\|_{H^1}} \right\}. \quad (378)$$

Since $\|u(T)\|_{H^1} \leq C_V \ell^{-1} K_1^{1/2}$,

$$\delta_T^V \geq c_V \min \left\{ 1, \frac{\ell^2}{\nu + 1 + C_V \ell^{-1} K_1^{1/2}} \right\}. \quad (379)$$

The bound does not deteriorate with the horizon. It crosses any alleged finite maximal time, and uniqueness glues the restarted curve to the old history. Induction through (350) identifies the entire right jet with the left endpoint jet.

The lossless recursion gives one further quantitative consequence. At $r = 2$, set

$$R_T := S_2^V(T), \quad L_T := \ell^{-2}(\nu + C_2 R_T). \quad (380)$$

Since

$$\sum_{a=0}^n \binom{n}{a} a!(n-a)! = (n+1)n!, \quad (381)$$

induction in (354) gives

$$U_{n,2}^V(T) \leq n! R_T L_T^n. \quad (382)$$

Taylor's formula with integral remainder and (382) give, whenever the segment from t to $t+h$ lies in $[0, T]$,

$$\left\| u(t+h) - \sum_{j=0}^N \frac{h^j}{j!} V_j(t) \right\|_{H^2} \leq 2R_T (L_T |h|)^{N+1}. \quad (383)$$

Hence the solution is real-time analytic as an H_σ^2 -valued curve with radius at least

$$\rho_T^V \geq \frac{1}{L_T} = \frac{\ell^2}{\nu + C_2 \ell^{-1} K_2(T)^{1/2}}. \quad (384)$$

The analytic-ODE continuation supplies the same radius at interval endpoints. The pressure recurrence has the same radius because its binomial sum is bounded by $C(n+1)! R_T^2 L_T^n$. This completes an all-data global three-dimensional application in which the full temporal tower requires no increase in datum depth. $\square$

28 What the applications have tested

The common construction has now survived four distinct changes to the equation. What remained fixed was not a particular estimate. In each case a datum-side producer first controlled every requested finite spatial row; the actual differentiated equation then generated the temporal and pressure rows; those rows agreed under restriction; their common endpoint satisfied the limiting recurrence; and a quantitative local theorem restarted the same history. The table records the part that genuinely changed.

Equation	Datum-side producer	Cost of one time row	Quantitative conclusion
Axisymmetric no swirl	Maximum principle for ω^θ/r , followed by the stretching and logarithmic gradient bounds	Two spatial derivatives	All smooth data on $\mathbb{R}^3$; complete jet and restart
Viscous-resistive MHD	Small critical Besov norm and integrable coupled Lipschitz action	Two spatial derivatives in both fields	Small critical data on $\mathbb{R}^3$; both field jets and both pressure conventions
Hyperdissipative flow	Kinetic energy plus the sharp fractional product estimate	2α spatial derivatives	All smooth data on $\mathbb{R}^3$ for $\alpha \geq 5/4$; real-order rectangles
Navier-Stokes-Voigt	Filtered energy hierarchy E_q	No increase in spatial order	All smooth periodic data; complete jet, restart, and time-analyticity radius

Table 1: The producer changes; finite compatibility, endpoint formation, and restart do not.

The applications reach farther than four global-existence statements. They show which parts of the proof are geometric, which depend on the order of dissipation, which survive coupling, and which improve when the time derivative is elliptically filtered. In every case the output remains a finite datum-resolved computation at each requested coordinate rather than an unquantified appeal to smoothness.

A The analytic facts used by the proof

The main argument uses the following analytic ingredients at several derivative orders. We collect them here so that every Sobolev, multiplier, trace, and local-construction step invoked in the body is proved or attached to a precise source.

A.1 Fourier convention, Sobolev spaces, and frequency cutoffs

For $f \in \mathcal{S}(\mathbb{R}^3)$, set

$$\widehat{f}(\xi) := (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix \cdot \xi} f(x) dx, \quad \langle \xi \rangle := (1 + |\xi|^2)^{1/2}.$$

With this normalization, Plancherel's theorem is an isometry. For every real s , define

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} |\widehat{f}(\xi)|^2 d\xi.$$

The completion defines $H^s(\mathbb{R}^3)$, and H_σ^s is its closed divergence-free subspace. All multipliers below act componentwise on vector fields, and repeated spatial indices are summed. For $s_1 \geq s_0$, the pointwise inequality between Fourier weights gives the continuous embedding

$$H^{s_1} \hookrightarrow H^{s_0}.$$

The homogeneous operator $\Lambda = (-\Delta)^{1/2}$ has symbol $|\xi|$. If $0 \leq \theta \leq k$, then

$$\left\| \Lambda^\theta f \right\|_2^2 = \int_{\mathbb{R}^3} |\xi|^{2\theta} |\widehat{f}(\xi)|^2 d\xi \leq \|f\|_{H^k}^2.$$

This comparison is what allows an inhomogeneous rectangle to control each homogeneous half-integer energy used in Part II. Every homogeneous quantity in the paper is evaluated on a function already known to lie in the indicated inhomogeneous Sobolev space, so no quotient by polynomials enters the argument.

The all-orders space is

$$H^\infty(\mathbb{R}^3) := \bigcap_{m \in \mathbb{N}_0} H^m(\mathbb{R}^3), \quad H_\sigma^\infty := H^\infty \cap H_\sigma^0.$$

Its topology is generated by the seminorms $f \mapsto \|f\|_{H^m}$. Thus $f_j \rightarrow f$ in H^∞ precisely when convergence holds in each fixed H^m . No sum over m , and no constant uniform in m , is part of this definition.

Choose a real, radial function $\chi \in C_c^\infty(\mathbb{R}^3)$ with $\chi = 1$ on $\{|\xi| \leq 1\}$ and $\chi = 0$ on $\{|\xi| \geq 2\}$, and define the low-frequency cutoff S_R by

$$\widehat{S_R f}(\xi) := \chi(\xi/R) \widehat{f}(\xi).$$

Dominated convergence gives $S_R f \rightarrow f$ strongly in every H^s . The same cutoffs act boundedly on every Bochner space $L^q(J; H^s)$ with $1 \leq q < \infty$, where the strong convergence follows first on simple functions and then by density. These properties justify the frequency truncations used in the critical-height and endpoint-trace calculations.

A.2 Dyadic shells and the Sobolev weight they carry

Part II resolves the critical height into dyadic shells, so the relation between a shell amplitude and a Sobolev norm has to be exact up to fixed partition constants. Let

$$\varphi(\xi) := \chi(\xi) - \chi(2\xi), \quad \varphi_j(\xi) := \varphi(2^{-j}\xi), \quad \widehat{P_j f} := \varphi_j \widehat{f}.$$

For $\xi \neq 0$, the telescoping sum satisfies

$$\sum_{j \in \mathbb{Z}} \varphi_j(\xi) = 1.$$

Each φ_j is supported where $2^{j-1} \leq |\xi| \leq 2^{j+1}$, and any fixed frequency belongs to only finitely many of these supports.

Lemma A.1 (Dyadic Sobolev equivalence). *For every $s \in \mathbb{R}$, there are constants $0 < c_s \leq C_s < \infty$, depending only on s and the chosen partition, such that*

$$c_s \|\Lambda^s f\|_2^2 \leq \sum_{j \in \mathbb{Z}} 2^{2js} \|P_j f\|_2^2 \leq C_s \|\Lambda^s f\|_2^2 \quad (385)$$

whenever either side is finite. If $K = 2^j$ and $A_K(f) := \frac{1}{2} \|P_j f\|_2^2$, then

$$\frac{1}{2} \|\Lambda^s f\|_2^2 \simeq_s \sum_K K^{2s} A_K(f).$$

Proof. Plancherel gives

$$\sum_j 2^{2js} \|P_j f\|_2^2 = \int_{\mathbb{R}^3} \left(\sum_j 2^{2js} |\varphi_j(\xi)|^2 \right) |\widehat{f}(\xi)|^2 d\xi.$$

On the support of φ_j , the quantities 2^j and $|\xi|$ differ by a factor between 1/2 and 2, so $2^{2js} \simeq_s |\xi|^{2s}$. Finite overlap gives the upper bound in (385). For the lower bound, Cauchy-Schwarz and the uniform overlap number give

$$1 = \left| \sum_j \varphi_j(\xi) \right|^2 \leq C \sum_j |\varphi_j(\xi)|^2 \quad (\xi \neq 0).$$

Multiplication by the comparable weights yields

$$|\xi|^{2s} \leq C_s \sum_j 2^{2js} |\varphi_j(\xi)|^2.$$

Integration against $|\widehat{f}|^2$ proves the lower bound. The single frequency $\xi = 0$ has Lebesgue measure zero and contributes nothing to the L^2 identity. $\square$

For the critical height $H = \frac{1}{2} \|\Lambda^{1/2} u\|_2^2$, the lemma gives

$$H(t) \simeq \sum_K K A_K(u(t)).$$

At the next half-integer height, the same shell amplitude carries the weight K^3 , and each further full spatial derivative multiplies that weight by K^2 . This is the precise frequency statement behind the vertical movement through the critical-height tower.

A.3 Spatial embeddings and interpolation

The whole-space Sobolev embedding used throughout the proof follows directly from Fourier inversion. If $|\alpha| \leq k$, then

$$\begin{aligned} \|\partial^\alpha f\|_\infty &\leq (2\pi)^{-3/2} \int_{\mathbb{R}^3} |\xi|^{|\alpha|} |\widehat{f}(\xi)| d\xi \\ &\leq C \|f\|_{H^s} \left(\int_{\mathbb{R}^3} |\xi|^{2|\alpha|} \langle \xi \rangle^{-2s} d\xi \right)^{1/2}. \end{aligned}$$

The last integral is finite under the strict inequality $s > |\alpha| + 3/2$. Consequently,

$$H^s(\mathbb{R}^3) \hookrightarrow C_b^k(\mathbb{R}^3) \quad \text{whenever} \quad s > k + \frac{3}{2}. \quad (386)$$

The finite- L^p version follows from the same dyadic decomposition. Let $2 \leq p \leq \infty$, $|\alpha| \leq k$, and $s > k + \frac{3}{2} - \frac{3}{p}$. Bernstein's inequality on the support of P_j gives^[20]

$$\|\partial^\alpha P_j f\|_p \leq C_{k,p} 2^{j(k+3/2-3/p)} \|P_j f\|_2 \quad (j \geq 0).$$

Put $\delta = s - k - \frac{3}{2} + \frac{3}{p} > 0$. Cauchy–Schwarz in j yields

$$\begin{aligned} \sum_{j \geq 0} \|\partial^\alpha P_j f\|_p &\leq C_{k,p} \sum_{j \geq 0} 2^{-j\delta} (2^{js} \|P_j f\|_2) \\ &\leq C_{s,k,p} \|f\|_{H^s}. \end{aligned}$$

The remaining low block $S_{1/2}f$ obeys the same bound directly by Bernstein's inequality. Thus

$$H^s(\mathbb{R}^3) \hookrightarrow W^{k,p}(\mathbb{R}^3) \quad \text{whenever} \quad 2 \leq p \leq \infty, \quad s > k + \frac{3}{2} - \frac{3}{p}. \quad (387)$$

The strict condition is a single sufficient range for every p ; no endpoint claim is needed in the body. In particular,

$$H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3).$$

This embedding turns the upper coordinate of the first concrete rectangle into an integrable Lipschitz coefficient, and it makes the local lifespan depend only on the base H^3 norm.

Two homogeneous estimates drive the first differentiated energy calculation. For $f \in C_c^\infty(\mathbb{R}^3)$, the whole-space Sobolev inequality and Hölder interpolation give^[22]

$$\|f\|_6 \leq C \|\nabla f\|_2, \quad \|f\|_r \leq \|f\|_2^\theta \|f\|_6^{1-\theta}, \quad \frac{1}{r} = \frac{\theta}{2} + \frac{1-\theta}{6}, \quad 2 \leq r \leq 6. \quad (388)$$

The interpolation estimate follows by writing $|f|^r = (|f|^2)^{\theta r/2} (|f|^6)^{(1-\theta)r/6}$ and applying Hölder with the conjugate exponents forced by the displayed relation. Density extends both estimates to the spaces in which they are used. Applied componentwise to ∇u , the Sobolev inequality gives

$$\|\nabla u\|_6 \leq C \|\nabla^2 u\|_2 = C \|\Delta u\|_2,$$

where the equality follows from Plancherel after summing all second derivatives.

The critical response uses the homogeneous endpoint

$$\|f\|_{L^3(\mathbb{R}^3)} \leq C \left\| \Lambda^{1/2} f \right\|_2.$$

For Schwartz functions this is the Hardy–Littlewood–Sobolev inequality applied to $f = \Lambda^{-1/2}(\Lambda^{1/2}f)$, and density gives the stated form^[22]. Applying it to u , ∇u , and Λu gives

$$\|u\|_3 \leq C \left\| \Lambda^{1/2} u \right\|_2, \quad \|\nabla u\|_3 + \|\Lambda u\|_3 \leq C \left\| \Lambda^{3/2} u \right\|_2.$$

These are the three factors used in the critical-response estimate (36).

A.4 Products, transport, and commutators

The multiplication estimates can be read directly from the Fourier convolution formula. For $s \geq 0$, the elementary weight inequality

$$\langle \xi \rangle^s \leq C_s (\langle \eta \rangle^s + \langle \xi - \eta \rangle^s)$$

and Young’s convolution inequality give

$$\|fg\|_{H^s} \leq C_s \left(\left\| \langle \cdot \rangle^s \widehat{f} \right\|_2 \|\widehat{g}\|_1 + \left\| \widehat{f} \right\|_1 \left\| \langle \cdot \rangle^s \widehat{g} \right\|_2 \right).$$

When $s > 3/2$, Cauchy–Schwarz gives $\left\| \widehat{f} \right\|_1 \leq C_s \|f\|_{H^s}$, because $\langle \xi \rangle^{-s} \in L^2(\mathbb{R}^3)$. Hence

$$\|fg\|_{H^s} \leq C_s \|f\|_{H^s} \|g\|_{H^s}, \quad s > \frac{3}{2}. \quad (389)$$

This is the inhomogeneous Sobolev-algebra estimate; its fractional and unequal-order variants follow from the same Littlewood–Paley calculus^[20].

The lowest rectangle requires a separate estimate because $H^1(\mathbb{R}^3)$ is below the algebra threshold. Interpolation between L^2 and L^6 gives $H^1 \hookrightarrow L^4$, so Hölder gives

$$\|fg\|_2 \leq \|f\|_4 \|g\|_4 \leq C \|f\|_{H^1} \|g\|_{H^1}. \quad (390)$$

For every integer $M \geq 1$, the algebra estimate at order $M + 1 > 3/2$ gives

$$\|fg\|_{H^M} \leq C_M \|f\|_{H^{M+1}} \|g\|_{H^{M+1}}.$$

Together with (390), this proves for all $M \in \mathbb{N}_0$ the rectangle product law

$$\|fg\|_{H^M} \leq C_M \|f\|_{H^{M+1}} \|g\|_{H^{M+1}}.$$

The two cases have different proofs, and keeping them separate prevents an algebra property from being attributed to H^1 .

Taking one more derivative before using the algebra gives the complete-row transport estimate

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^m} \leq C_m \|f\|_{H^{m+2}} \|g\|_{H^{m+2}}, \quad m \in \mathbb{N}_0.$$

For the base local construction, the sharper placement is

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^2} \leq C \|f\|_{H^3} \|g\|_{H^3}. \quad (391)$$

Indeed, (389) at $s = 3$ controls $f \otimes g$ in H^3 , after which divergence lowers the order by one and the Leray projection preserves it.

For higher-order energy estimates, the undifferentiated transport term must be kept apart from the commutator containing the differentiated coefficients. Write $\mathcal{J}^m = (1 - \Delta)^{m/2}$. The Kato–Ponce commutator theorem gives, for every integer $m \geq 3$,

$$\|[\mathcal{J}^m, f \cdot \nabla]g\|_2 \leq C_m (\|\nabla f\|_\infty \|g\|_{H^m} + \|f\|_{H^m} \|\nabla g\|_\infty). \quad (392)$$

This is the exact specialization of the standard commutator estimate used here^[23]. When $f = g = w$, it becomes

$$\|[\mathcal{J}^m, w \cdot \nabla]w\|_2 \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}.$$

The integer multi-index form used in Section 4 is the same estimate before the derivatives are collected:

$$\sum_{|\alpha| \leq m} |([\partial^\alpha, w \cdot \nabla]w, \partial^\alpha w)_2| \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}^2.$$

The embedding (386) supplies the L^∞ factors in every application made in the paper.

A.5 Riesz transforms, the Leray projection, and normalized pressure

For $\xi \neq 0$, the Riesz transform R_j has symbol $i\xi_j/|\xi|$, and the Leray projection has matrix symbol

$$\widehat{\mathbb{P}}(\xi) = I - \frac{\xi \otimes \xi}{|\xi|^2}.$$

Their values at $\xi = 0$ may be fixed arbitrarily. Since these symbols are bounded, Plancherel gives, for every $s \in \mathbb{R}$,

$$\|R_j f\|_{H^s} \leq \|f\|_{H^s}, \quad \|\mathbb{P}f\|_{H^s} \leq C \|f\|_{H^s}. \quad (393)$$

The same bounds hold for the vector and tensor fields used below. These are the Sobolev instances of the order-zero multiplier theorem^[24].

The pressure formula, including its sign, follows from the unprojected momentum equation. If $\nabla \cdot u = 0$, then

$$(u \cdot \nabla)u_i = \partial_j(u_j u_i).$$

Taking the divergence cancels the time and viscous terms and gives

$$-\Delta p = \partial_i \partial_j (u_i u_j). \quad (394)$$

The symbol of $R_i R_j$ is $-\xi_i \xi_j / |\xi|^2$, so

$$p = R_i R_j (u_i u_j) \quad (395)$$

solves (394). If two such pressures belong to $L^2(\mathbb{R}^3)$, their difference h is a harmonic L^2 function. The distributional identity $|\xi|^2 \widehat{h} = 0$ says that the L^2 function $\widehat{h}$ is supported at the single point $\xi = 0$, and hence $h = 0$. Requiring the pressure to belong to any nonnegative inhomogeneous Sobolev space consequently fixes both the spatial constant and any time-dependent constant.

Let $F = \nabla \cdot (u \otimes u)$. Because $-\Delta p = \nabla \cdot F$, the Fourier symbols of the two sides give

$$\mathbb{P}F = F + \nabla p. \quad (396)$$

Thus the projected equation and the normalized pressure-complete equation are equivalent in every Sobolev class used in the paper. Temporal and spatial derivatives commute with the Riesz transforms and with $\mathbb{P}$, so the same normalization passes through every row of the mixed jet.

A.6 The two Leibniz rules behind the mixed jet

The temporal recurrence and the mixed spatial recurrence are generated by ordinary differentiation of one product. Writing both combinatorial factors once prevents either factor from being hidden when a pressure or transport row is used later.

Lemma A.2 (Mixed Leibniz identity). *Let $n \in \mathbb{N}_0$, let α be a spatial multi-index, and let f, g be smooth enough for the derivatives below to exist. Then*

$$\partial_t^n \partial_x^\alpha (fg) = \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (\partial_t^a \partial_x^\beta f) (\partial_t^{n-a} \partial_x^{\alpha-\beta} g). \quad (397)$$

The same identity holds for the Sobolev rows used in this paper.

Proof. The one-variable Leibniz rule gives

$$\partial_t^n (fg) = \sum_{a=0}^n \binom{n}{a} (\partial_t^a f) (\partial_t^{n-a} g).$$

For each summand, the multi-index spatial rule gives

$$\partial_x^\alpha (FG) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (\partial_x^\beta F) (\partial_x^{\alpha-\beta} G).$$

Spatial and temporal derivatives commute, so applying the second identity to every term in the first produces (397). For the Sobolev rows used here, spatial smoothing gives the classical identity, and the product estimates above pass every term to the limit in the fixed target topology. $\square$

Applied to the product in (395) and followed by $R_i R_j$, the lemma gives

$$\partial_t^n \partial_x^\alpha p = R_i R_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (\partial_t^a \partial_x^\beta u_i) (\partial_t^{n-a} \partial_x^{\alpha-\beta} u_j),$$

which is the formula used for the pressure coordinates in (29) and (42). Applied to $u \cdot \nabla u$, it gives the corresponding transport rows. Every term at a fixed (n, α) uses only temporal orders at most n and spatial orders at most $|\alpha| + 1$, which is the finite dependency count required by each rectangle.

B Time regularity beyond the middle trace

Lemma 12.1 already proves that the two outer rows $L^2 H^{s+1}$ and $L^2 H^{s-1}$ meet in one strong H^s trace. This appendix records the shifted pairing behind that argument, the Banach-valued fundamental theorem of calculus used to identify time derivatives, and the higher time-Sobolev embedding used for the complete jet.

B.1 Shifted duality

The Hilbert triple is

$$H^{s+1} \hookrightarrow H^s \hookrightarrow H^{s-1}.$$

For $f \in H^{s-1}$ and $g \in H^{s+1}$, define the outer-space pairing with H^s as pivot by

$$\langle f, g \rangle_{H^{s-1}, H^{s+1}, H^s} := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} \widehat{f}(\xi) \cdot \overline{\widehat{g}(\xi)} d\xi.$$

The exact factorization

$$\langle \xi \rangle^{2s} = \langle \xi \rangle^{s-1} \langle \xi \rangle^{s+1}$$

and Cauchy–Schwarz give

$$\left| \langle f, g \rangle_{H^{s-1}, H^{s+1}, H^s} \right| \leq \|f\|_{H^{s-1}} \|g\|_{H^{s+1}}. \quad (398)$$

This pairing is the exact one used both in the terminal trace and in the linear H^3 estimate of Appendix C.

B.2 The Banach-valued fundamental theorem of calculus

The time-jet construction first identifies a derivative in distributions and then needs to know that the corresponding row is a genuine time-differentiable curve. The following elementary result is the bridge.

Lemma B.1 (Banach-valued fundamental theorem of calculus). *Let X be a Banach space, let $J = (a, b)$ be a bounded interval, and suppose $v, g \in L^1(J; X)$ satisfy $\partial_t v = g$ in distributions. Then v has a unique absolutely continuous representative on $[a, b]$, still denoted by v , and*

$$v(t) - v(r) = \int_r^t g(s) ds \quad (a \leq r \leq t \leq b). \quad (399)$$

In particular, if $g \in C([a, b]; X)$, then $v \in C^1([a, b]; X)$ and its classical derivative is g .

Proof. Fix $t_0 \in J$ and define the Bochner integral

$$G(t) := \int_{t_0}^t g(s) ds.$$

The map G is absolutely continuous, belongs to $L^1(J; X)$, and has distributional derivative g . Thus $h := v - G$ belongs to $L^1(J; X)$ and has distributional derivative zero.

Let ρ_ε be a scalar time mollifier and set $h_\varepsilon = \rho_\varepsilon * h$ on the interior interval $J_\varepsilon = (a + \varepsilon, b - \varepsilon)$. Convolution commutes with the distributional derivative, so $(h_\varepsilon)' = 0$ and h_ε is an X -valued constant on J_ε . On any compact interval $K \Subset J$, the functions h_ε converge to h in $L^1(K; X)$. Since each h_ε is constant on K , the identity

$$|K| \|h_\varepsilon - h_\delta\|_X = \|h_\varepsilon - h_\delta\|_{L^1(K; X)}$$

shows that these constants form a Cauchy family. Their limit is a vector $c_K \in X$, and $h = c_K$ almost everywhere on K . Overlapping compact intervals force the constants to agree, so one vector $c \in X$ satisfies $h = c$ almost everywhere on J . Consequently, $v = c + G$ almost everywhere, and this formula supplies the absolutely continuous representative and identity (399). Two continuous representatives agreeing almost everywhere agree everywhere, which proves uniqueness. If g is continuous, the ordinary differentiation theorem for the Bochner integral gives $G' = g$ at every time. $\square$

B.3 Higher time regularity

After the compatible rectangles have been intersected, the paper also uses the ordinary embedding of a finite time-Sobolev row into classical time derivatives.

Lemma B.2 (Time-Sobolev embedding with values in a Hilbert space). *Let X be a Hilbert space, let $r \geq 1$ be an integer, and define*

$$H^r(J; X) := \left\{ v \in L^2(J; X) : \partial_t^j v \in L^2(J; X), \ 1 \leq j \leq r \right\}.$$

If $a \in \mathbb{N}_0$ and $r > a + \frac{1}{2}$, then

$$H^r(J; X) \hookrightarrow C^a(\bar{J}; X). \tag{400}$$

Proof. Since r and a are integers, the strict inequality says $a \leq r - 1$. For each $0 \leq j \leq a$, the functions $\partial_t^j v$ and $\partial_t^{j+1} v$ belong to $L^2(J; X)$, and hence to $L^1(J; X)$ because J is bounded. Lemma B.1 gives a continuous representative of $\partial_t^j v$, and it identifies the derivative of the j -th representative with the $(j + 1)$ -st. Induction from $j = 0$ through $j = a$ yields $v \in C^a(\bar{J}; X)$. The embedding is continuous by the closed graph theorem, or directly from (399) after choosing a time at which the pointwise norm is bounded by the L^2 average. $\square$

Taking $X = H^m(\mathbb{R}^3)$ in (400) and then applying (386) converts the all-orders Bochner control into the mixed classical regularity used in the endpoint and local-jet arguments.

C Local construction, pressure recovery, and restart

The endpoint argument has to restart the same pressure-normalized equation from the terminal velocity. This section uses the time tools collected in Appendix B to construct that local solution in the base space H_σ^3 , prove that its lifespan depends only on the base norm and the viscosity, and retain every higher order already present in the datum. The result is the local theorem stated as Proposition 1.1.

C.1 The linear H^3 estimate

Let $J_\delta = [t_0, t_0 + \delta]$ with $0 < \delta \leq 1$, and consider

$$z_t - \nu \Delta z = F, \quad z(t_0) = a,$$

where $a \in H^3$ and $F \in L^2(J_\delta; H^2)$. For a smooth solution, take the H^3 energy pairing and place the force in the shifted H^2, H^4 duality from (398). This gives

$$\frac{1}{2} \frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 = \operatorname{Re} \langle F, z \rangle_{H^2, H^4; H^3}. \quad (401)$$

The inhomogeneous Fourier weights satisfy the exact identity

$$\|z\|_{H^4}^2 = \|z\|_{H^3}^2 + \|\nabla z\|_{H^3}^2. \quad (402)$$

Using shifted duality, (402), and Young's inequality in (401) gives

$$\frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 \leq C_\nu \left(\|z\|_{H^3}^2 + \|F\|_{H^2}^2 \right).$$

Gronwall's inequality, time integration, and (402) now yield

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right).$$

The equation gives $z_t = \nu \Delta z + F \in L^2(J_\delta; H^2)$, and hence

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} + \|z_t\|_{L^2(J_\delta; H^2)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right). \quad (403)$$

The constant is uniform for $0 < \delta \leq 1$. Its viscosity dependence can be kept explicit. Define

$$L_1(\nu) := e^{\nu/2} \max\{1, \nu^{-1/2}\},$$

$$L_\nabla(\nu) := \max \left\{ \left(\frac{1 + \nu e^\nu}{\nu} \right)^{1/2}, \left(\frac{e^\nu + \nu^{-1}}{\nu} \right)^{1/2} \right\},$$

and

$$C_{\text{lin}}(\nu) := 1 + L_1(\nu) + (1 + \nu)(L_1(\nu)^2 + L_\nabla(\nu)^2)^{1/2}. \quad (404)$$

Indeed, Young's inequality in (401) gives

$$\frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 \leq \nu \|z\|_{H^3}^2 + \nu^{-1} \|F\|_{H^2}^2.$$

Gronwall on an interval of length at most one gives the $L_1(\nu)$ coefficient for the uniform H^3 norm. Integration gives the $L_\nabla(\nu)$ coefficient for the gradient term. The exact splitting (402) and $z_t = \nu\Delta z + F$ then show that $C_{\text{lin}}(\nu)$ is one admissible constant on the right of (403).

For completeness, this estimate also constructs the nonsmooth solution. Apply the cutoff S_R to the datum and force, solve the resulting frequency-truncated equation, and use (403) on differences. The truncated solutions are Cauchy in all three spaces on the left of (403). Their limit solves the equation in distributions, and Lemma 12.1 with $m = 3$ supplies the strong H^3 trace and the value $z(t_0) = a$. Uniqueness follows by applying the same estimate to the difference of two solutions.

Taking the spatial Fourier transform and solving the scalar ordinary differential equation at each frequency gives the equivalent Duhamel formula

$$z(t) = e^{\nu(t-t_0)\Delta}a + \int_{t_0}^t e^{\nu(t-s)\Delta}F(s)ds.$$

The plus sign records the sign of the force F ; the Navier–Stokes force inserted below is negative.

C.2 The nonlinear fixed point and the base lifespan

Applying the Leray projection to the velocity equation gives the mild map

$$\Phi(w)(t) := e^{\nu(t-t_0)\Delta}a - \int_{t_0}^t e^{\nu(t-s)\Delta}\mathbb{P}\nabla \cdot (w \otimes w)(s)ds. \quad (405)$$

Set

$$X_\delta := C(J_\delta; H_\sigma^3) \cap L^2(J_\delta; H^4), \quad \|w\|_{X_\delta} := \|w\|_{CH^3} + \|w\|_{L^2H^4}.$$

All abbreviated time norms in this subsection are taken over J_δ . Let C_{quad} be a valid dimension-dependent constant in the local product estimate (391). It gives

$$\|\mathbb{P}\nabla \cdot (w \otimes w)\|_{L^2H^2} \leq C_{\text{quad}}\delta^{1/2} \|w\|_{CH^3}^2, \quad (406)$$

and the decomposition

$$w \otimes w - v \otimes v = (w - v) \otimes w + v \otimes (w - v)$$

gives

$$\|\mathbb{P}\nabla \cdot (w \otimes w - v \otimes v)\|_{L^2H^2} \leq C_{\text{quad}}\delta^{1/2} (\|w\|_{CH^3} + \|v\|_{CH^3}) \|w - v\|_{CH^3}. \quad (407)$$

Estimate (403) applied to (405) yields

$$\|\Phi(w)\|_{X_\delta} \leq C_{\text{lin}}(\nu) \|a\|_{H^3} + C_{\text{lin}}(\nu)C_{\text{quad}}\delta^{1/2} \|w\|_{X_\delta}^2$$

and

$$\|\Phi(w) - \Phi(v)\|_{X_\delta} \leq C_{\text{lin}}(\nu)C_{\text{quad}}\delta^{1/2} (\|w\|_{X_\delta} + \|v\|_{X_\delta}) \|w - v\|_{X_\delta}.$$

If $a = 0$, the zero solution is the unique fixed point. For $a \neq 0$, take $R = 2C_{\text{lin}}(\nu) \|a\|_{H^3}$ and choose $\delta \leq 1$ so that

$$4C_{\text{lin}}(\nu)C_{\text{quad}}\delta^{1/2}R \leq 1.$$

Then Φ maps the closed X_δ -ball of radius R into itself and has Lipschitz constant at most $1/2$ there. Banach's fixed-point theorem produces a solution $w \in X_\delta$. The choice above permits the quantitative lower bound

$$\delta(\nu, \|a\|_{H^3}) \geq \min \{1, c_\nu(1 + \|a\|_{H^3})^{-2}\}. \quad (408)$$

Here one may take

$$c_\nu := (8C_{\text{quad}}C_{\text{lin}}(\nu)^2)^{-2}. \quad (409)$$

Only the H^3 size of the datum and the fixed viscosity enter this lifespan.

The fixed point is unique throughout X_δ . If $w, v \in X_\delta$ have the same datum, divide J_δ into finitely many intervals whose lengths make the coefficient obtained from (407) and (403) smaller than one. The difference vanishes on the first interval. Its common endpoint becomes zero initial data on the next interval, and finite induction gives $w = v$ throughout J_δ . The heat semigroup and $\mathbb{P}$ preserve divergence-free fields, so $w(t) \in H_\sigma^3$. The projected equation and (406) also give

$$w_t \in L^2(J_\delta; H^2),$$

which completes the base regularity in (3).

C.3 Recovering the normalized pressure

The fixed point satisfies

$$w_t + \mathbb{P}\nabla \cdot (w \otimes w) = \nu \Delta w.$$

Define

$$q := R_i R_j (w_i w_j).$$

Because $w \in C(J_\delta; H^3)$ and H^3 is an algebra, the Riesz bounds give $q \in C(J_\delta; H^3)$. With $F = \nabla \cdot (w \otimes w)$, identity (396) gives

$$\mathbb{P}F = F + \nabla q.$$

Thus the projected fixed point solves

$$w_t + (w \cdot \nabla)w + \nabla q = \nu \Delta w, \quad \nabla \cdot w = 0.$$

The condition $q(t) \in H^3(\mathbb{R}^3)$ fixes the whole-space normalization at each time. Any second pressure in the same class has harmonic L^2 difference and hence coincides with q . Uniqueness of the projected fixed point consequently gives uniqueness of the pressure-complete pair.

C.4 Persistence on the base lifespan

Fix an integer $m \geq 3$ and first take smooth data. Apply $\mathcal{J}^m = (1 - \Delta)^{m/2}$ to the projected equation and pair with $\mathcal{J}^m w$. The operators $\mathcal{J}^m$, $\mathbb{P}$, and Δ commute. Since $\mathcal{J}^m w$ is divergence-free,

$$((w \cdot \nabla) \mathcal{J}^m w, \mathcal{J}^m w)_2 = 0.$$

After this cancelling transport term is separated, the energy identity is

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 = -\operatorname{Re}([\mathcal{J}^m, w \cdot \nabla]w, \mathcal{J}^m w)_2.$$

The commutator estimate (392) gives

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}^2. \quad (410)$$

This is the local form of the higher-order continuation estimate (14). The base solution belongs to $C(J_\delta; H^3)$, and (386) puts ∇w in $C(J_\delta; L^\infty)$. Thus the coefficient in (410) is integrable on the whole base interval. Gronwall gives

$$\|w(t)\|_{H^m}^2 \leq \|a\|_{H^m}^2 \exp\left(2C_m \int_{t_0}^t \|\nabla w(s)\|_\infty ds\right). \quad (411)$$

The dissipation term in (410), together with the weight splitting at order m , gives

$$w \in L^2(J_\delta; H^{m+1}).$$

The product estimate then puts $\mathbb{P}\nabla \cdot (w \otimes w)$ in $L^2(J_\delta; H^{m-1})$, so the equation gives

$$w_t \in L^2(J_\delta; H^{m-1}).$$

Lemma 12.1 supplies $w \in C(J_\delta; H^m)$.

For a general datum $a \in H_\sigma^m$, let $a^{(k)} = S_k a$. The data converge to a in H^m , and their H^3 norms are uniformly bounded. The lifespan (408) hence places every smooth solution $w^{(k)}$ on one common interval determined by the base norm. The contraction estimate makes $w^{(k)}$ converge to the H^3 fixed point in $CH^3 \cap L^2 H^4$. Estimate (411) gives a uniform $L^\infty H^m$ bound, while (410) gives a uniform $L^2 H^{m+1}$ bound and the equation gives a uniform $L^2 H^{m-1}$ bound for the time derivative. Weak compactness and the already known H^3 limit identify all three weak limits with the same base solution w . Lemma 12.1 gives its H^m -continuous representative. The energy inequality passed to the limit gives

$$\limsup_{t \downarrow t_0} \|w(t)\|_{H^m} \leq \|a\|_{H^m},$$

while weak convergence to the prescribed distributional trace gives the reverse lower bound. Uniform convexity of H^m then yields $w(t) \rightarrow a$ strongly in H^m as $t \downarrow t_0$. This completes persistence for nonsmooth H^m data without changing the base lifespan.

If $a \in H_\sigma^\infty$, repeat the argument at each fixed m . The solutions obtained at different orders coincide by the H^3 uniqueness already proved. One solution consequently satisfies

$$w \in C(J_\delta; H^m) \cap L^2(J_\delta; H^{m+1}), \quad w_t \in L^2(J_\delta; H^{m-1}) \quad (m \in \mathbb{N}, m \geq 3). \quad (412)$$

The constants may depend on m , while the interval remains the one selected by the H^3 datum.

C.5 The pressure-complete time jet

All spatial orders in (412) allow the equation to generate temporal rows without leaving the construction interval. Set $V_0 = w$. Given rows $V_0, \dots, V_n$ that are continuous in every finite Sobolev space, define

$$Q_n := R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j} \in C(J_\delta; H^m) \quad (m \in \mathbb{N}_0) \quad (413)$$

and

$$V_{n+1} := \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n \in C(J_\delta; H^m) \quad (m \in \mathbb{N}_0). \quad (414)$$

The membership statements follow from the product estimates above after choosing two extra spatial orders for diffusion and one extra order for transport and pressure. They are asserted one target m at a time, so no constant uniform across the tower is required.

The recursive symbols are now identified with the actual derivatives. At $n = 0$, the pressure-complete equation gives $\partial_t V_0 = V_1$ in distributions. Since V_1 is continuous in every target H^m , Lemma B.1 gives $V_0 \in C^1(J_\delta; H^m)$ and identifies its classical derivative with V_1 . Suppose inductively that $V_j = \partial_t^j w$ for $0 \leq j \leq n$. Differentiate the pressure formula and momentum equation n times in distributions. The temporal instance of Lemma A.2 produces the two binomial sums in (413) and (414), hence $\partial_t V_n = V_{n+1}$. Another application of the Banach-valued fundamental theorem promotes that distributional derivative to a classical derivative in every fixed H^m . Induction gives

$$\partial_t^n w, \partial_t^n q \in C(J_\delta; H^m) \quad (n, m \in \mathbb{N}_0).$$

This argument constructs derivatives at each finite order; it makes no assertion that the resulting formal Taylor series converges in time.

Evaluation at $t = t_0$ shows that the complete right-hand jet is determined by the single datum a :

$$V_0(t_0) = a,$$

$$Q_n(t_0) = R_i R_j \sum_{b=0}^n \binom{n}{b} V_{b,i}(t_0) V_{n-b,j}(t_0),$$

and

$$V_{n+1}(t_0) = \nu \Delta V_n(t_0) - \sum_{b=0}^n \binom{n}{b} (V_b(t_0) \cdot \nabla) V_{n-b}(t_0) - \nabla Q_n(t_0).$$

These are the normalized right-hand recurrences compared with the terminal left jet in (130) and (134). The local construction starts with the same pressure gauge and the same equation-generated derivatives that arrive from the old interval.

C.6 Maximal histories and the H^3 alternative

Starting from t_0 , join successive local intervals whenever the current solution has an H^3 endpoint value. Uniqueness identifies the solutions on every overlap, so their union is one pressure-normalized history on a maximal interval $[t_0, T_{\max})$.

Assume $T_{\max} < \infty$ and

$$\sup_{t_0 \leq t < T_{\max}} \|w(t)\|_{H^3} \leq M.$$

The lifespan bound (408) supplies the same positive number

$$\delta_M := \min\{1, c_\nu(1 + M)^{-2}\}$$

whenever the local theorem is restarted from a time $t < T_{\max}$. Choose $t > T_{\max} - \delta_M/2$ and start the construction from the actual datum $w(t)$. The new local solution exists past $T_{\max}$, and uniqueness identifies it with the maximal history on their overlap. This contradicts maximality. Hence

$$T_{\max} < \infty \implies \limsup_{t \uparrow T_{\max}} \|w(t)\|_{H^3} = \infty.$$

If the initial datum belongs to H^m for some $m \geq 3$, the persistence argument keeps that order on every local piece of the maximal history. A bounded H^m norm bounds the H^3 norm, so the same late-slice restart proves the corresponding H^m alternative under its datum hypothesis. The construction, pressure recovery, persistence, and continuation argument together prove Proposition 1.1.

At the endpoint in Part IV, the compatible intersection supplies $u_* \in H_\sigma^\infty$, and in particular $u_* \in H_\sigma^3$. The construction above produces a normalized solution on a right-hand interval whose length depends only on ν and $\|u_*\|_{H^3}$. The pressure-complete jet calculation shows that its right derivatives agree with the already reconstructed left jet. The matching jets make the piecewise pair classical across T_* , while local uniqueness preserves the original pair on its old interval.

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